Implementation report

Four applications are deployed on App Service. Each one pulls its image from Azure Container Registry. Traffic Manager reports every endpoint Online, and the storefront is live on the UI hostname.

Document control

Product	PetStore — catalog, orders, and administration
Date	23 October 2026
Resource group	Petstore-Resource-Group
Subscription	Petstore subscription
App Service plan	ASP-PetstoreResourceGroup-9881 · Basic B3 · Central India
Registry	petstoreazurecontainerregistry.azurecr.io
Evidence	Running shop and Azure portal screenshots. Secret values are not printed.

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16 Azure Cosmos DB
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18 App Service scale-out
19 Azure Function HTTP trigger
20 Microsoft Entra External ID
21 Order item reserver
22 Azure Container Apps

Deployed applications

Web app	Image pulled from the registry	State
product-service-web-app	product-service-image · port 8081	Running, endpoint Online
order-service-wep-app	order-service-image · port 8083	Running, endpoint Online
pet-service-web-app	pet-service-image · port 8082	Running, endpoint Online
petstoreui-web-app	petstore-ui-image · port 80	Running, shop live, endpoint Online

1. What this project covers

PetStore is a small shop in which every pet is a product for sale. A visitor can browse, filter, and open a pet without signing in. Buying, order history, and administration are signed-in flows. The

browser talks to two APIs. A third service shares the same database and the same secret source.

Area	What is in place	Shown by
Storefront	Catalog, category chips, details, cart drawer, checkout	Figure 1, Figure 2
Customer account	Separate shop session, order history, logout	Figure 3
Administration	Categories, pets with a required photo, all orders, delete	Figure 4, Figure 5, Figure 6
Database	Azure Database for PostgreSQL Flexible Server, database `petstore`	Figure 7, Figure 8
Images	Azure Blob Storage account `petstore`, container `pet-container`	Figure 9, Figure 10
Secrets	Key Vault `db-credentials-petstore`. No database password, connection string, or account key in application YAML	Figure 11, Figure 12
Container images	Azure Container Registry `petstoreazurecontainerregistry`. GitHub Actions pushes four images on every push to `main`	Figure 13, Figure 14, Figure 15, Figure 16
Hosting	Four Linux web apps deployed on plan `ASP-PetstoreResourceGroup-9881`, Basic B3, Central India. Each app pulls its image from the container registry. The live shop is Figure 22	Figure 17 through Figure 23
Routing	Four Traffic Manager profiles, routing method Priority. Online means that deployed app answered the health probe	Figure 24 through Figure 28

product-service

Port 8081. Pets for sale, categories, and image upload. Reads the database username and password, plus the blob connection string and account key, from Key Vault.

order-service

Port 8083. Register, login, checkout, and orders. Issues the JWT. Makes sure the admin account exists. Reads the database username and password from Key Vault.

pet-service

Port 8082. Pet API on the same database. Reads the database username and password from the same vault. The shop UI sells through the product service.

The React application runs on port 3000. Swagger is available at `http://localhost:8081/swagger-ui.html` and `http://localhost:8083/swagger-ui.html`.

2. How the pieces fit

The browser keeps the cart in local storage. Login is issued by the order service. The product service trusts the same JWT. Shop and admin sessions are stored separately, so signing in as a customer does not sign the administrator out.

Caller	Calls	For
Storefront and admin UI	product-service `:8081`	Categories, pets, photos
Storefront and admin UI	order-service `:8083`	Register, login, checkout, orders
All three services	Key Vault `db-credentials-petstore`	Database username and password
product-service	Blob container `pet-container`	Upload, replace, and delete pet photos

Caller	Calls	For
All three services	PostgreSQL `petstore-server-name`	Database `petstore`
GitHub Actions on `main`	Container Registry `petstoreazurecontainerregistry`	Build and store the four application images
App Service	Those four images	Public HTTPS sites for the shop and the three APIs
Traffic Manager	The App Service hostnames	DNS answer for an endpoint whose health probe is passing

Application accounts

These are shop accounts stored in the database. They are not the Azure or database secrets.

Username	Password	Role
demo	demo123	USER
admin	admin123	ADMIN

> On startup, Hibernate creates any missing tables (`ddl-auto: update`). The order service creates the admin account when that username is missing. It does not reset an existing admin password.

Sections 10 and 11 are the hosted form of this same layout. App Service runs the four images. Traffic Manager watches those hostnames and answers DNS with an endpoint that is Online.

3. Storefront

The public shop is the home page. There is no login wall. Category chips are built from the pets already loaded, so a guest can filter by All, Cat, Dog, or Parrot. Each card shows the photo, category, name, description, and price, with Details and Add to cart. A new pet cannot be saved without an image, which is why every card in the catalog has a photograph.

Add to cart asks a guest to sign in. For a signed-in shopper it opens the cart drawer on the same page. Checkout needs a shop login, then clears the cart and opens that customer's orders.

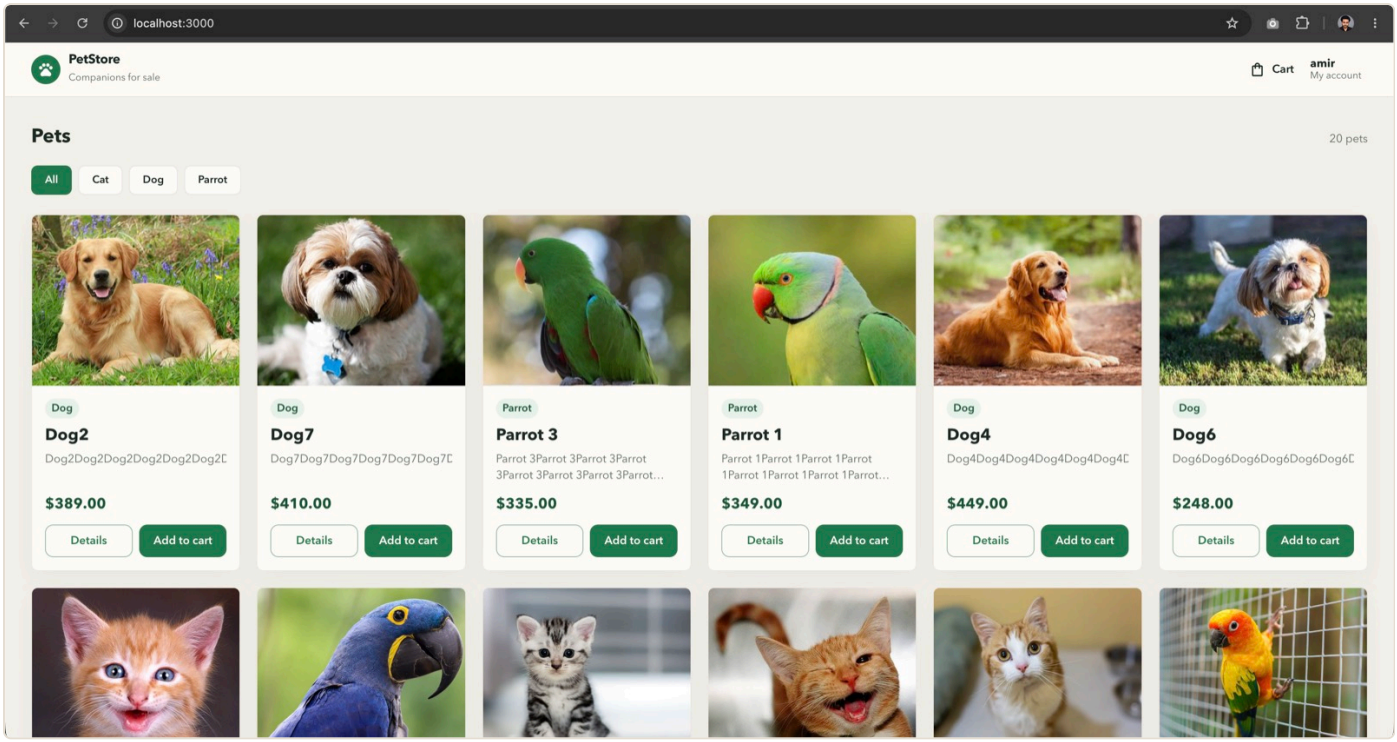

Figure 1. Storefront at `localhost:3000`. Twenty pets, category filters, and a signed-in shopper named amir. The header shows Cart and My account. There is no admin button on the shop.

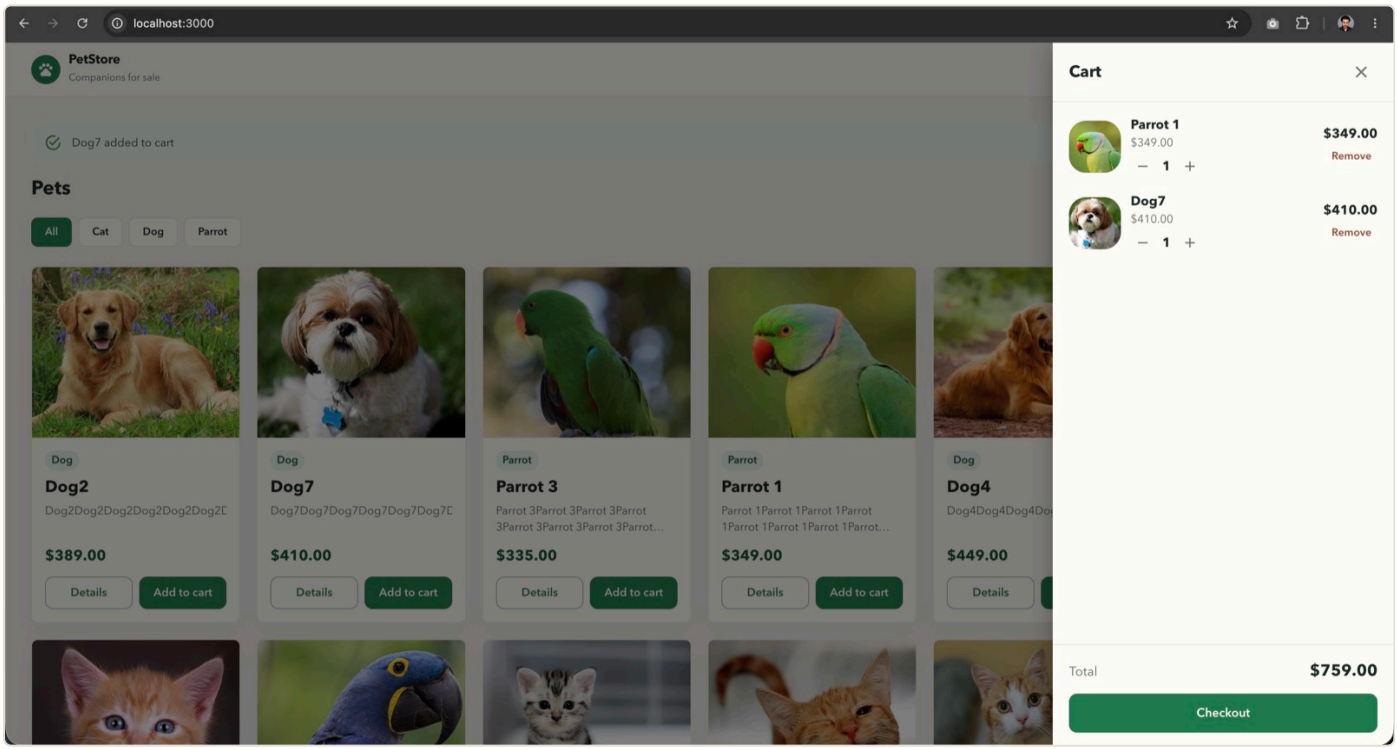

Figure 2. Cart drawer after adding Dog7. Parrot 1 and Dog7 are listed with quantity controls. The total is $759.00 and Checkout stays on this page until the order is placed.

4. Customer orders

A signed-in customer opens My account and sees only their own orders. The page is a data grid with search, column controls, and paging. Columns are Order, Items, Total, Status, and Placed. Customers cannot delete an order from this page. Delete stays on the admin orders screen.

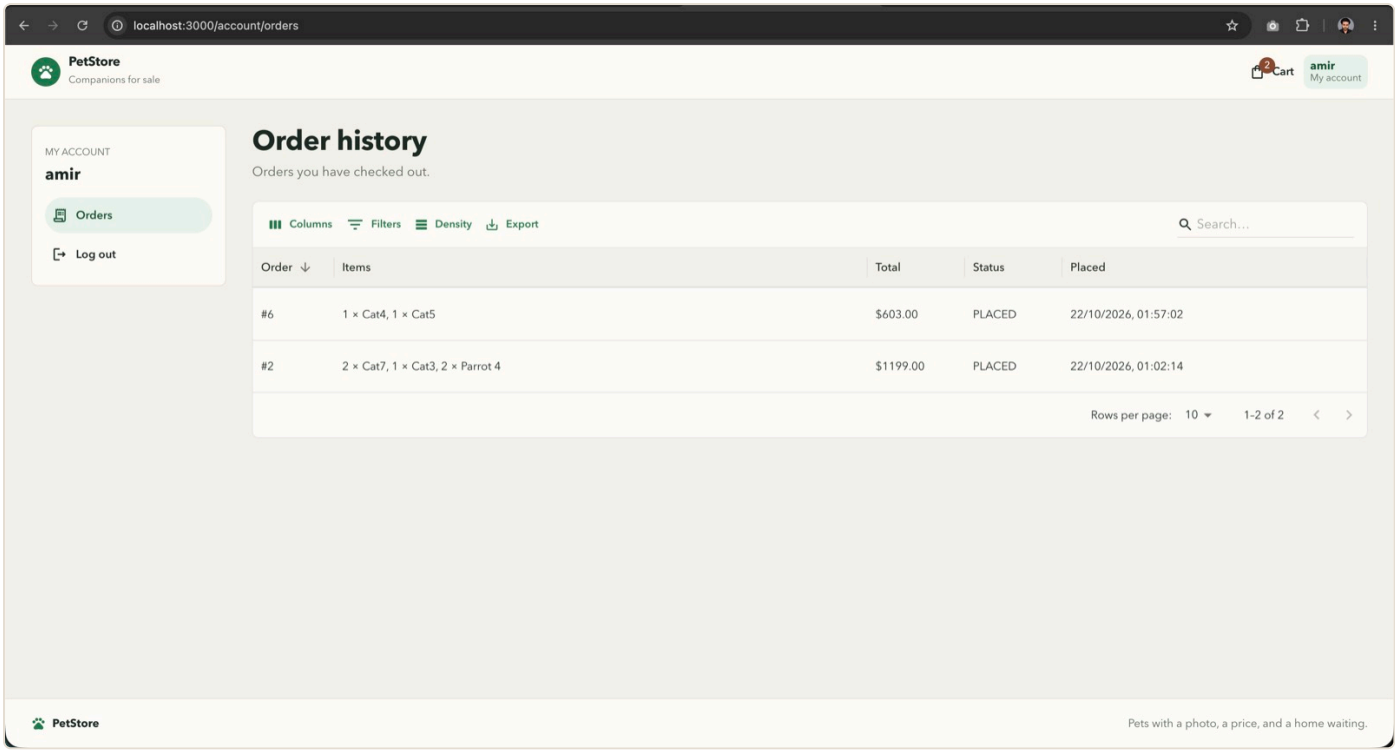

Figure 3. Account page for amir. Order #6 is two cats for $603.00. Order #2 is three pets for $1,199.00. Both are PLACED. Log out is in the account menu, not in the shop header.

5. Administration

Administration lives at `/admin` and uses its own session. The navigation is Categories, Pets, and Orders. The signed-in administrator in these screenshots is `admin`. View store returns to the public shop without clearing the admin session.

Categories

Categories are created here before they can be chosen on a pet. The screenshot shows Cat, Dog, and Parrot, which are the same chips on the storefront. Each row can be deleted. The grid has search, filters, density, and export.

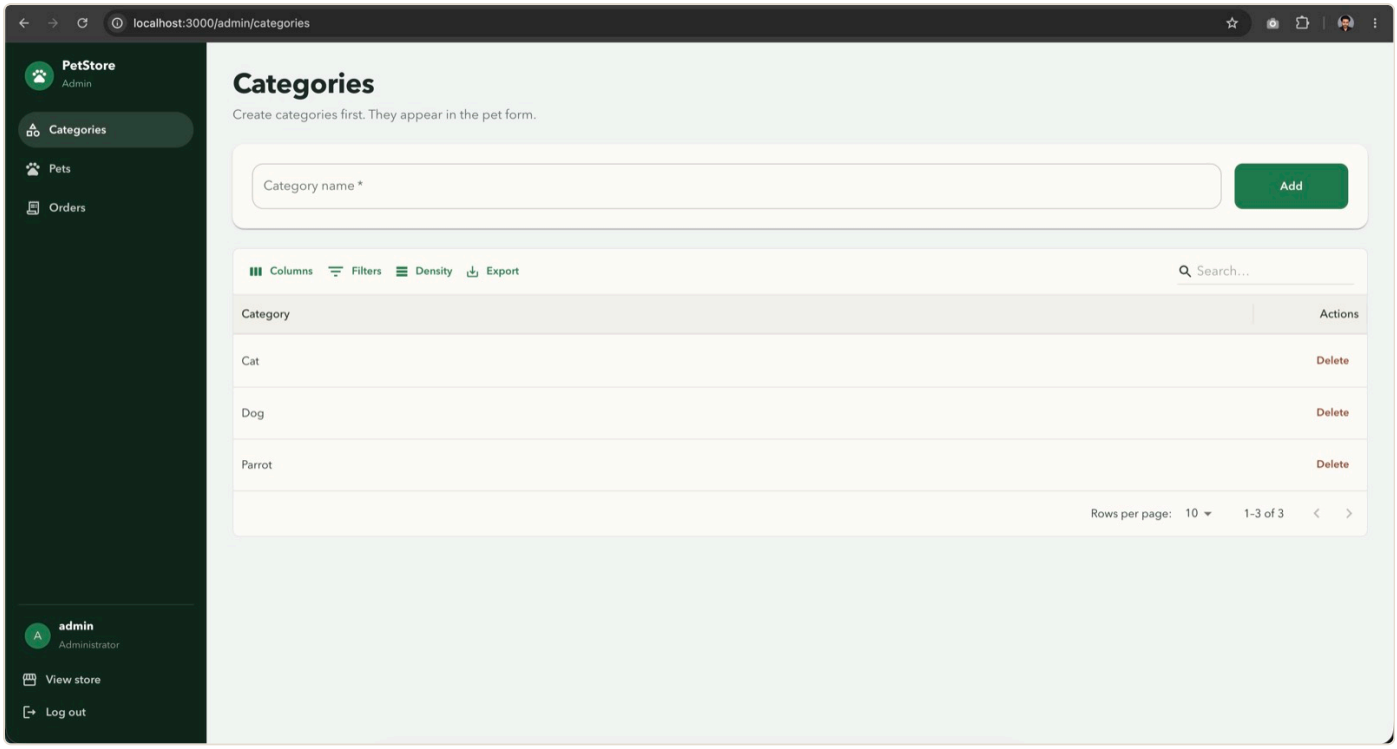

Figure 4. Admin categories. Three categories are in use. The add form is on the same page.

Pets

The pet list is the catalog the shop sells. Columns are Photo, Name, Category, Description, Price, and Actions. Edit and Delete are on each row. A new pet requires a photo. Uploads go to `pet-container`. Replacing a photo deletes the previous blob. Deleting a pet deletes its blob as well.

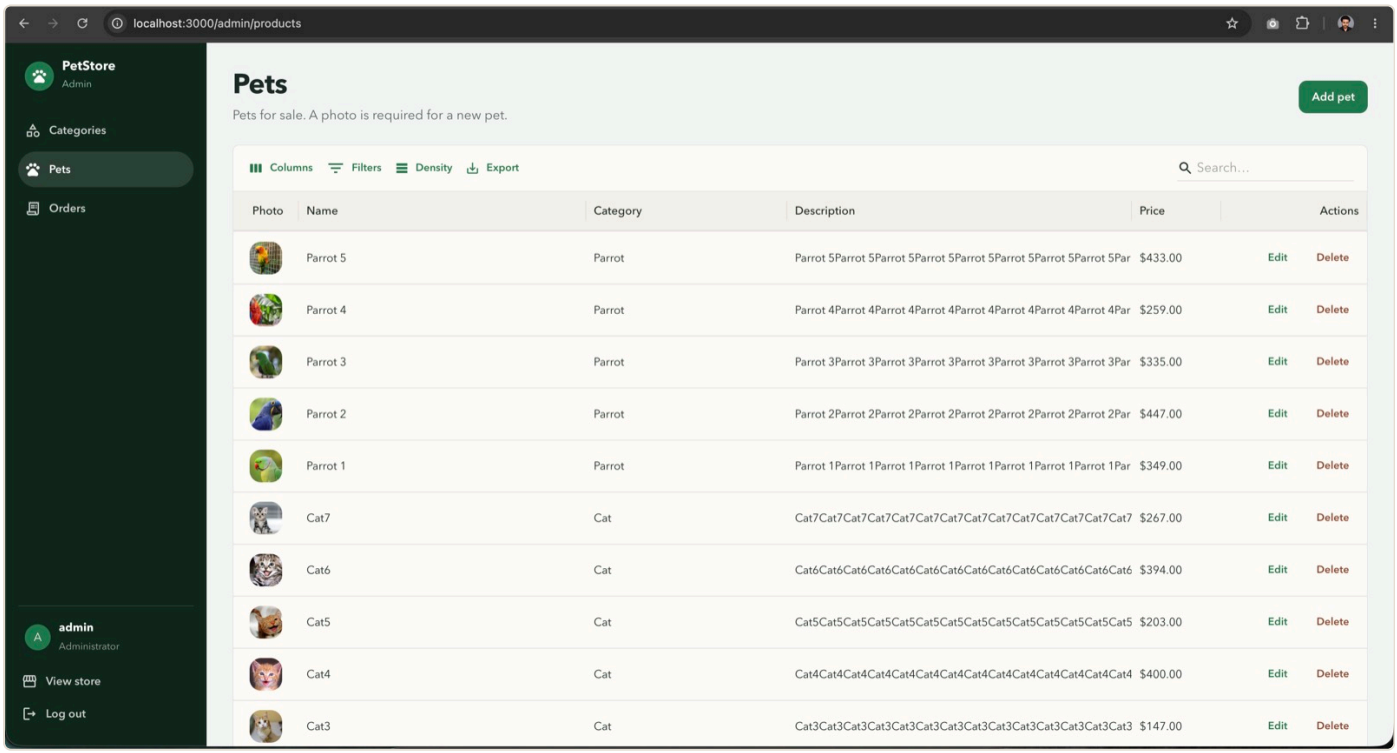

Figure 5. Admin pets. Parrots and cats are listed with the photograph stored in Blob Storage, a category, a price, and Edit / Delete.

Orders

Administrators see every order, including the customer id. The same orders that appear on a customer's account page appear here, with an extra Customer column and a Delete action. Delete is allowed only

for the ADMIN role.

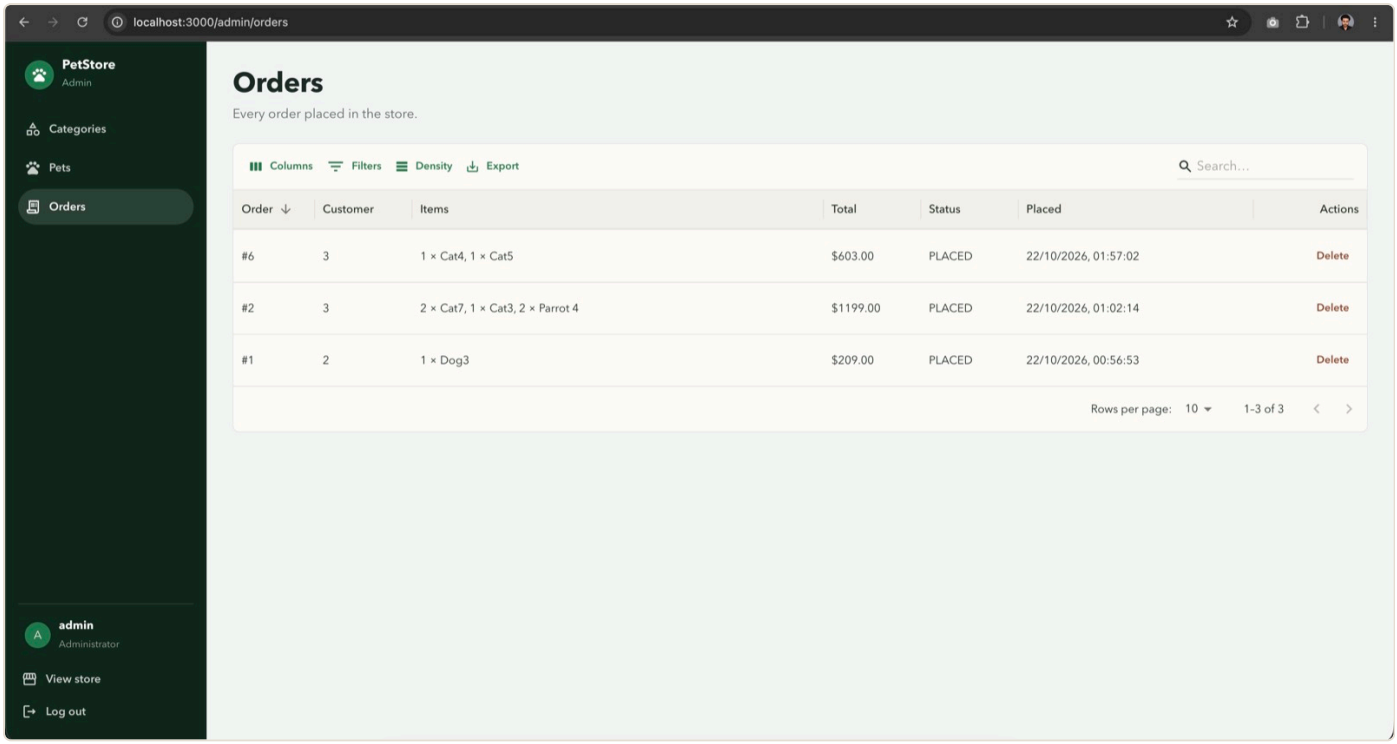

Figure 6. Admin orders. Orders #6 and #2 belong to customer 3. Order #1 belongs to customer 2. Each row can be deleted.

6. Azure Database for PostgreSQL

All three services use one database, `petstore`, on an Azure Database for PostgreSQL flexible server. The server is in the Petstore resource group and is Ready.

Setting	Value
Server	petstore-server-name
Endpoint	petstore-server-name.postgres.database.azure.com
Subscription	Petstore subscription · b5c67bf9-6f36-431b-82ea-a27d96f61190
Resource group	Petstore-Resource-Group
Location	Central India
PostgreSQL	16
Compute	Burstable B2s · 2 vCores · 4 GiB RAM · 32 GiB storage
Administrator login	amirrahi29
High availability	Not enabled. This tier is aimed at development and test.

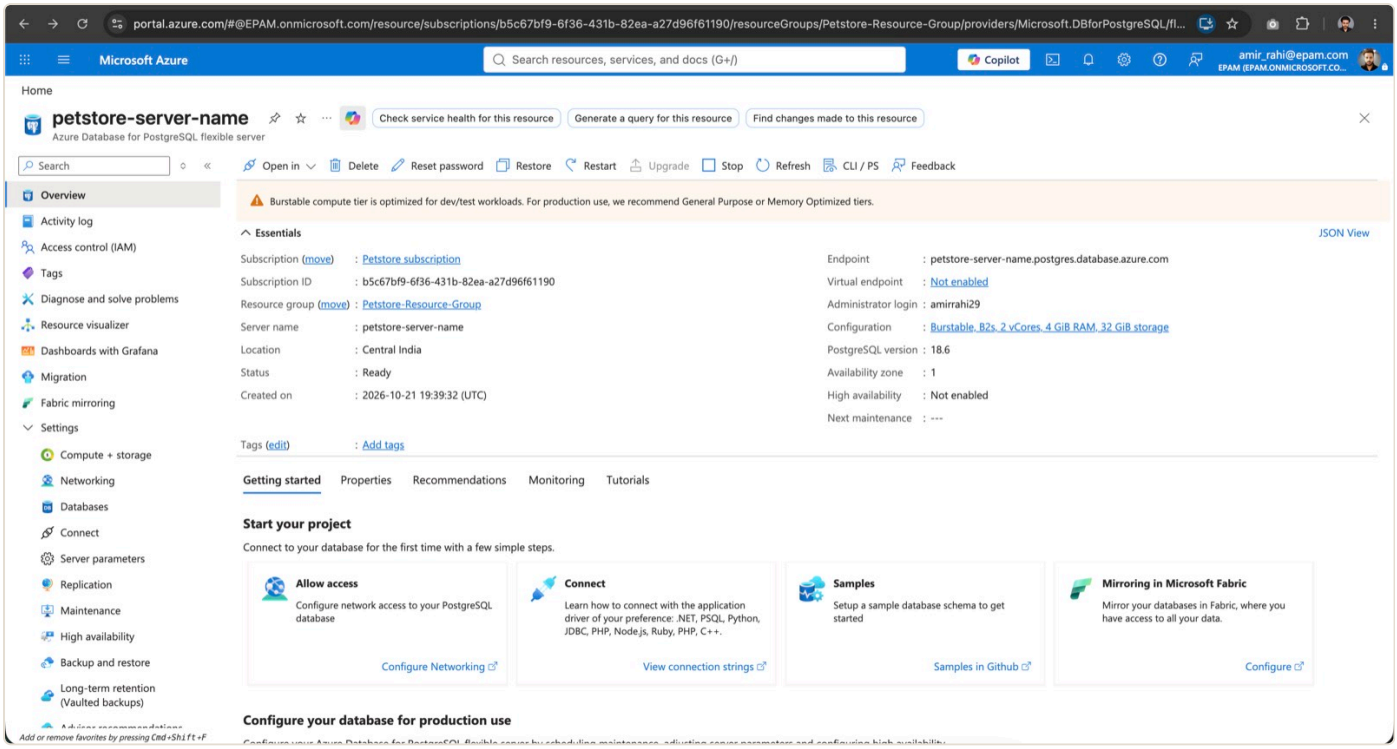

Figure 7. Flexible server overview. Status is Ready. The JDBC URL in each service points at this endpoint and the `petstore` database.

pgAdmin is connected to that server as `amirrahi29`. The `public` schema holds the application tables: `categories`, `order_items`, `orders`, `pets`, `products`, `roles`, and `users`. The products query in the screenshot returns the same catalog the storefront shows, twenty rows, with category and description filled in.

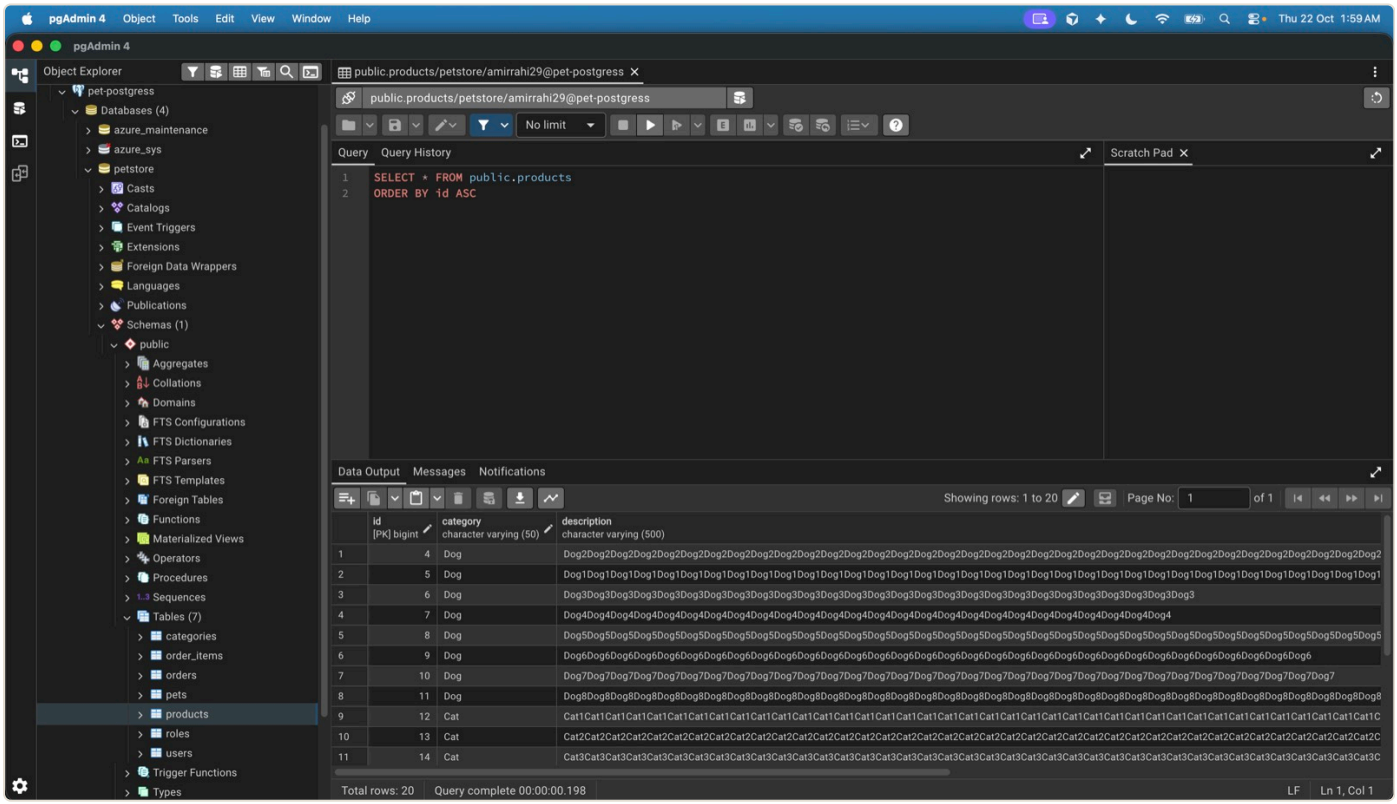

Figure 8. Database `petstore` on server pet-postgres. `SELECT * FROM public.products ORDER BY id` returns the pets for sale.

> The database username and password are not written in the service YAML files. Each service reads the Key Vault secrets named `username` and `password` when it starts, then uses them as `spring.datasource.username` and `spring.datasource.password`.

7. Azure Blob Storage

Pet photographs are files in the storage account `petstore`. The application container is `pet-container`. A second container, `$logs`, is the storage log container and is not used by the shop.

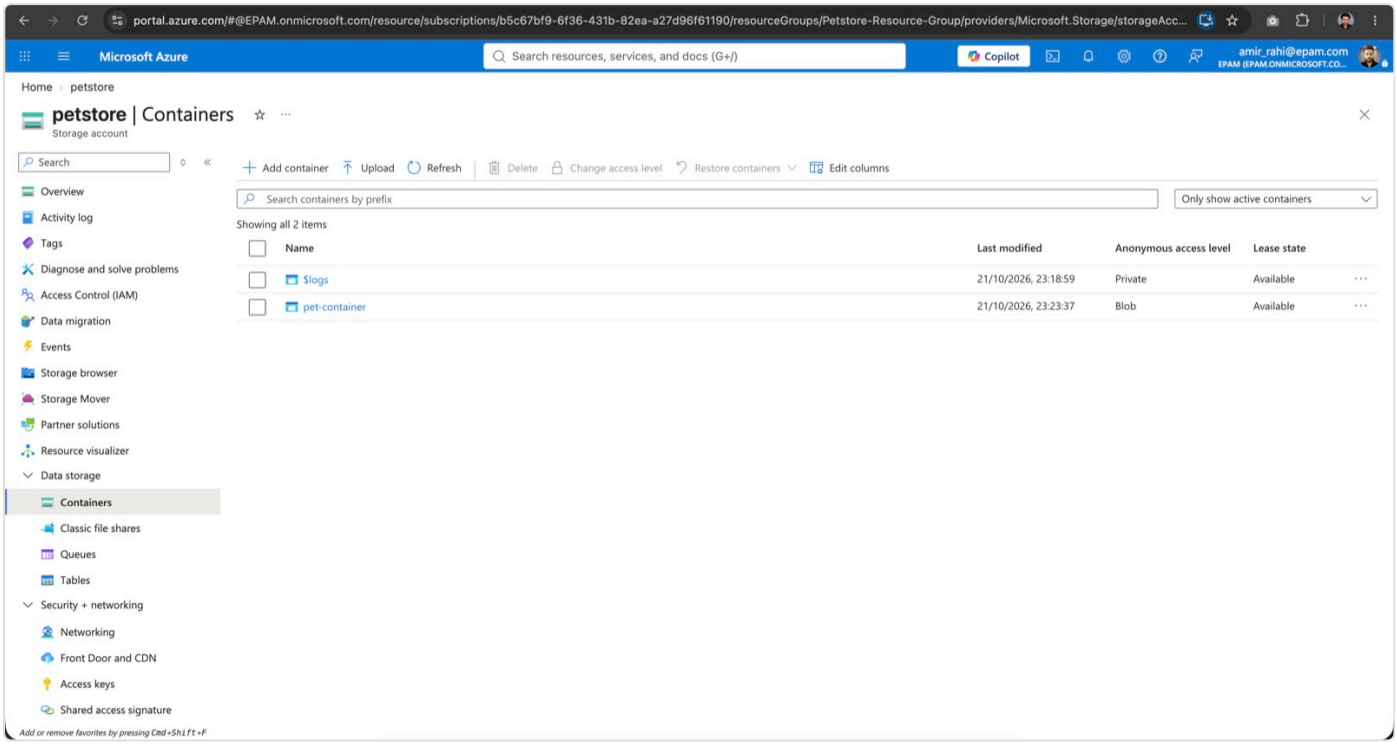

Figure 9. Containers on storage account petstore. `pet-container` is the container named in the product service. Its anonymous access level in the portal is Blob.

Each upload is stored under a generated file name. The database keeps the blob URL. When the API returns a pet, it adds a short-lived shared access signature so the page can show the image. The expiry configured for that link is 60 minutes. The storage connection string and account key come from the Key Vault secrets `connection-string` and `account-key`. The account name, blob endpoint, container name, and SAS lifetime stay in configuration because they are not secret values.

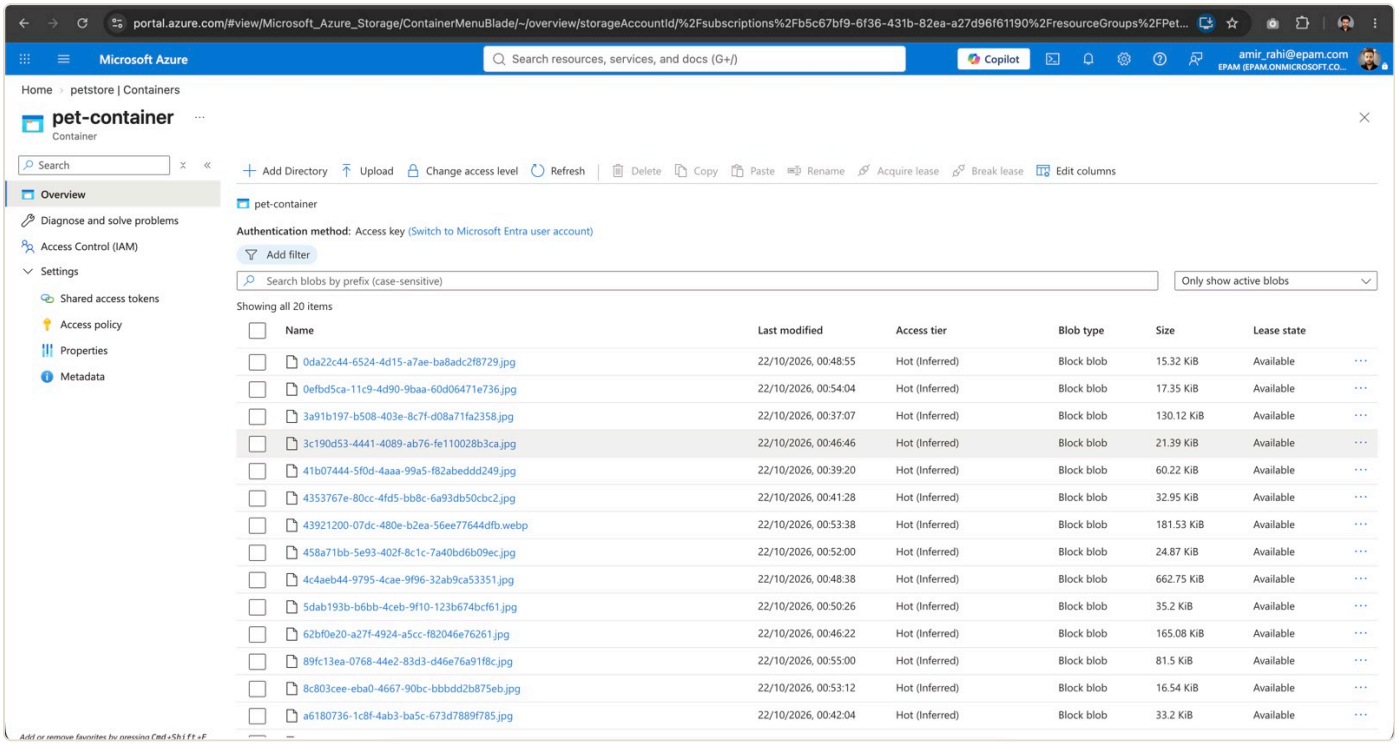

Figure 10. `pet-container` holding the pet images. Twenty blobs are listed, created when pets were saved from the admin form.

8. Azure Key Vault

One vault holds the values the services used to keep in YAML. It is `db-credentials-petstore`, in Petstore-Resource-Group, location India South Central, on the Petstore subscription.

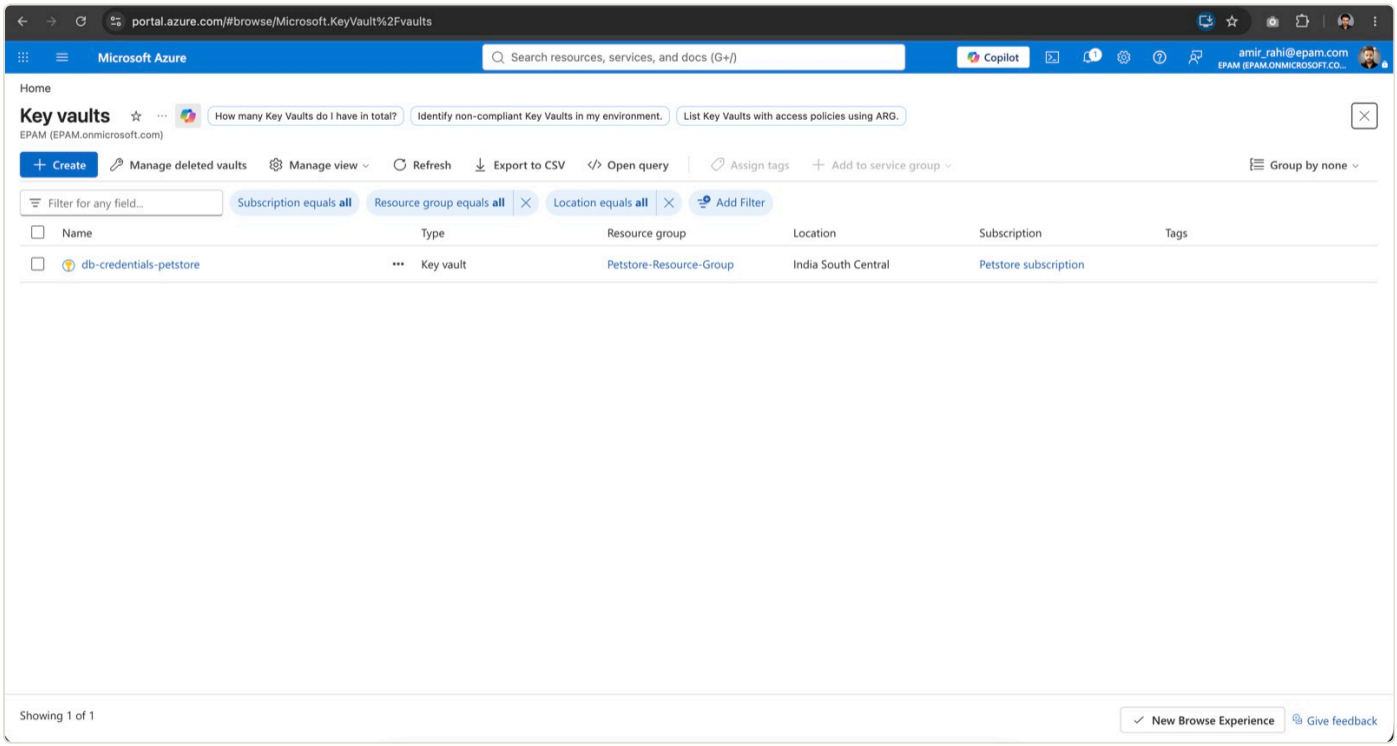

Figure 11. The vault in the subscription. One vault is in use for this project.

Four secrets are enabled. None of the secret values are stored in the repository.

Secret	Used as	Services
`username`	Database username	product, order, pet

Secret	Used as	Services
`password`	Database password	product, order, pet
`connection-string`	Blob Storage connection string	product
`account-key`	Blob Storage account key	product

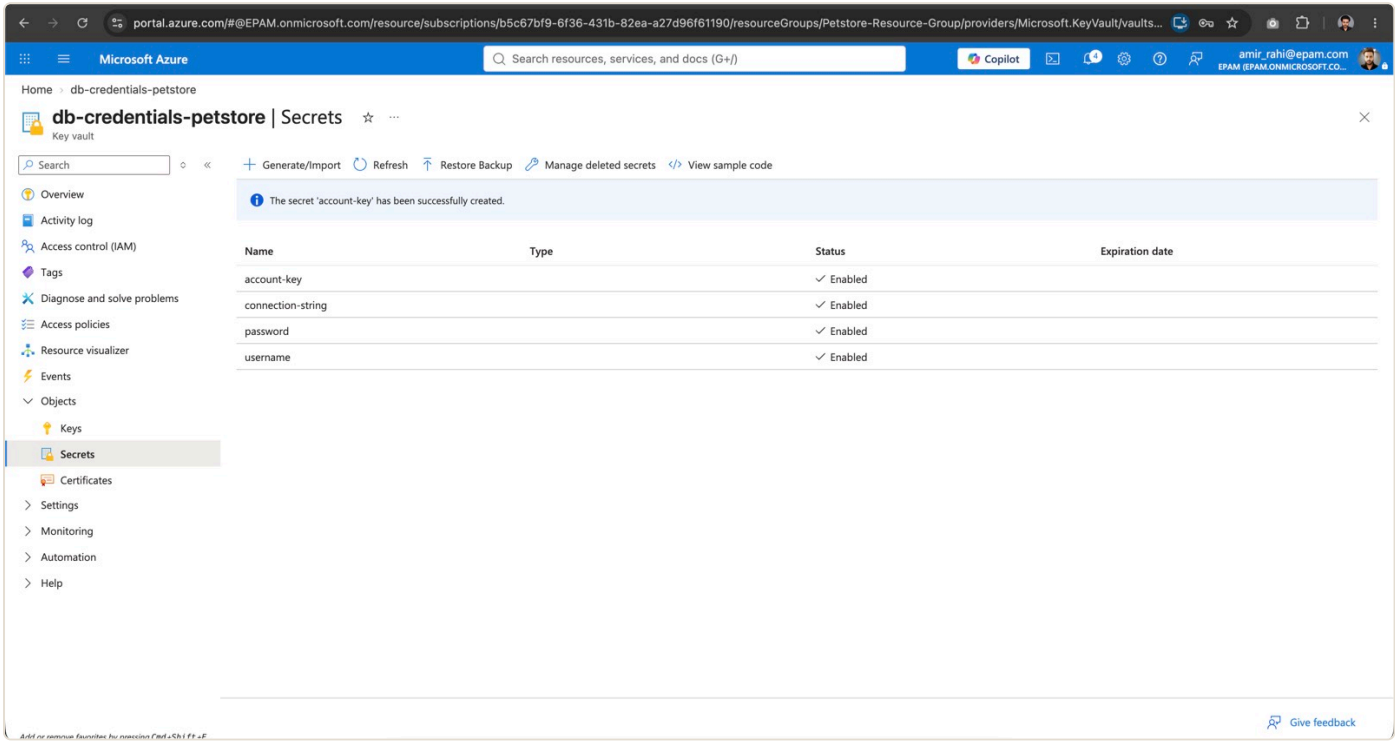

Figure 12. Secrets on `db-credentials-petstore`. All four are Enabled. The portal confirms `account-key` was created.

How a service reads them

The vault URL is the only Key Vault setting in configuration: `https://db-credentials-petstore.vault.azure.net/`. At startup each service signs in with `DefaultAzureCredential` and copies the secret values into the Spring environment before the database connection is opened. On a developer machine that credential is an Azure CLI login for an account that has Get permission on secrets. No client secret is written into the YAML.

> A service that starts without an Azure login stops with a clear error: it could not read the secrets from this vault. It does not fall back to a password in a file. Sign in with `az login` as a principal that can get secrets, then start the jars again.

9. Azure Container Registry

Each application already has a Dockerfile. A push to the `main` branch of `amirrahi29/Petstore-Microservice-App-SpringBoot-React` builds all four images and stores them in Azure Container Registry. The registry login server is `petstoreazurecontainerregistry.azurecr.io`.

Setting	Value
Registry	petstoreazurecontainerregistry
Login server	petstoreazurecontainerregistry.azurecr.io
Resource group	Petstore-Resource-Group

Setting	Value
Subscription	Petstore subscription · b5c67bf9-6f36-431b-82ea-a27d96f61190
Location	India South Central
SKU	Basic. Provisioning succeeded on 22 October 2026.
Sign-in from GitHub	OpenID Connect. The workflow does not store a registry password.

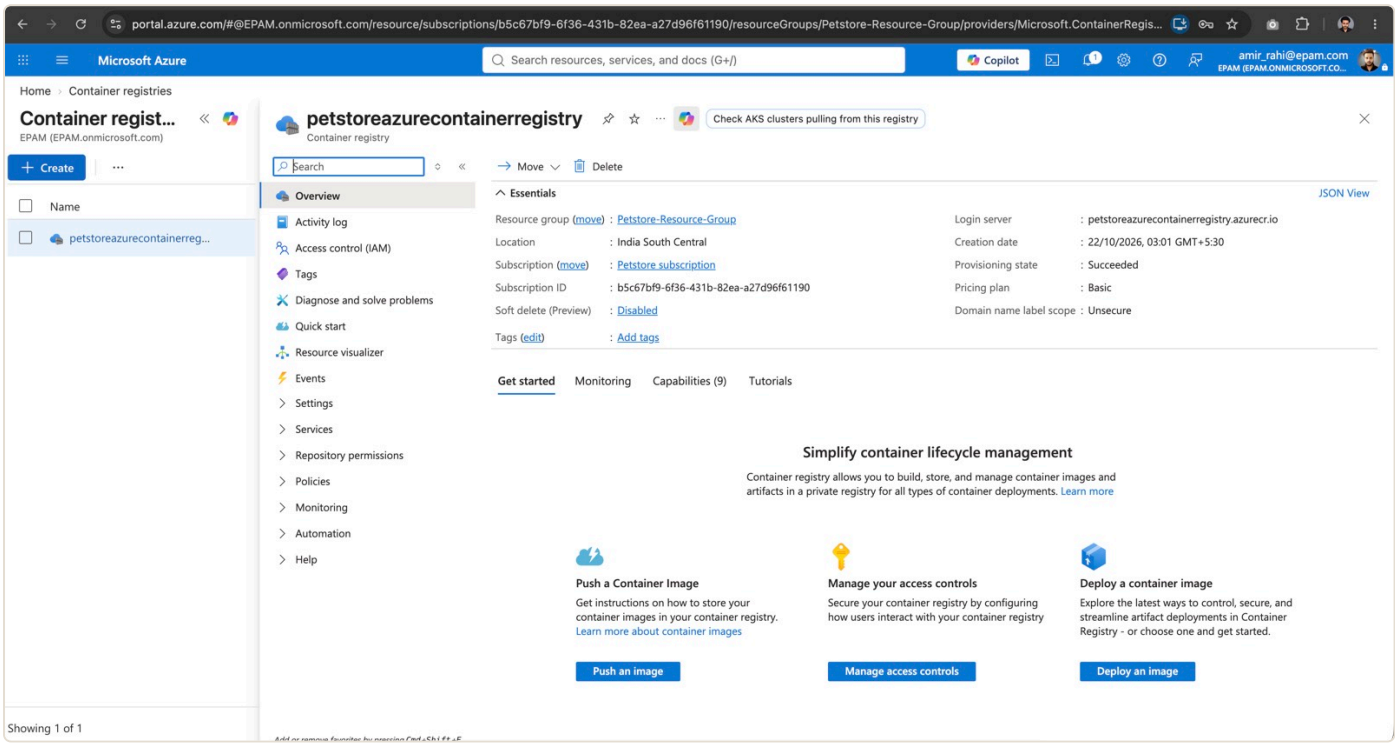

Figure 13. Registry overview. One registry is in the Petstore resource group, on the Basic plan, in India South Central.

The repository name is the service name with `-image` on the end. Each repository has two tags: `latest`, and the git commit that built it.

Application	Image	Port inside the image
product-service	product-service-image	8081
order-service	order-service-image	8083
pet-service	pet-service-image	8082
petstore-ui	petstore-ui-image	80

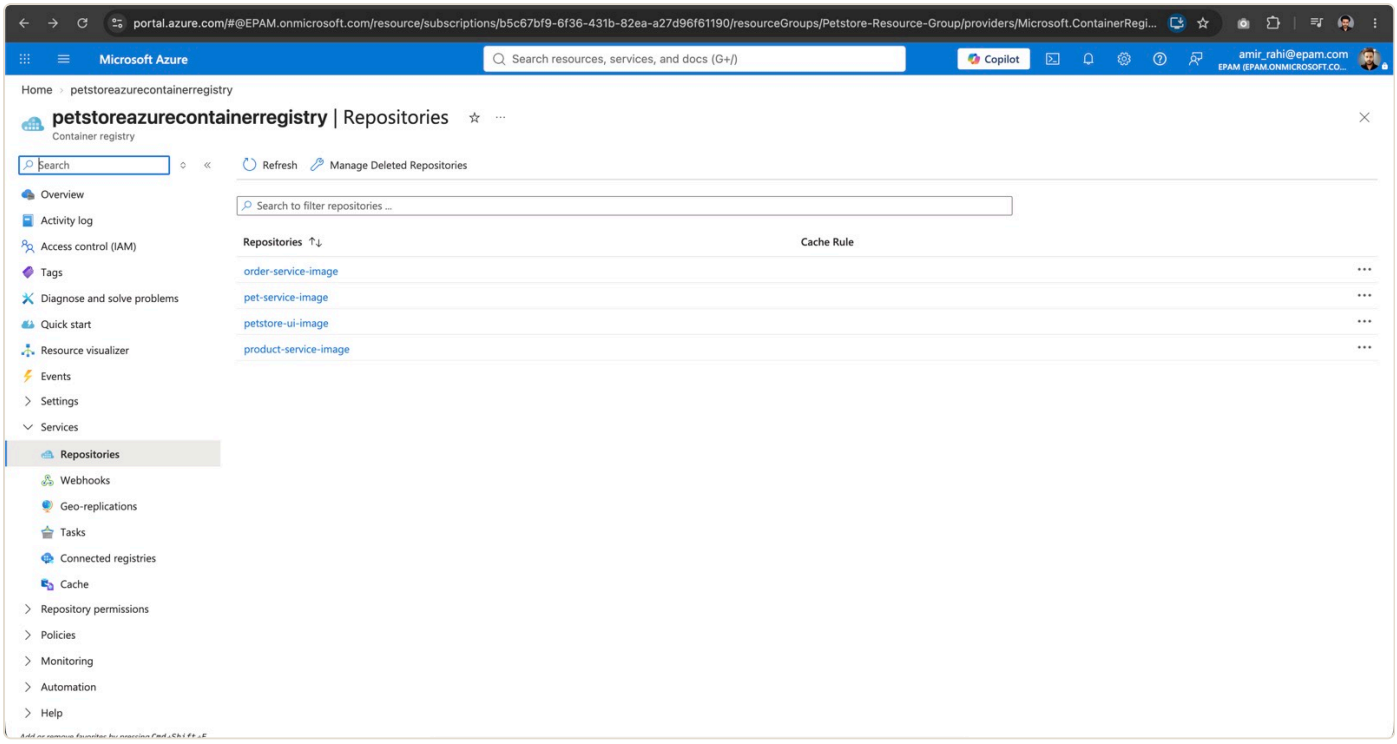

Figure 14. Repositories in the registry: `order-service-image`, `pet-service-image`, `petstore-ui-image`, and `product-service-image`.

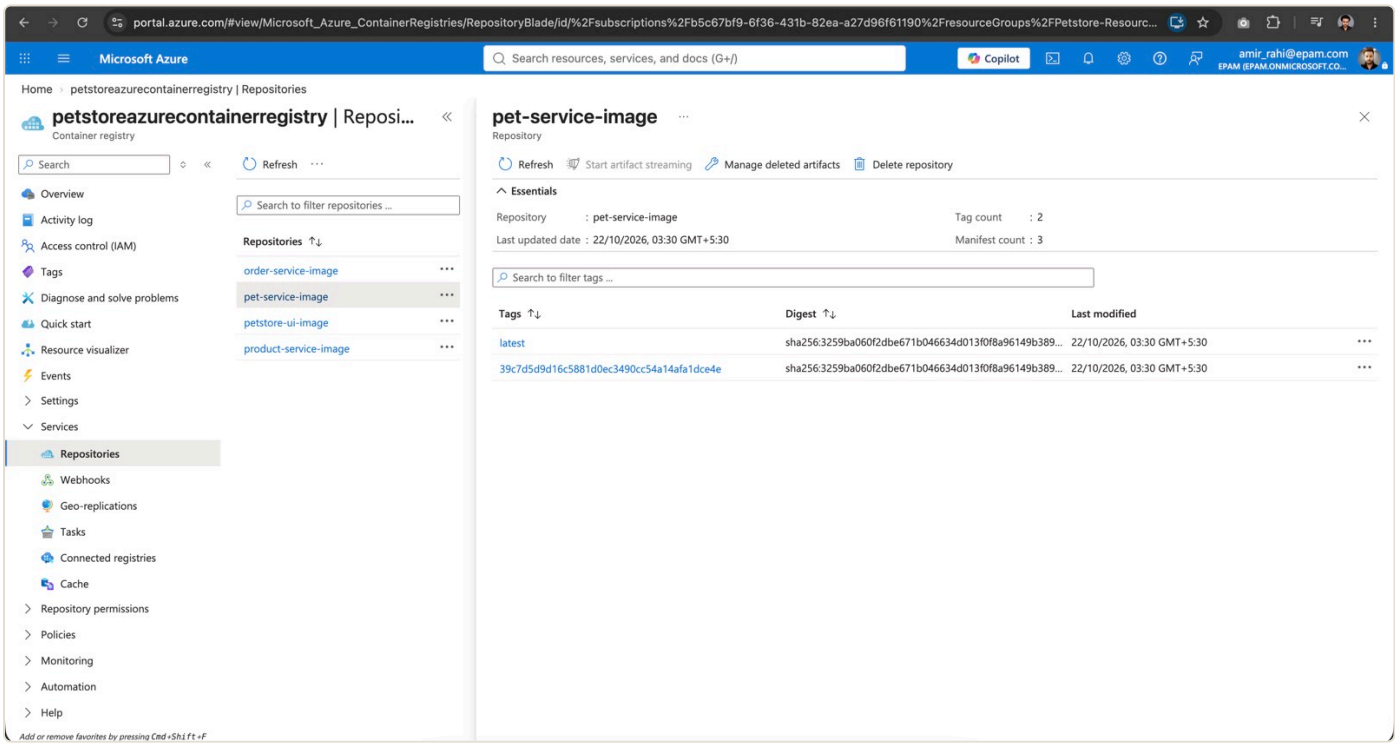

Figure 15. `pet-service-image` has two tags, `latest` and the commit `39c7d5d`. The same pair is published for the other three images.

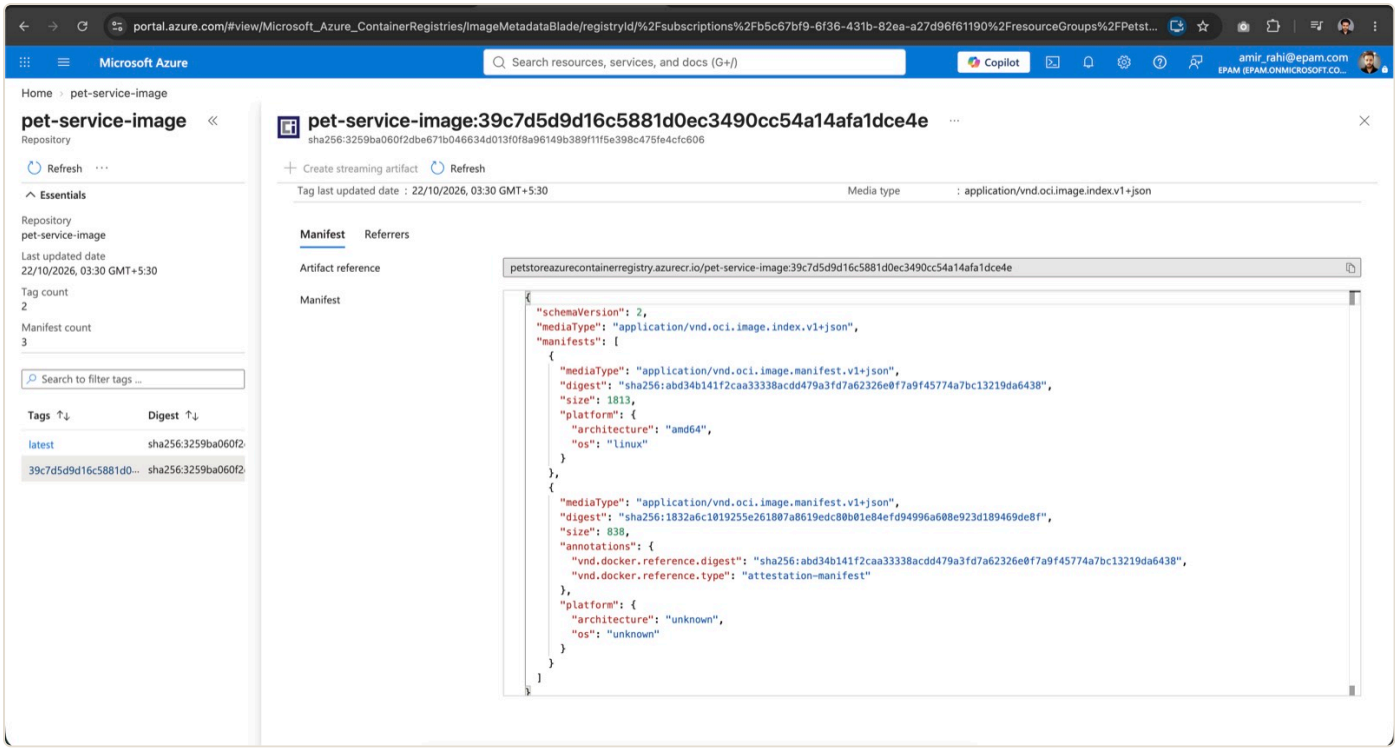

Figure 16. Manifest for `petstoreazurecontainerregistry.azurecr.io/pet-service-image:39c7d5d9d16c5881d0ec3490cc54a14afa1dce4e`. The image is a Linux amd64 OCI index.

> The four workflow files live in `.github/workflows`. A push to `main` starts all of them. While a build is running, GitHub shows a yellow status on the commit. It turns green when that image is in the registry. Section 10 is the next step: each web app pulls its image from this registry and runs it.

10. Azure App Service

The four applications are deployed. App Service runs them as Linux web apps in `Petstore-Resource-Group`, on one plan, `ASP-PetstoreResourceGroup-9881`, in Central India. The plan is Basic, size B3, with one instance. Nothing is copied up as a jar or a zip. The publishing model is Container, so each app pulls its image from `petstoreazurecontainerregistry.azurecr.io`, the registry in section 9.

Deployed, in this report, means three things that the screenshots show together. The app list says Running. Each overview says runtime status Healthy, with the container image coming from that registry. A browser opened on the public hostname renders the shop. Figure 22 is that last check for the UI: the catalog, with photos and prices, is the deployed `petstore-ui-image`, not a local dev server.

Web app	What it serves	Port	Default hostname
product-service-web-app	Catalog, categories, pet photos	8081	product-service-web-app-g7bhexgnaje3brbw.centralindia-01.azurewebsites.net
order-service-wep-app	Register, login, checkout, orders	8083	order-service-wep-app-huhhatejg5adc0cr.centralindia-01.azurewebsites.net
pet-service-web-app	Pet API on the same database	8082	pet-service-web-app-adgkgbhmfchtfge0.centralindia-01.azurewebsites.net
petstoreui-web-app	Storefront and the admin UI	80	petstoreui-web-app-e3cahqc4bad5dthh.centralindia-01.azurewebsites.net

The order app keeps the name `order-service-wep-app` that was used when it was created. That spelling is the resource name and the start of its hostname. Each public URL is HTTPS on the default domain

above.

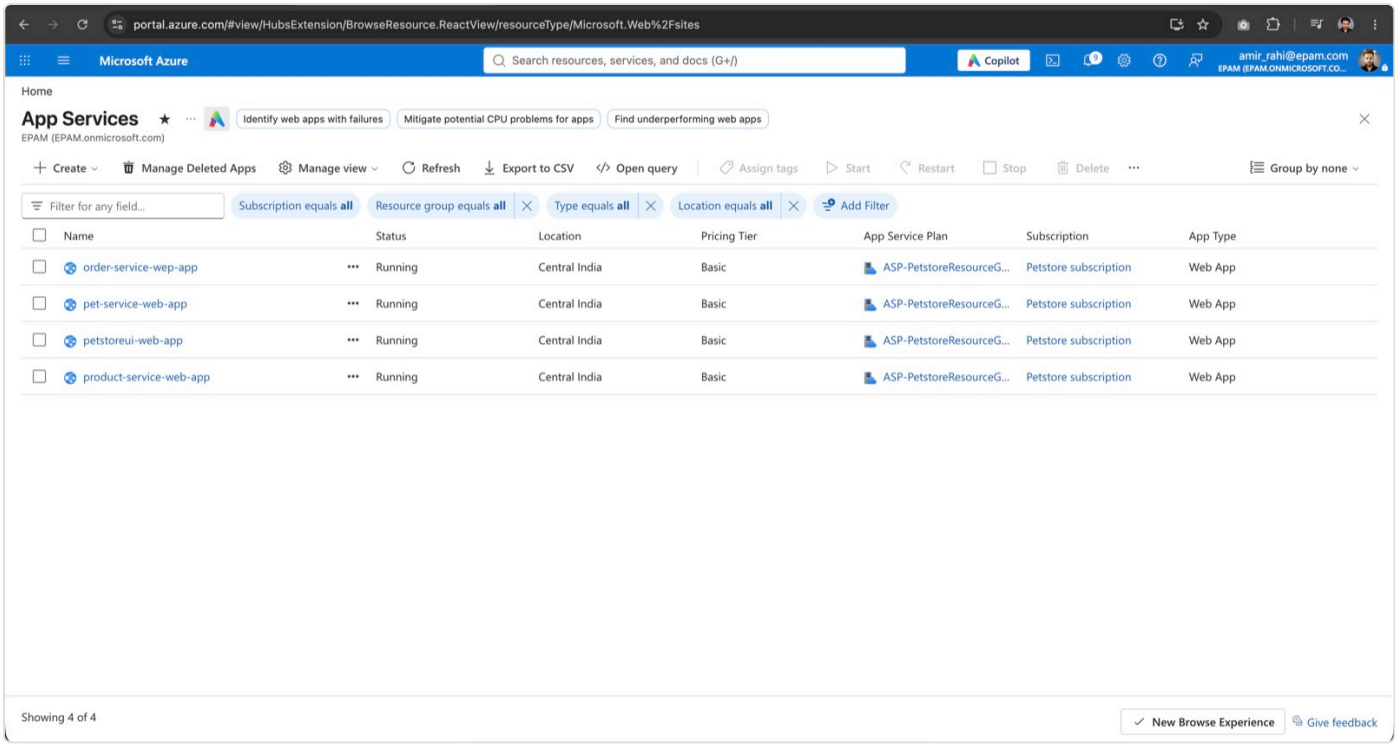

Figure 17. The deployment list. All four web apps are Running in Central India, on plan `ASP-PetstoreResourceGroup-9881`, pricing tier Basic, subscription Petstore subscription. Running is the platform state after App Service has pulled the image and started the container.

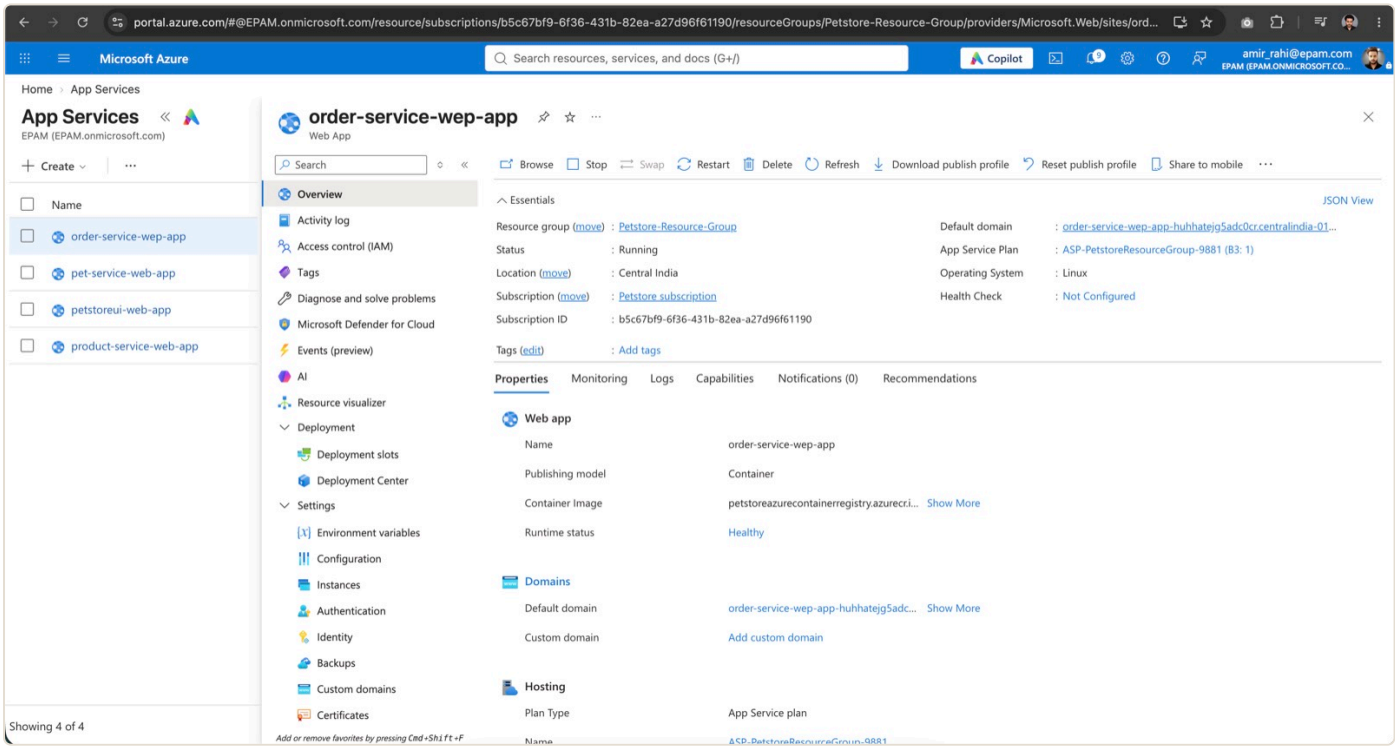

Figure 18. `order-service-wep-app` is deployed. Publishing model Container, operating system Linux, runtime status Healthy. The container image line is the order image App Service pulled from the registry. The default domain is the public hostname in the table.

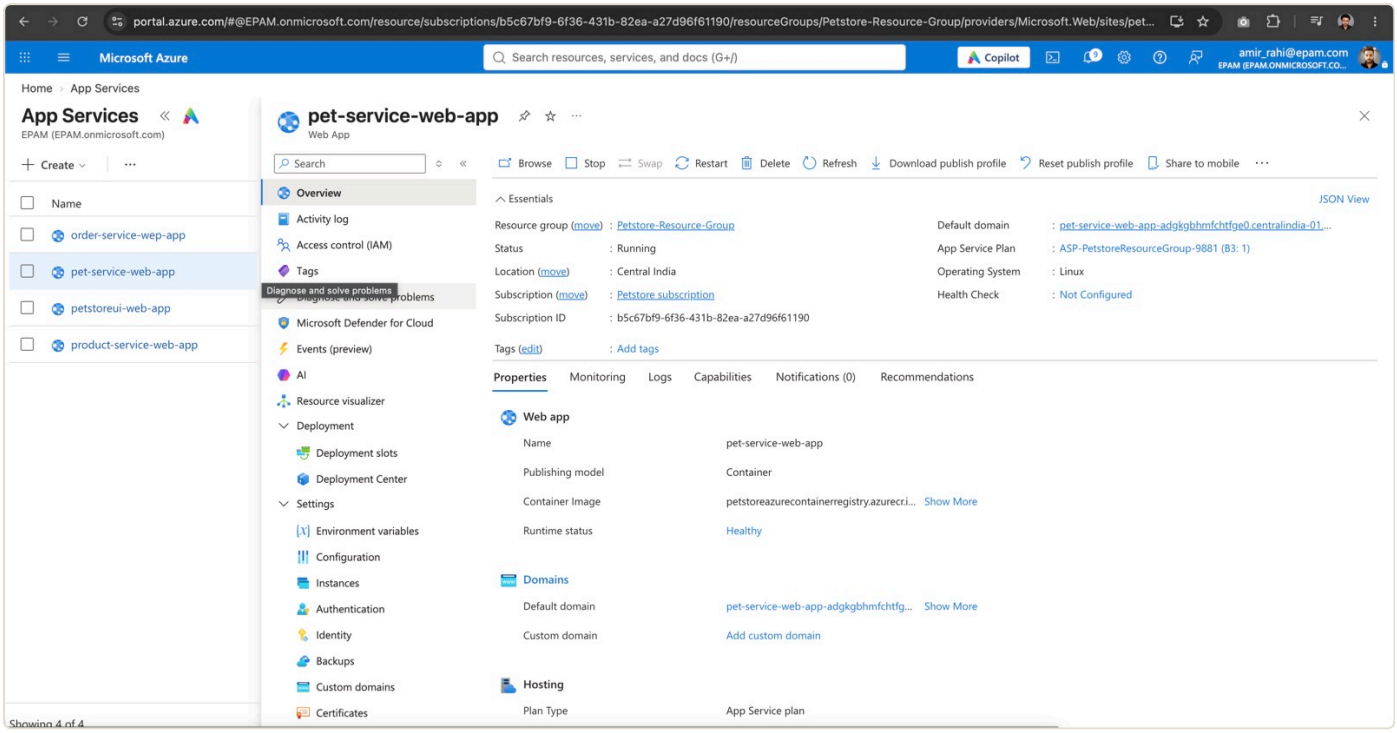

Figure 19. `pet-service-web-app` is deployed on the same plan and in the same region. Runtime status Healthy. Its container image is pulled from `petstoreazurecontainerregistry`, the same way as the order app.

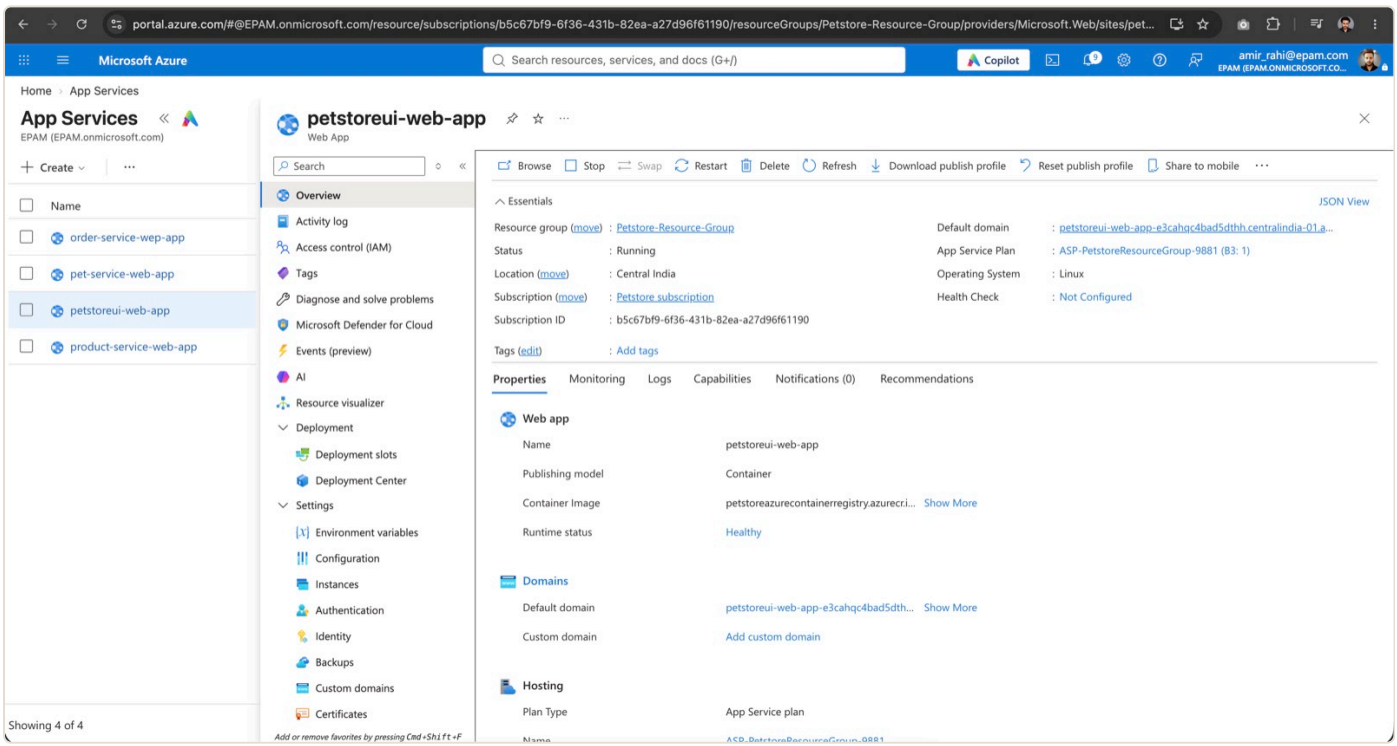

Figure 20. `petstoreui-web-app` is deployed. Runtime status Healthy, Linux, publishing model Container. This is the app whose hostname serves Figure 22 and Figure 23. App Service pulled `petstore-ui-image` from the registry to run it.

Deployment Center is where the pull is recorded. On `product-service-web-app` the main container is named `product-service`. Source is Azure Container Registry, image `product-service-image`, tag `5a99f3e102c5612cccff9ac7330e5eb319ffe8db`, port 8081. Continuous deployment for that main container is turned on, so App Service pulls this image from the registry when the app runs. The order, pet, and UI apps are wired the same way: one main container, image from this registry, port 8083, 8082, or 80.

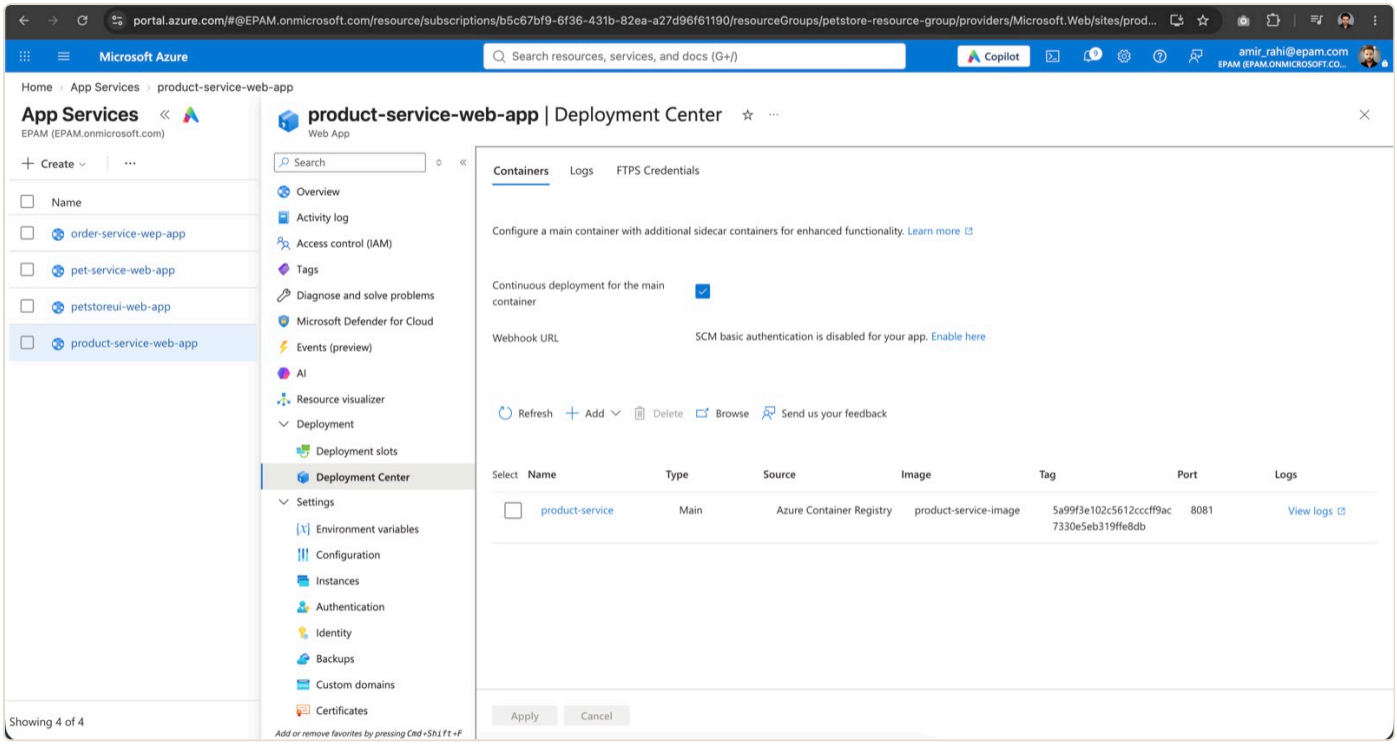

Figure 21. The ACR pull for the deployed product app. Source Azure Container Registry, image `product-service-image`, tag `5a99f3e102c5612cccff9ac7330e5eb319ffe8db`, port 8081. Continuous deployment for the main container is on.

Figure 22 is the deployed shop, opened in a browser. The address bar is the App Service hostname, so this page is `petstoreui-web-app` after it pulled `petstore-ui-image`. The catalog matches section 3: no login wall, category chips, prices, photos, Details, and Add to cart, with 20 pets. Figure 23 is the same deployment at `/admin/products`. The signed-in administrator is `admin`, and the grid has photo, category, price, edit, and delete. Together these two figures show that the container App Service pulled is the application, and that it is serving traffic.

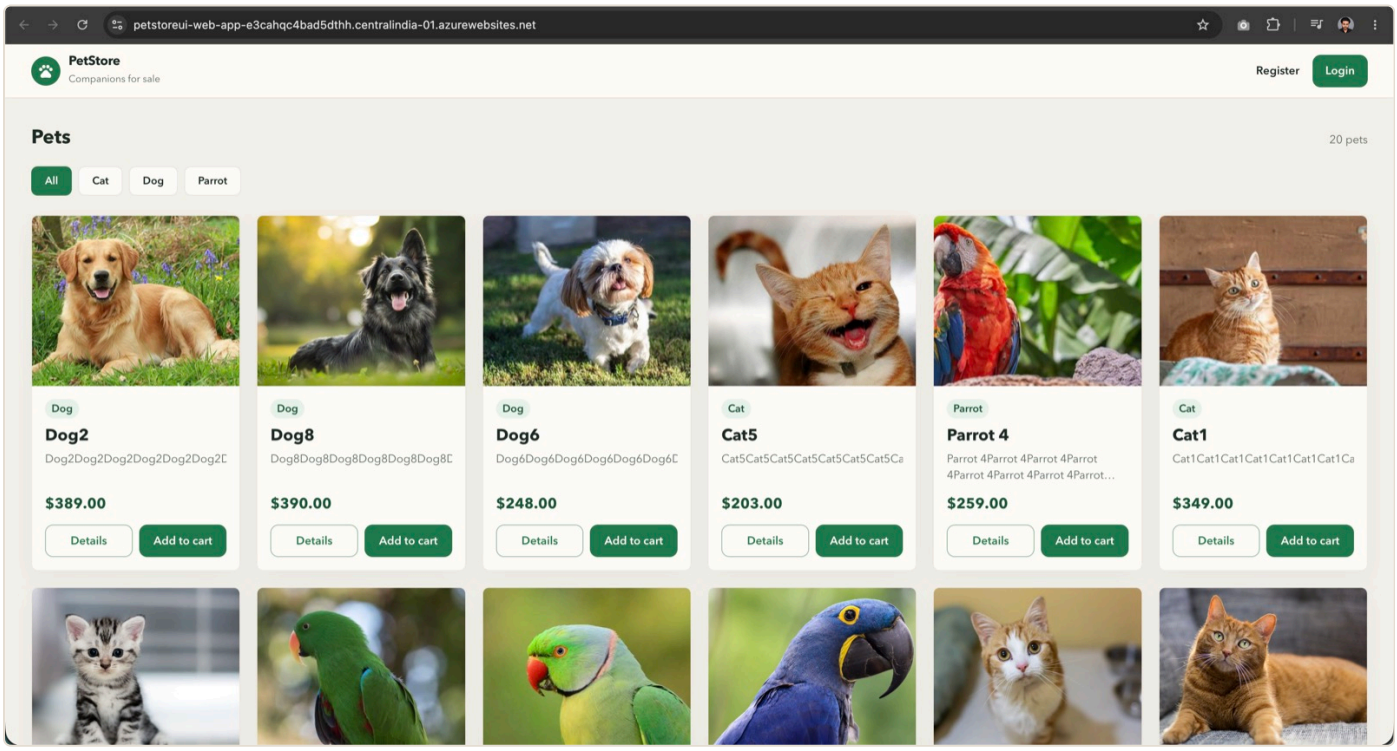

Figure 22. The deployed UI, live. `https://petstoreui-web-app-e3cahqc4bad5dthh.centralindia-01.azurewebsites.net` serves the catalog from the image App Service pulled from ACR. Twenty pets, filters for All, Cat, Dog, and Parrot.

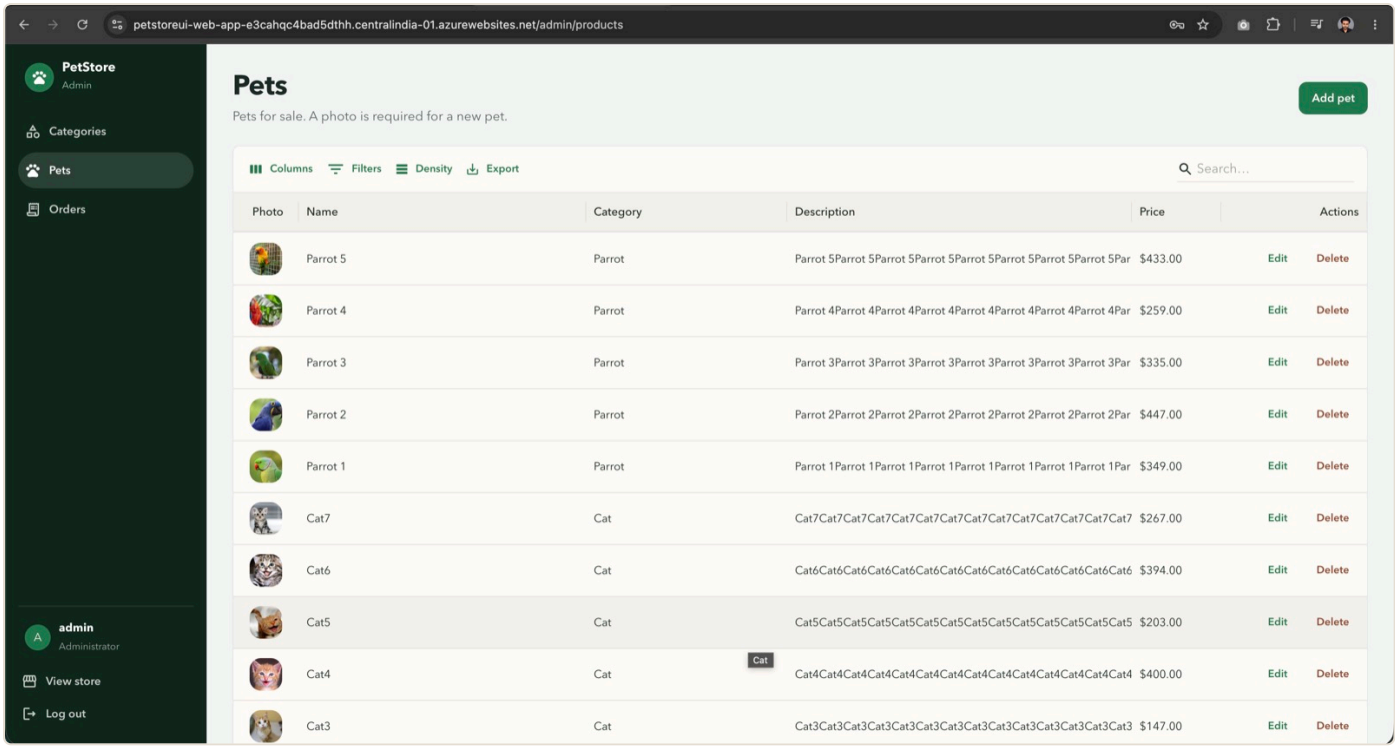

Figure 23. The same deployed app, path `/admin/products`, signed in as `admin`. Administration is running on App Service, on the image pulled from the registry.

The UI image is built with the product and order hostnames in the table, so the browser on this shop calls those App Service URLs. The three Java apps still read `db-credentials-petstore` before they open PostgreSQL. On App Service that read uses the app's managed identity. A Healthy runtime status means the container finished that startup and is serving its port.

> The App Service Health Check setting on these overview blades is Not Configured. That setting is separate from runtime status. Endpoint health for DNS routing is the Traffic Manager probe in the next section.

11. Azure Traffic Manager

Traffic Manager answers DNS for a profile name such as `ui-traffic-manager.trafficmanager.net`. The client then connects to the address it was given. The profile chooses an endpoint whose health probe is passing. These four profiles use Priority routing. Each profile holds one External endpoint, aimed at one of the deployed web apps in section 10.

Monitor status Online means that deployed app answered. The probe calls the App Service hostname over HTTPS, and Online is the result when the container App Service pulled from ACR is up. It is the same fact as Figure 22, read from the routing side: the UI endpoint is Online because `petstoreui-web-app` is deployed and serving. The product, order, and pet endpoints are Online for the same reason on their own apps.

Profile	DNS name	Endpoint	Web app
ui-traffic-manager	ui-traffic-manager.trafficmanager.net	ui-service	petstoreui-web-app
product-traffic-manager	product-traffic-manager.trafficmanager.net	product-service	product-service-web-app
petstore-traffic-manager	petstore-traffic-manager.trafficmanager.net	order-service	order-service-wep-app
pet-traffic-manager	pet-traffic-manager.trafficmanager.net	pet-service	pet-service-web-app

All four profiles sit in `Petstore-Resource-Group`, on the Petstore subscription, with status Enabled and routing method Priority. The endpoint type is External. Each one targets the default hostname of the deployed app, the hostname App Service published after the ACR pull. Each endpoint is Enabled, and Monitor status is Online. The probe is HTTPS on port 443, path `/`. A success response from that deployed app is what the portal marks Online. A failed probe stops that endpoint from being offered.

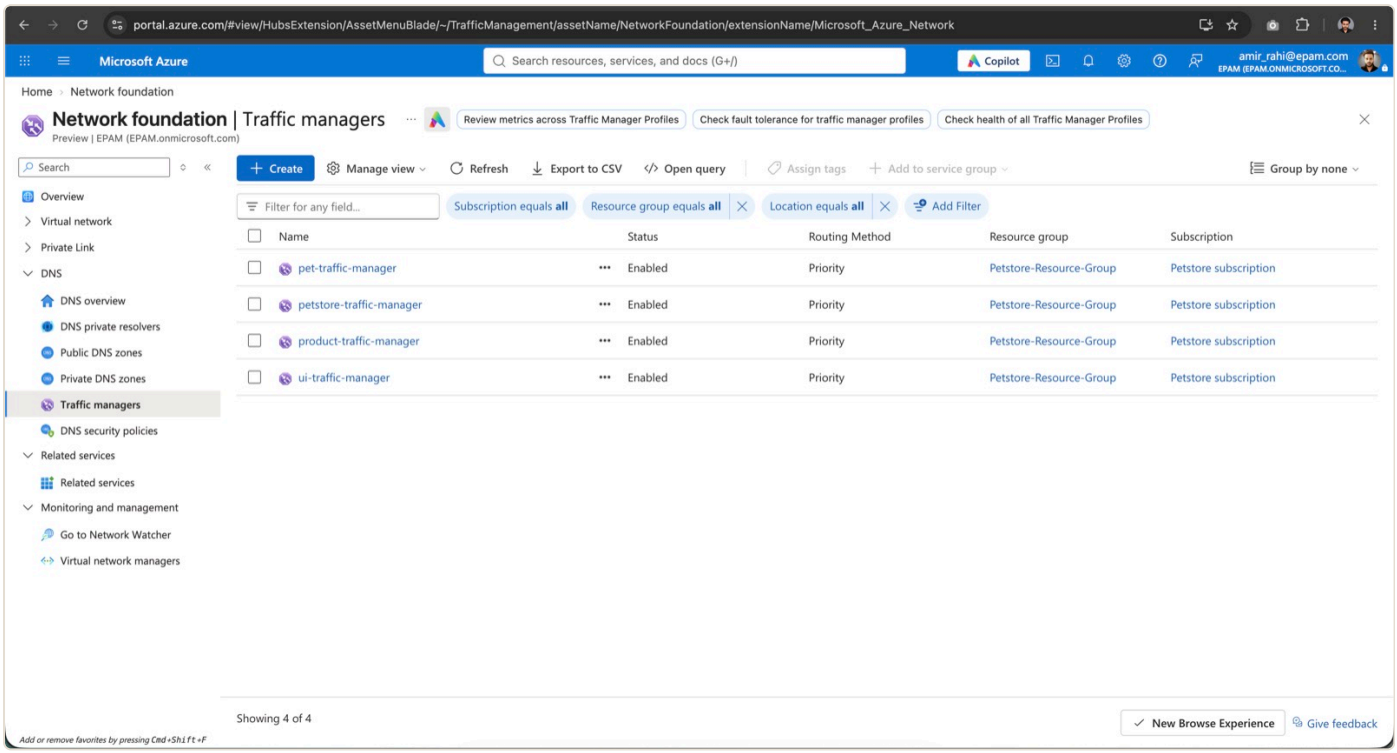

Figure 24. Four profiles in front of the four deployed apps. Status Enabled, routing method Priority, resource group Petstore-Resource-Group. Each profile's endpoint is one App Service hostname.

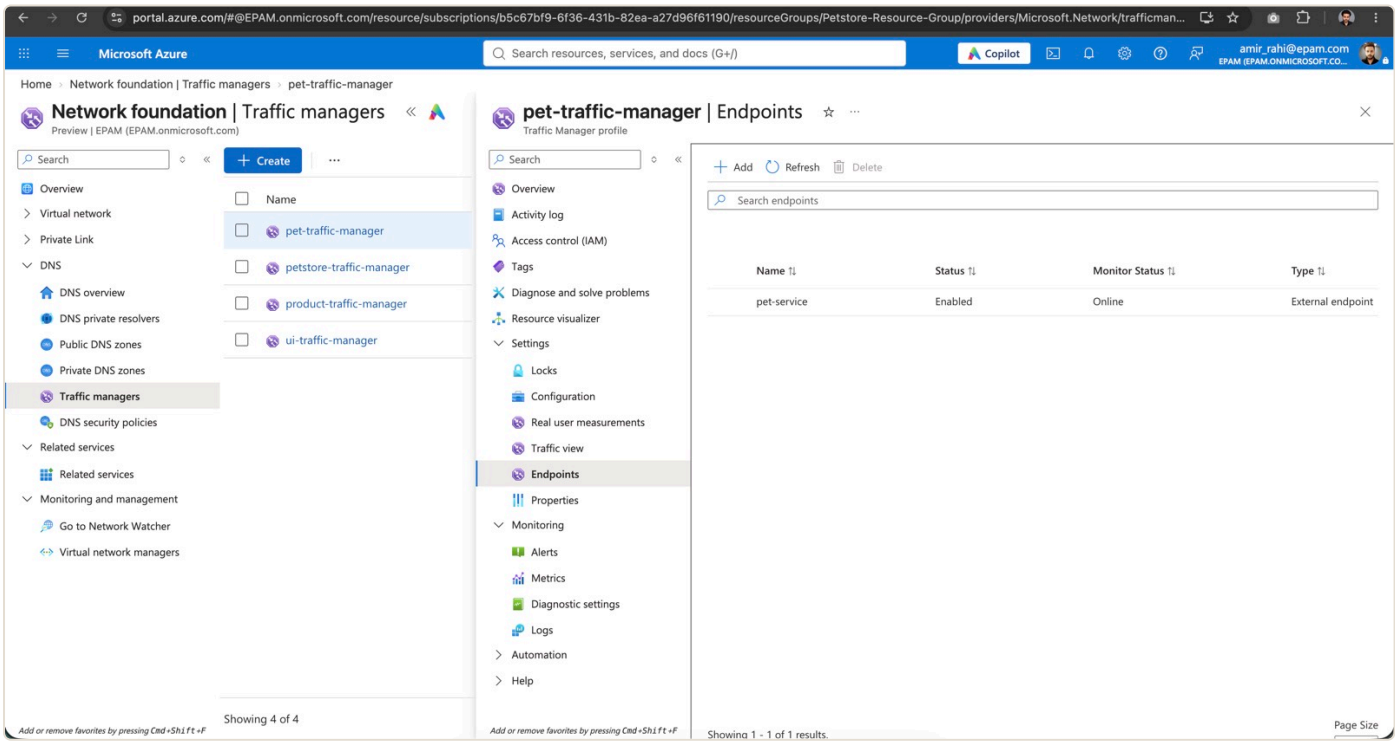

Figure 25. `pet-traffic-manager`. Endpoint `pet-service` is Enabled and Online. Online means the deployed `pet-service-web-app`, the container pulled from ACR, answered the health probe.

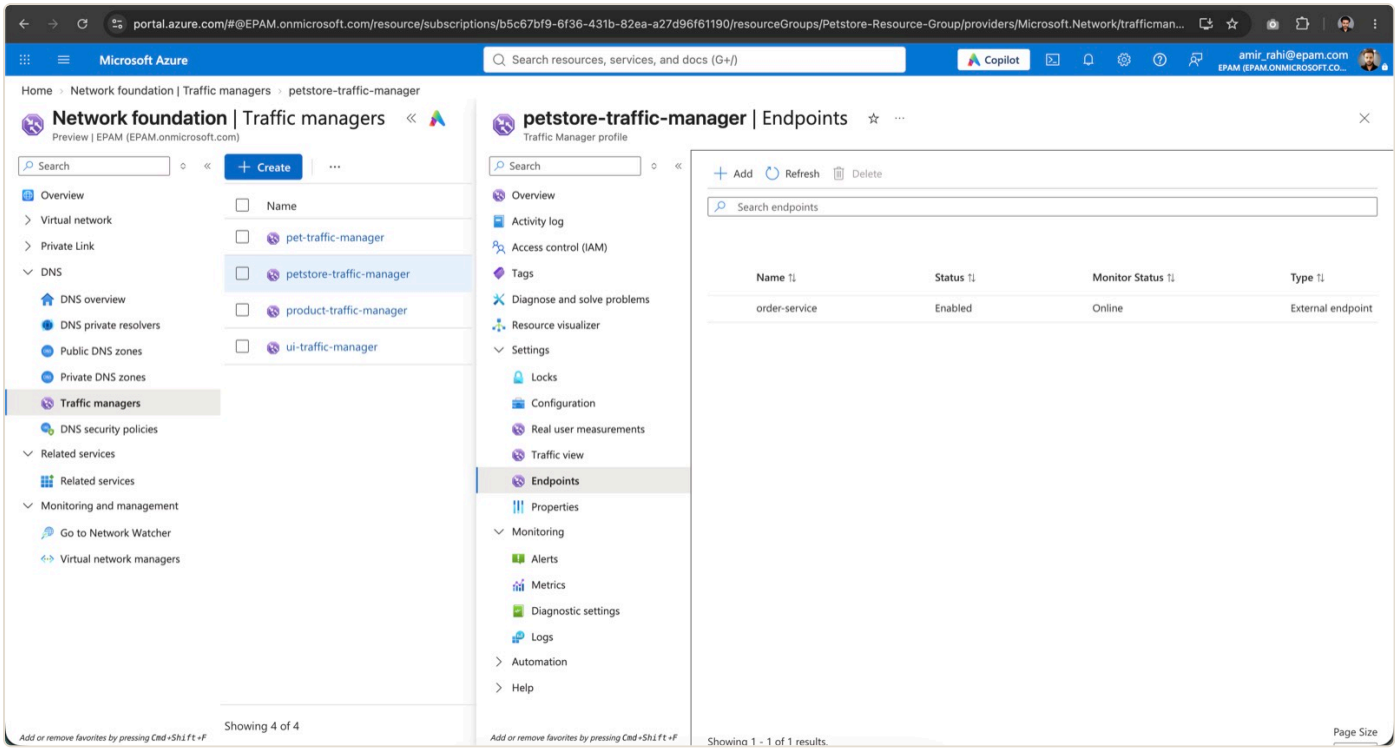

Figure 26. `petstore-traffic-manager`. Endpoint `order-service` is Enabled and Online. The probe reached the deployed `order-service-wep-app`.

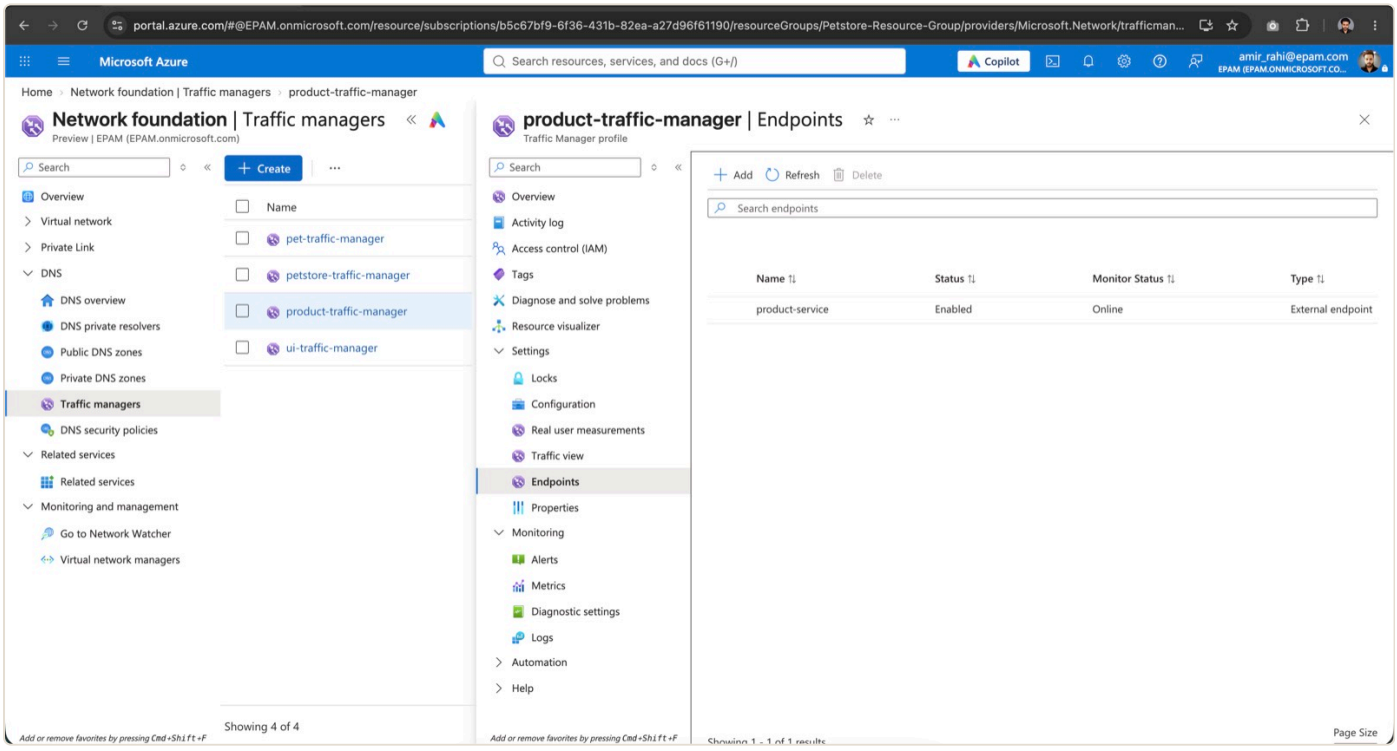

Figure 27. `product-traffic-manager`. Endpoint `product-service` is Enabled and Online. The probe reached the deployed `product-service-web-app`, the app whose ACR image and tag are in Figure 21.

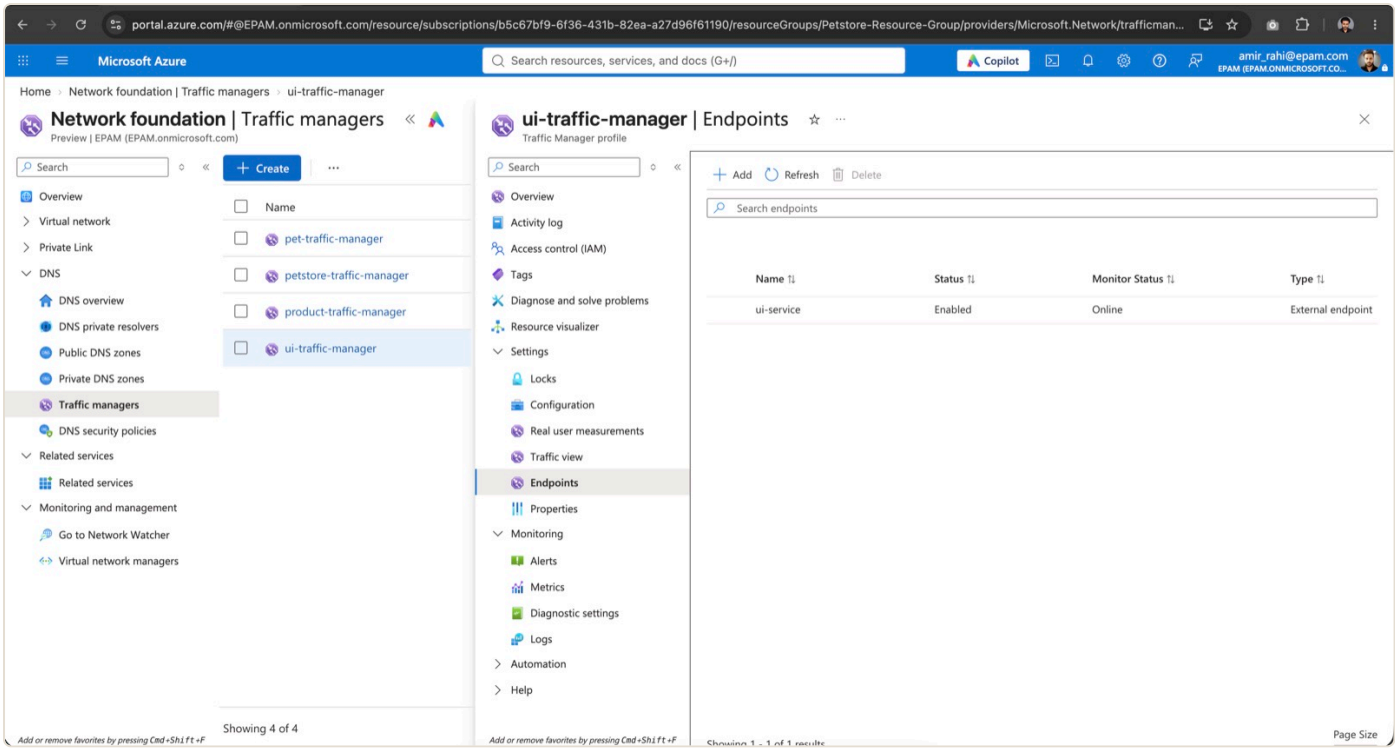

Figure 28. `ui-traffic-manager`. Endpoint `ui-service` is Enabled and Online. Online here is the deployed storefront: the same `petstoreui-web-app` that serves Figure 22.

Priority with a single Online endpoint sends every DNS answer for that profile to that app. A second endpoint, at a worse priority, would be used only after this one stopped passing the probe. These profiles do not have a second endpoint, and every app is in Central India, so the screenshots show a healthy route rather than a failover between regions.

The shop in Figure 22 and Figure 23 is opened on the App Service hostname. From there the page calls the product and order hostnames in section 10. Traffic Manager's record of this setup is the health of each endpoint, and DNS lookups of the `trafficmanager.net` names. The apps themselves answer on the default domains listed above. Those domains are what the probes check.

12. Azure App Configuration

The storefront hostnames are kept in Azure App Configuration, store `petstore-appconfig`, in the same resource group. Configuration explorer, filtered to label `petstore-ui`, shows three keys. The portal hides the values. The keys are the three API addresses the UI build reads.

Key	Label	Last modified
VITE_ORDER_API	petstore-ui	26/10/2026, 22:07:14
VITE_PET_API	petstore-ui	26/10/2026, 22:07:14
VITE_PRODUCT_API	petstore-ui	26/10/2026, 22:07:13

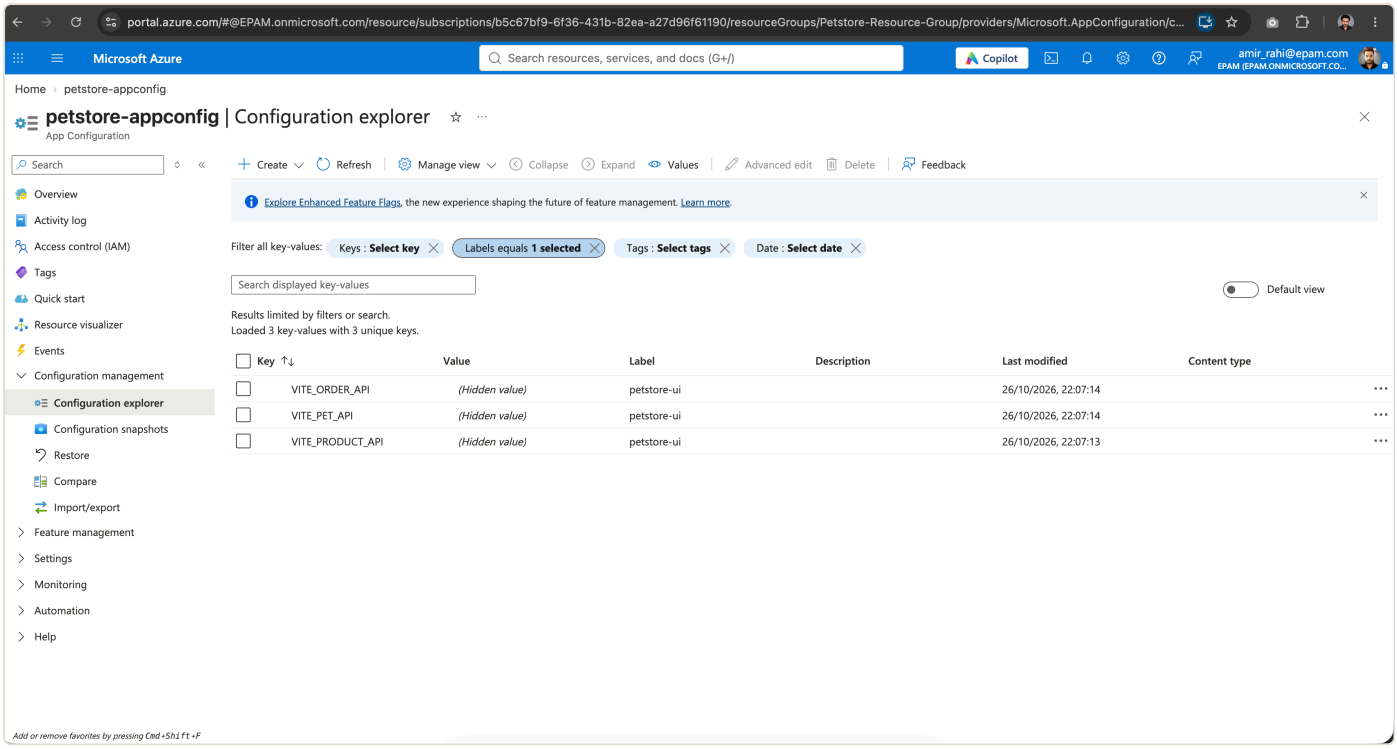

Figure 29. `petstore-appconfig`, Configuration explorer. Label equals `petstore-ui`. Three keys are loaded: `VITE_ORDER_API`, `VITE_PET_API`, and `VITE_PRODUCT_API`. Each value is hidden in the portal.

13. Deployment slots

`petstoreui-web-app` has a deployment slot beside production. The Deployment slots blade lists the production slot and `petstoreui-web-app-staging`. Both sit on plan `ASP-PetstoreResourceGroup-9881`. In this screenshot production is Stopped and holds 100 percent of the traffic. The staging slot is Running and holds 0 percent. Swap is available on the same blade. The app list on the left still shows the four web apps from section 10.

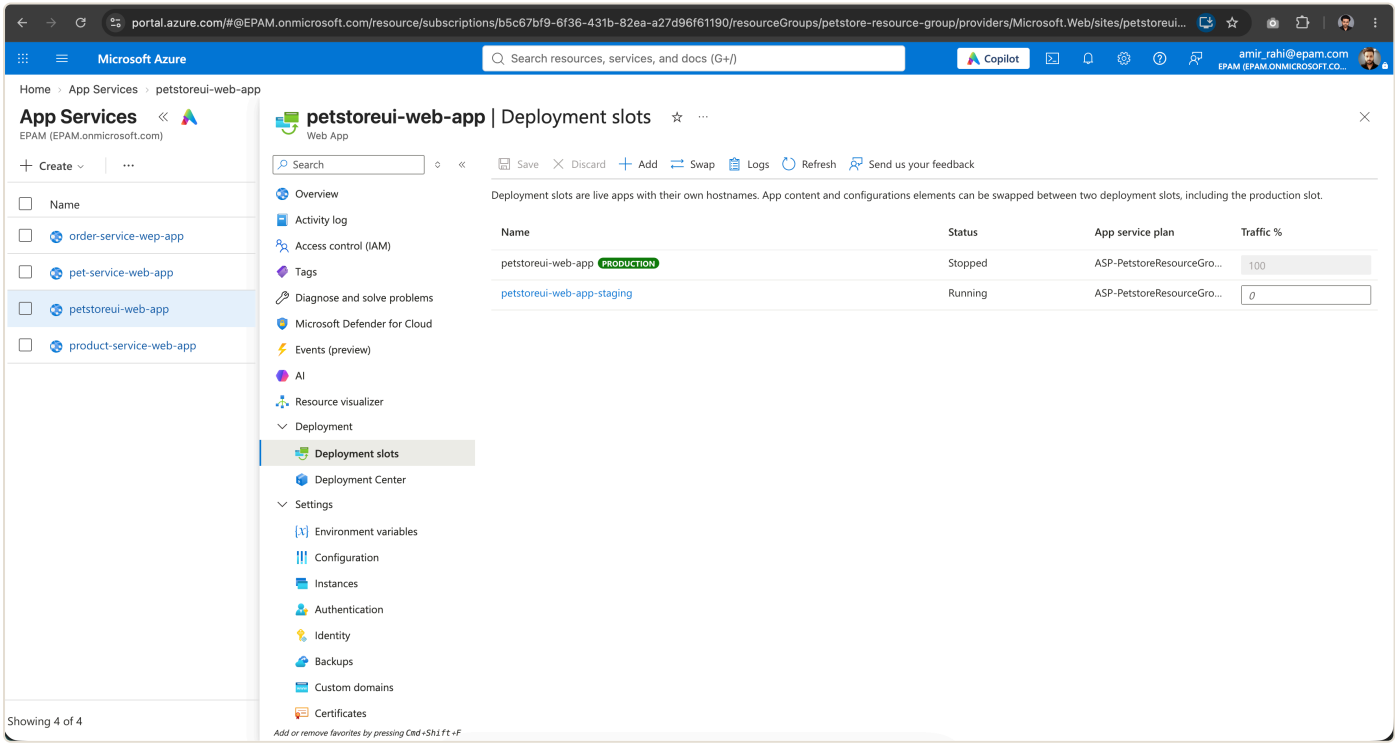

Figure 30. Deployment slots on `petstoreui-web-app`. Production is Stopped with traffic 100. `petstoreui-web-app-staging` is Running with traffic 0.

14. Pet Store monitoring

The deployed shop was opened on its App Service hostname with the browser network panel recording the page load. The document, the script, and the stylesheet return HTTP 200. The catalog call `products?page=0&size=50` is still pending, and the page stays on Loading with empty cards.

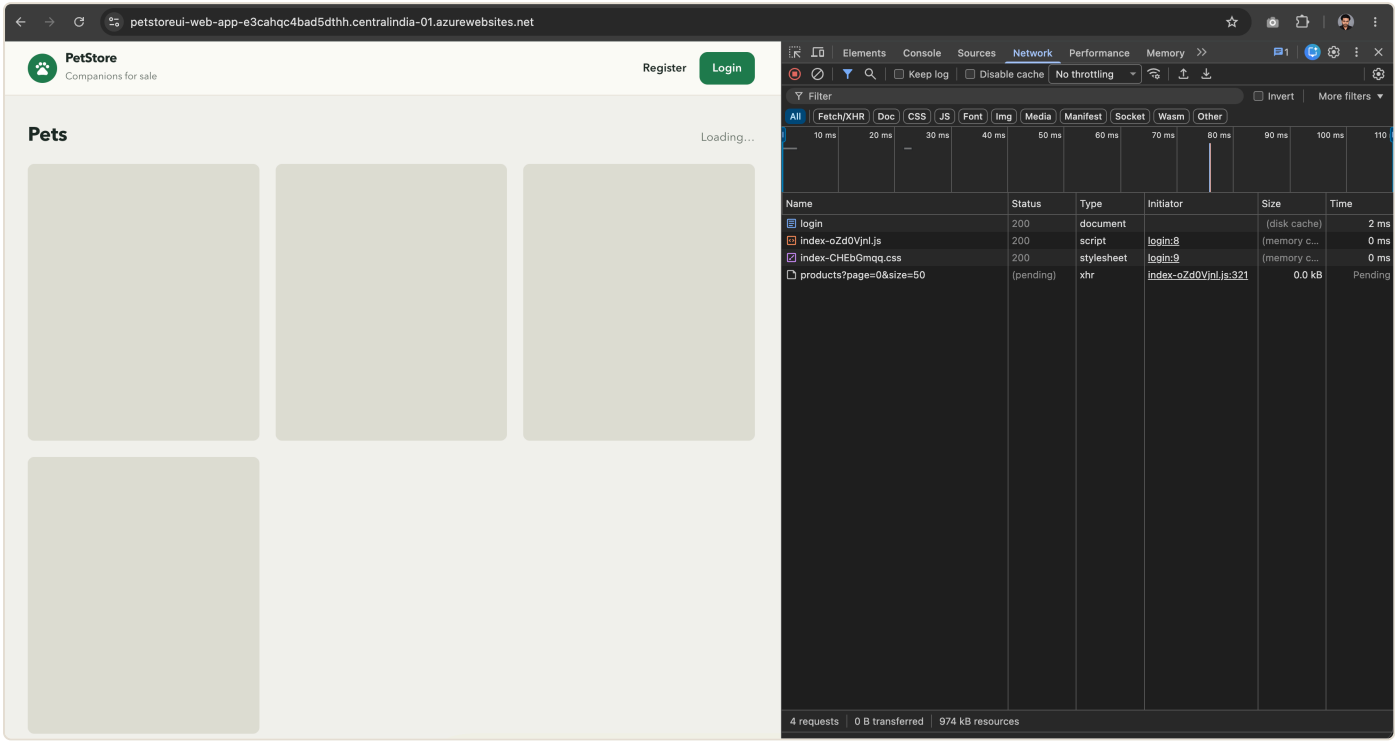

Figure 31. `petstoreui-web-app-e3cahqc4bad5dthh.centralindia-01.azurewebsites.net`. Login, the index script, and the stylesheet are 200. The products request is pending, so the Pets grid is still Loading.

15. Application Insights

Live metrics for `petstore-app-insights` show the product service as it runs on App Service. The blade is the last 60 seconds. Incoming charts are request rate, request duration, and request failure rate. Outgoing charts are dependency call rate, dependency duration, and dependency failure rate. Overall health shows committed memory, CPU total, and exception rate. Sample telemetry on the right is the log line behind a spike. Servers lists each process with requests, failed requests, dependencies, CPU, and committed memory.

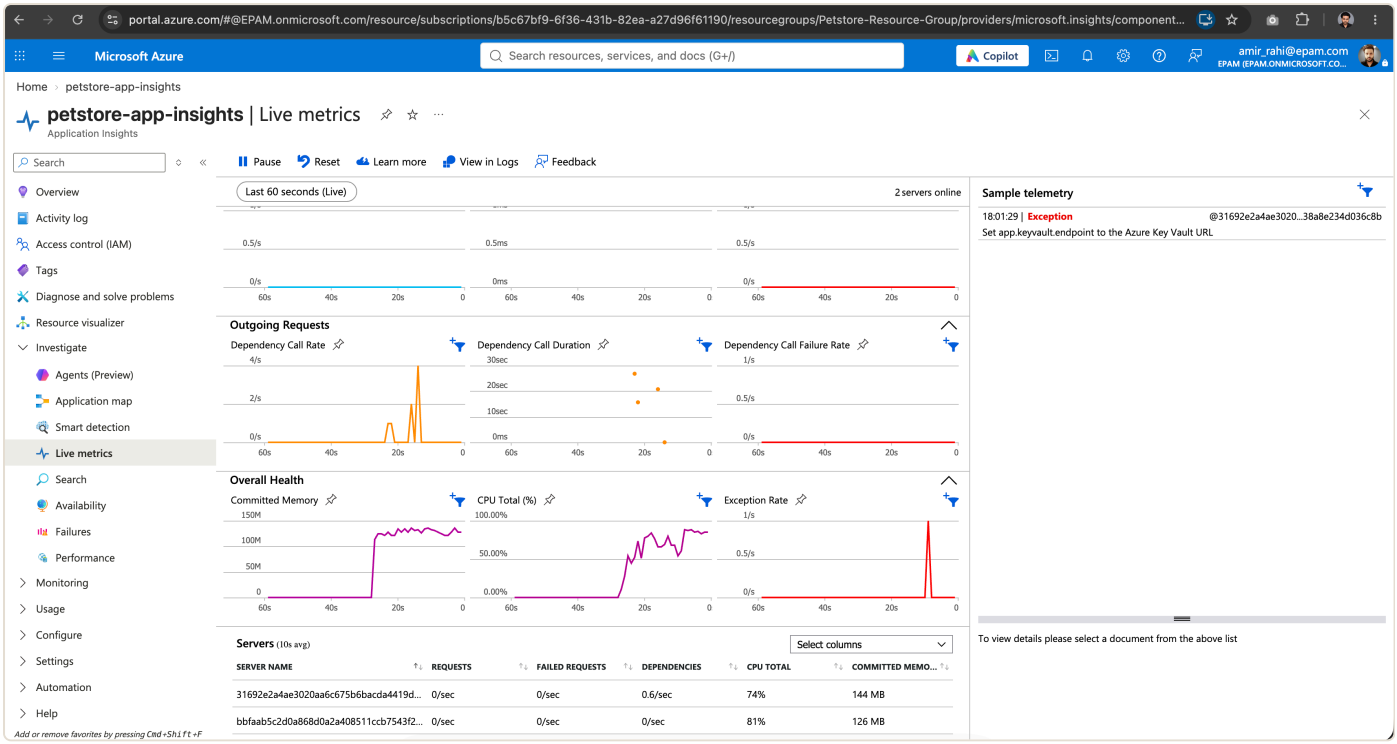

Figure 32. Live metrics, two servers online, at 18:01. Sample telemetry is the exception "Set app.keyvault.endpoint to the Azure Key Vault URL". Request charts are at zero. Committed memory and CPU are plotted, and the servers table shows about 74 percent and 81 percent CPU with 144 MB and 126 MB committed.

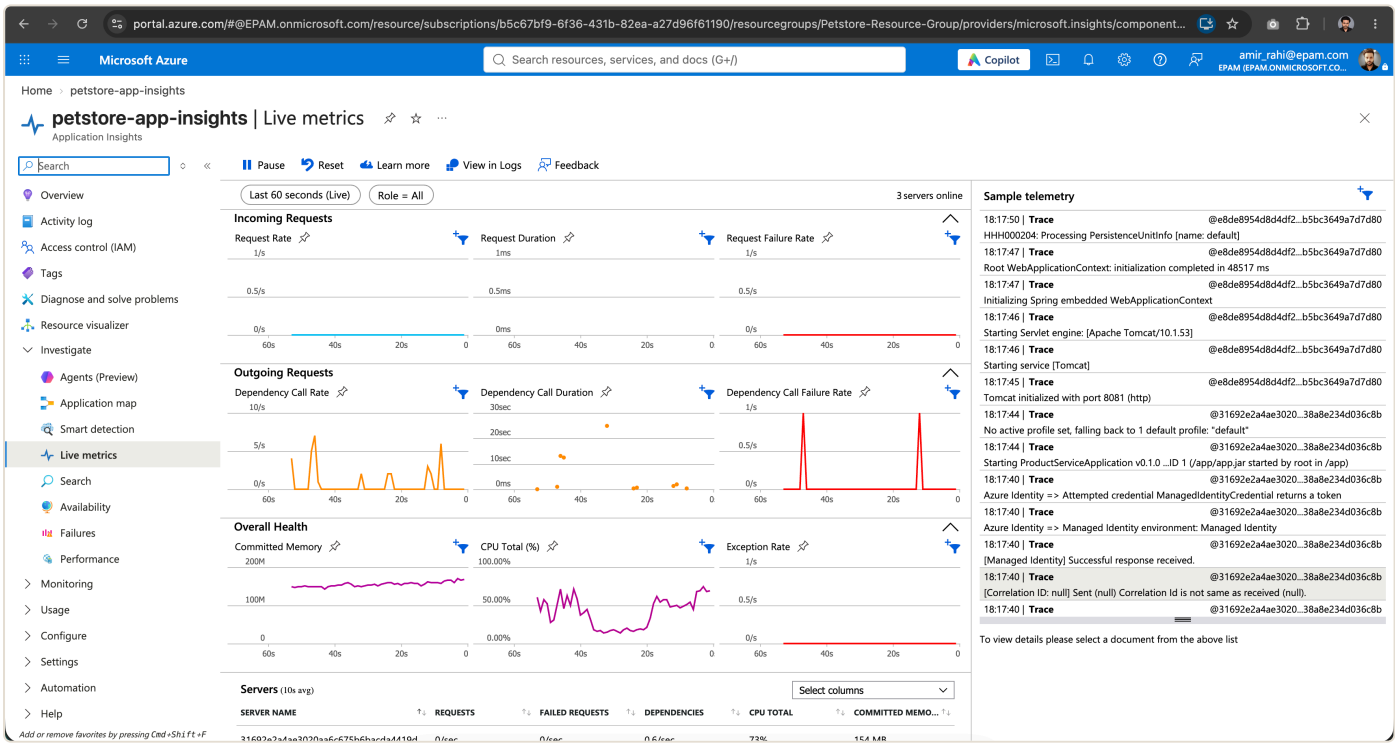

Figure 33. The same Live metrics blade, three servers online, at 18:17. Traces record Spring context initialization, Tomcat on port 8081, `ProductServiceApplication` started, and a managed-identity token. CPU and committed memory stay on the overall-health row. Dependency failure rate spikes once in the minute.

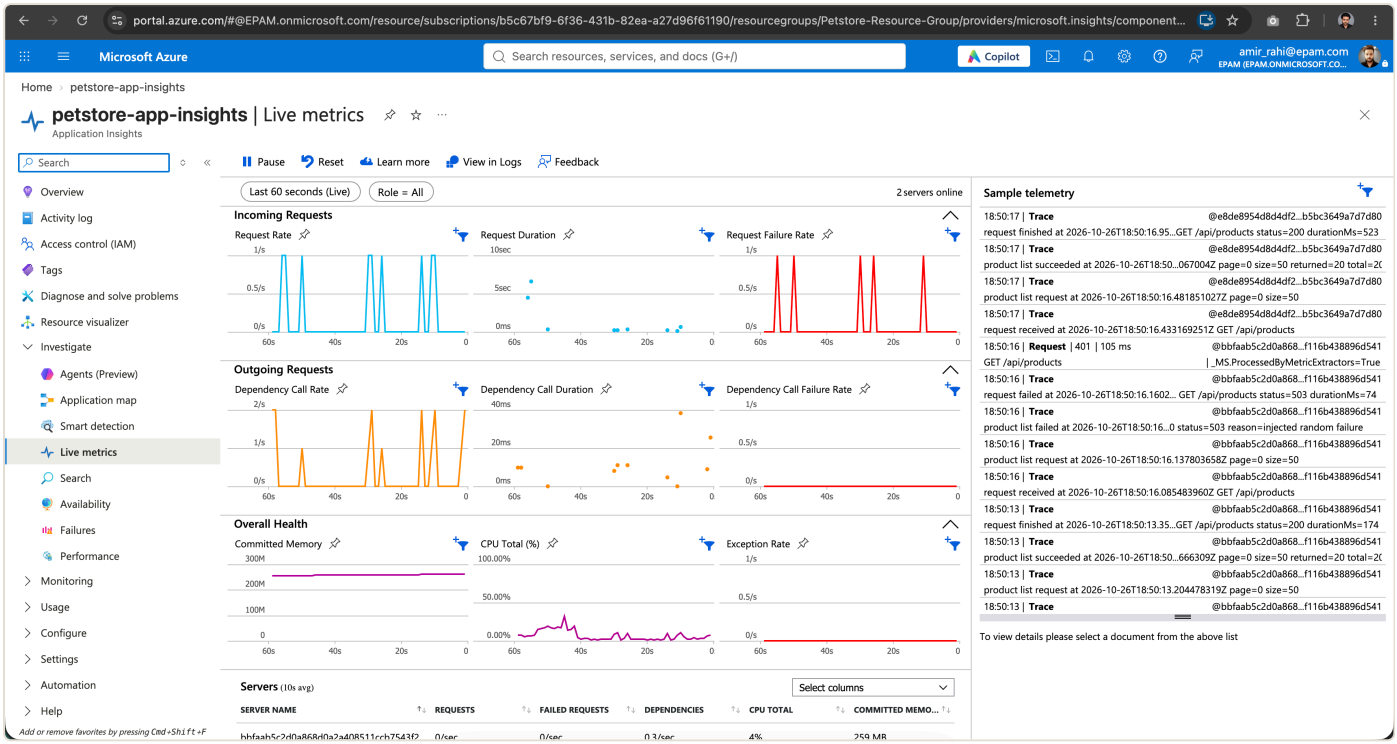

Figure 34. Live metrics at 18:50, two servers online. Traces for `GET /api/products` include status 200 and status 503 with the reason "injected random failure". One request line is HTTP 401. Request failure rate spikes with those 503s. Committed memory holds near 250 MB and CPU stays low.

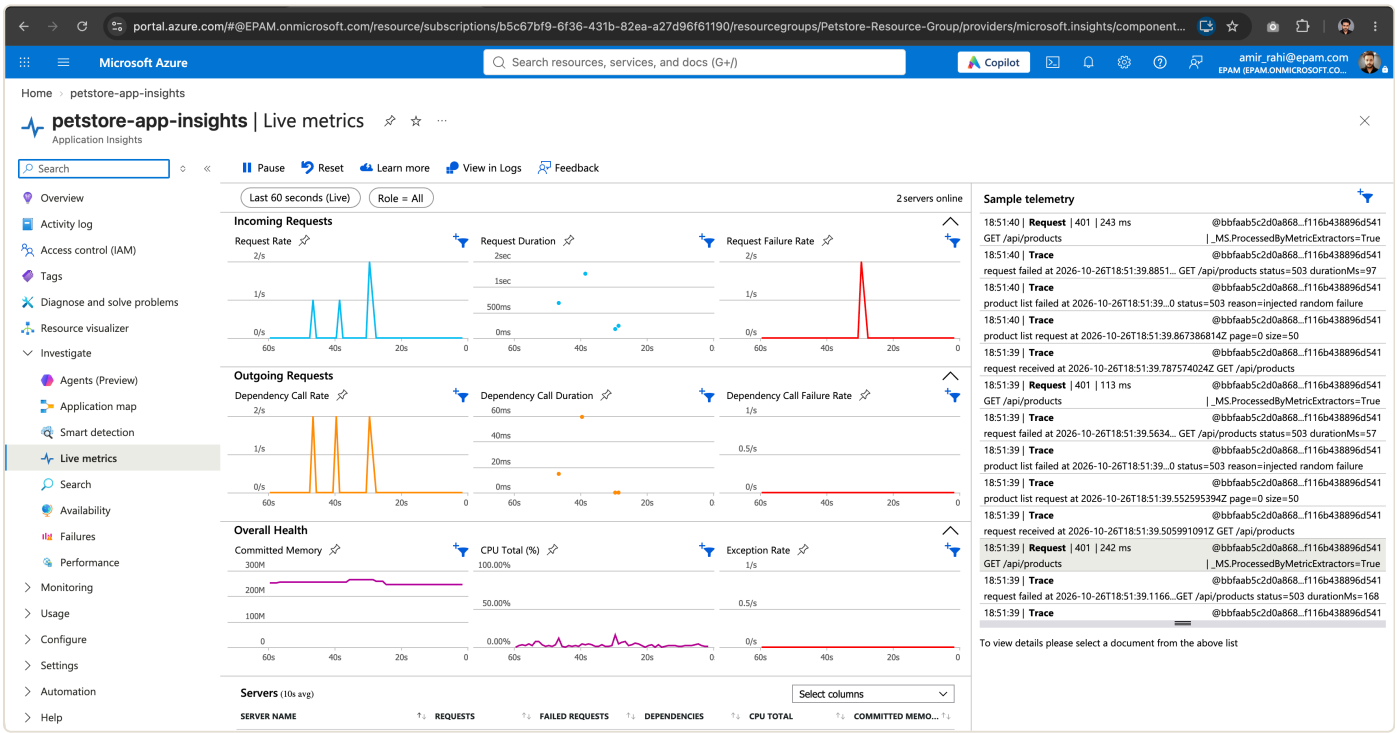

Figure 35. Live metrics at 18:51, two servers online. Request failure rate spikes at the right edge. Sample telemetry shows `GET /api/products` failing with status 503, "injected random failure", and one request recorded as 401 in 243 ms. Committed memory and CPU remain on the overall-health charts.

16. Azure Cosmos DB

The shop cart and a copy of the catalog sit in Azure Cosmos DB for NoSQL, account `petstore-cosmos`, database `petstore`. PostgreSQL remains the store for products, categories, orders, and users. The order service writes the signed-in shopper's cart to container `cart`. One document per user uses the user id as both `id` and `userId`, and the lines live in `items`. The product service copies each saved pet into container `products`. Data Explorer shows both containers under the same database.

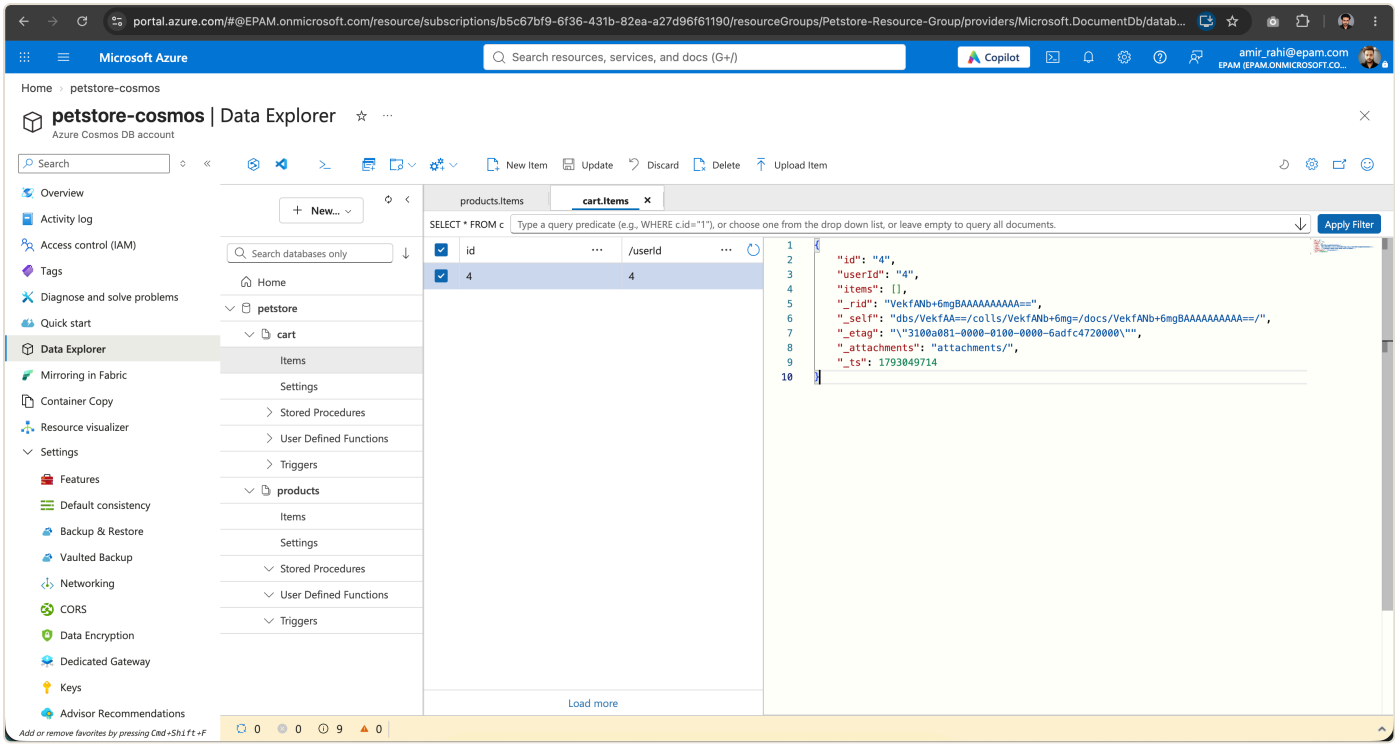

Figure 36. Data Explorer on `petstore-cosmos`. Database `petstore`, container `cart`, document id 4. `userId` is 4 and `items` is an empty array.

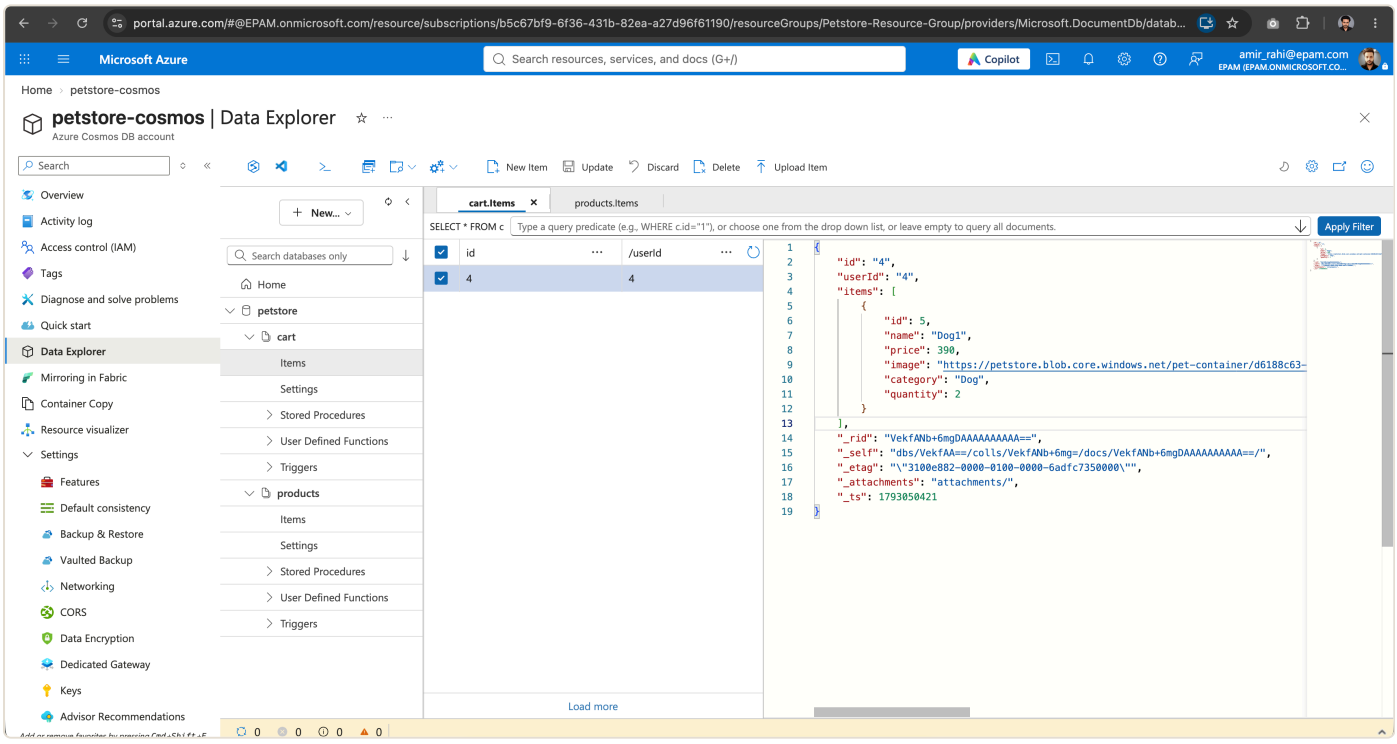

Figure 37. The same cart document after a shop add. `items` holds Dog1, price 390, category Dog, quantity 2, with the photo URL in Blob Storage. Container `products` is listed beside `cart`.

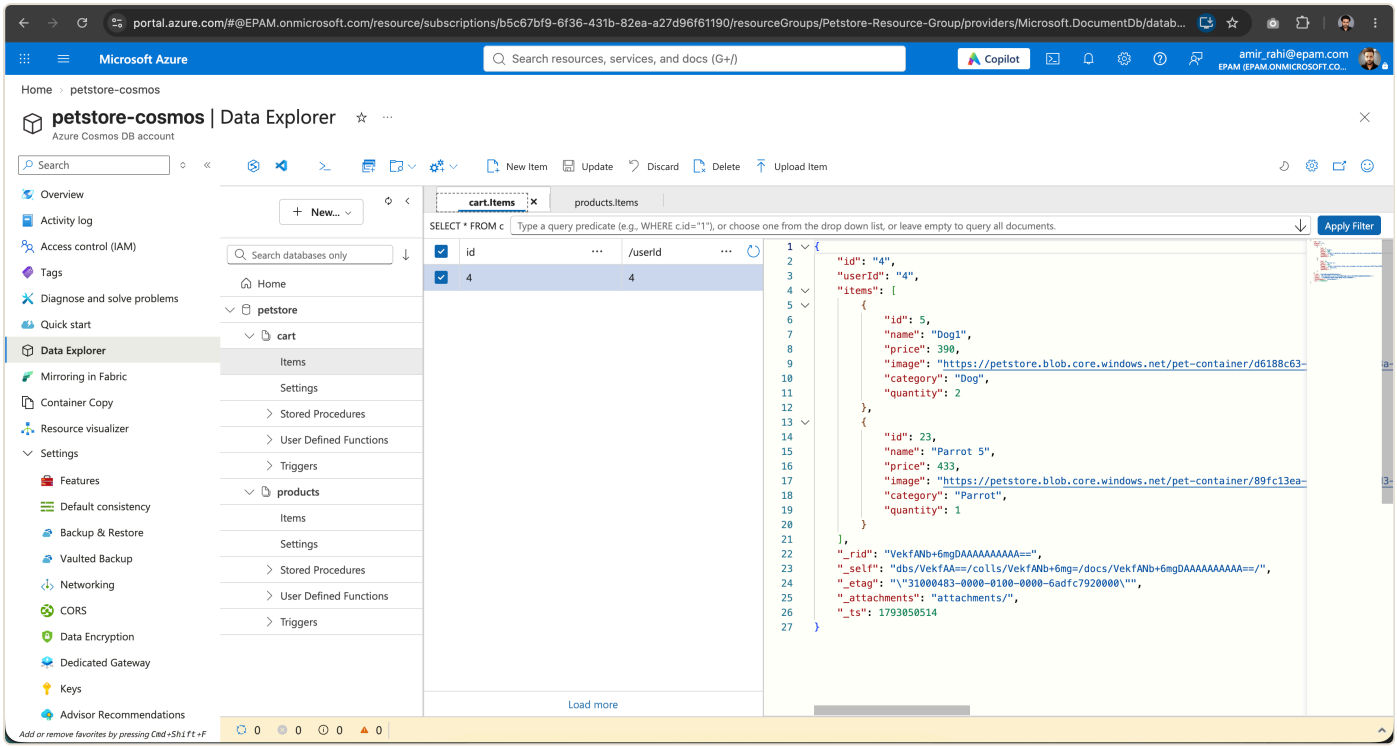

Figure 38. Cart document 4 with two lines: Dog1, quantity 2, and Parrot 5, price 433, quantity 1. The query is `SELECT * FROM c on cart`.

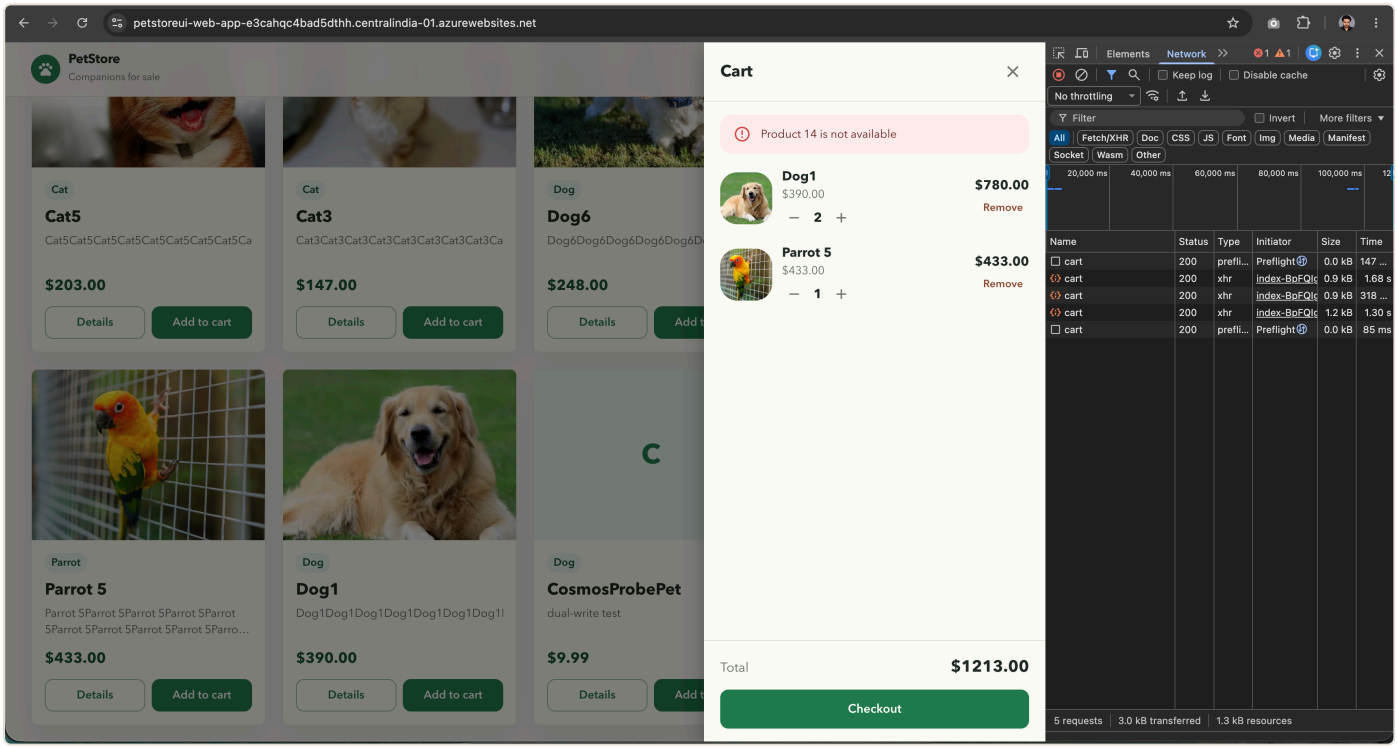

Figure 39. The deployed shop at `petstoreui-web-app-e3cahqc4bad5dthh.centralindia-01.azurewebsites.net`. The cart drawer lists Dog1 at quantity 2 and Parrot 5 at quantity 1. The total is $1,213.00. The network panel shows the cart calls returning 200. The catalog also includes CosmosProbePet, the pet written through the product copy.

17. PostgreSQL products, later query

Section 6 showed the flexible server and the `petstore` database. A later query in pgAdmin reads the same table after more pets were saved. The public schema lists `categories`, `order_items`, `orders`, `pets`, `products`, `roles`, and `users`.

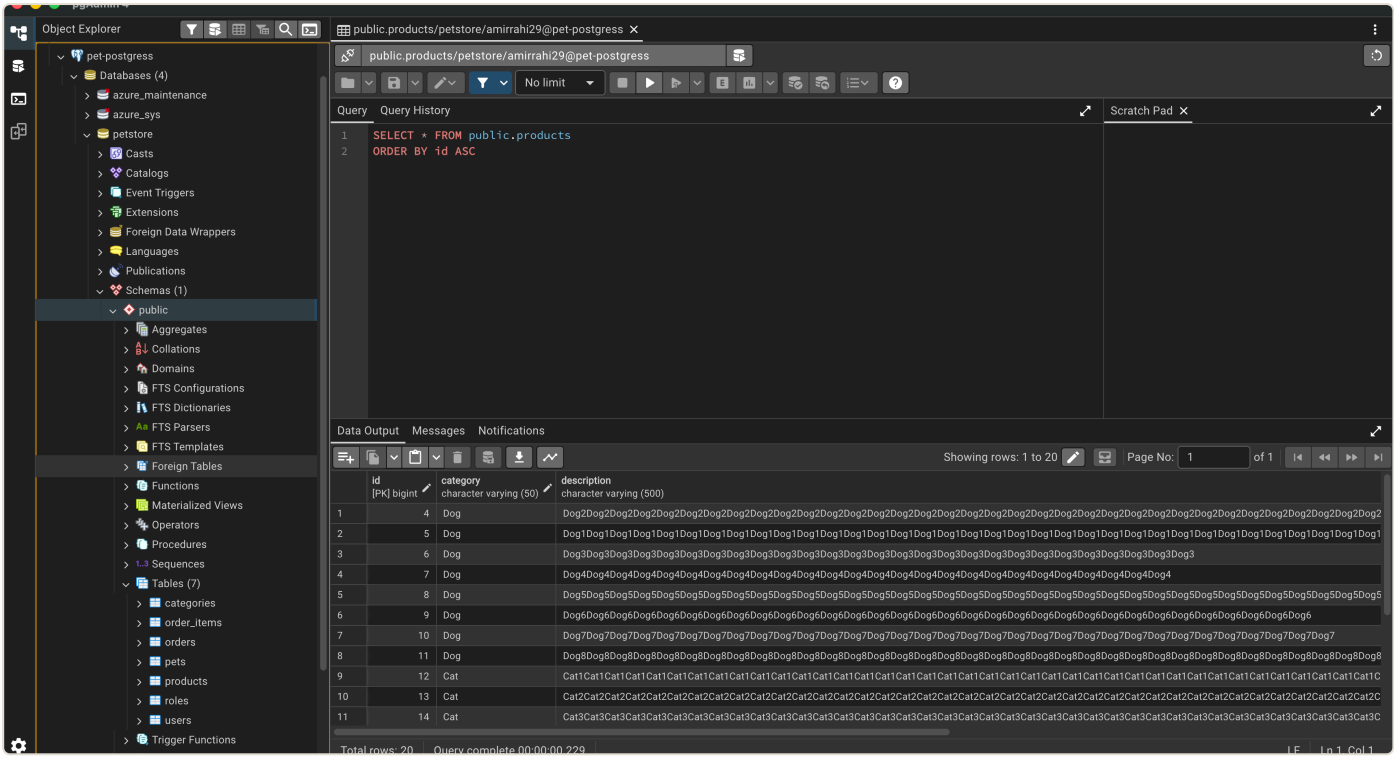

Figure 40. pgAdmin, database `petstore` on `pet-postgres`. `SELECT * FROM public.products ORDER BY id ASC` returns 20 rows, ids 4 through 13 on the first page, categories Dog and Cat.

18. App Service scale-out

Plan `ASP-PetstoreResourceGroup-9881` can add instances from the Scale out blade. The selected scale-out method is Rules Based, so the instance count follows user-defined rules on a schedule or a metric. Manual would hold a constant count. Automatic scale-out is not available on this plan.

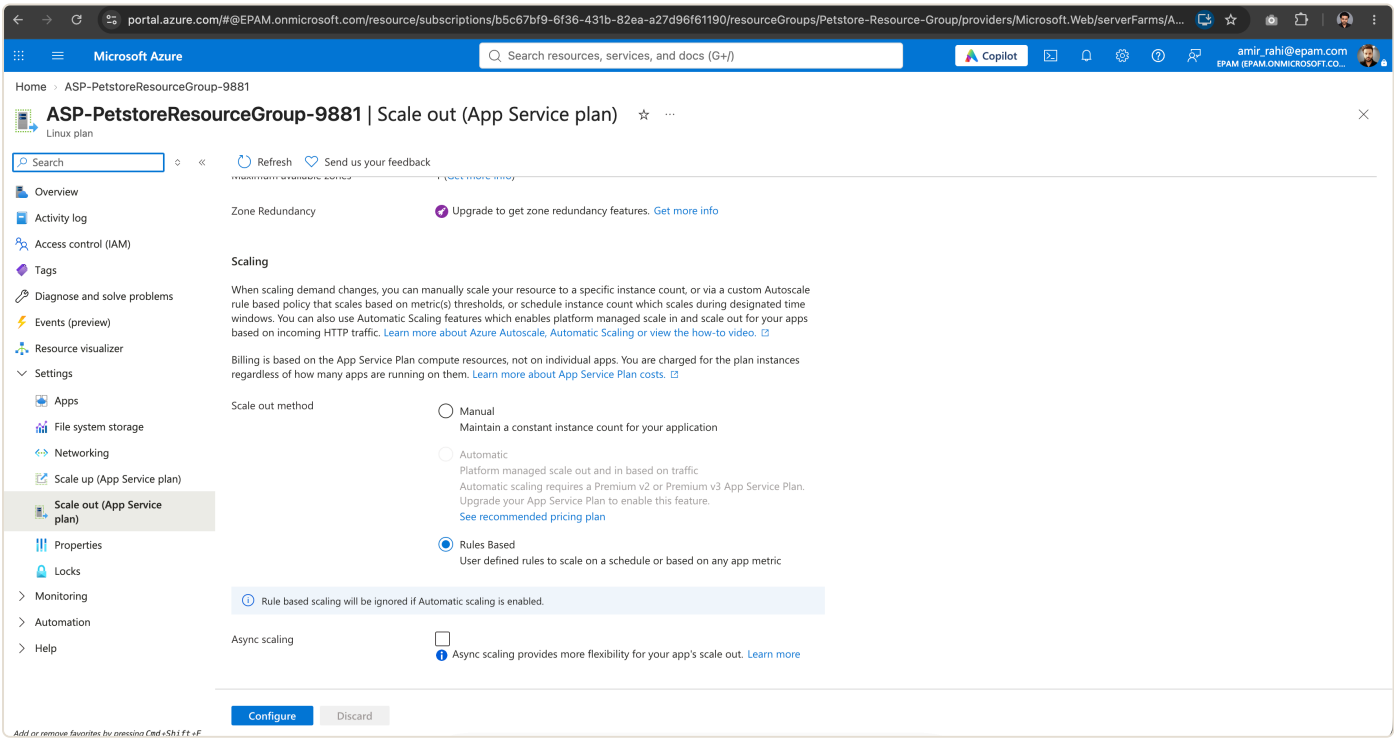

Figure 41. Scale out for `ASP-PetstoreResourceGroup-9881`. Rules Based is selected. The note on the blade says rule-based scaling is ignored if Automatic scaling is enabled.

19. Azure Function HTTP trigger

Function App `petstore-function` in `petstore-resource-group` hosts the HTTP trigger `petstore-http`. The function is packaged as `petstore-http-1.0-SNAPSHOT.jar`, entry point `com.petstore.Function.run`. `function.json` binds an anonymous HTTP trigger named `req` that accepts GET and POST. Code + Test runs the function in the portal and returns the response beside the log.

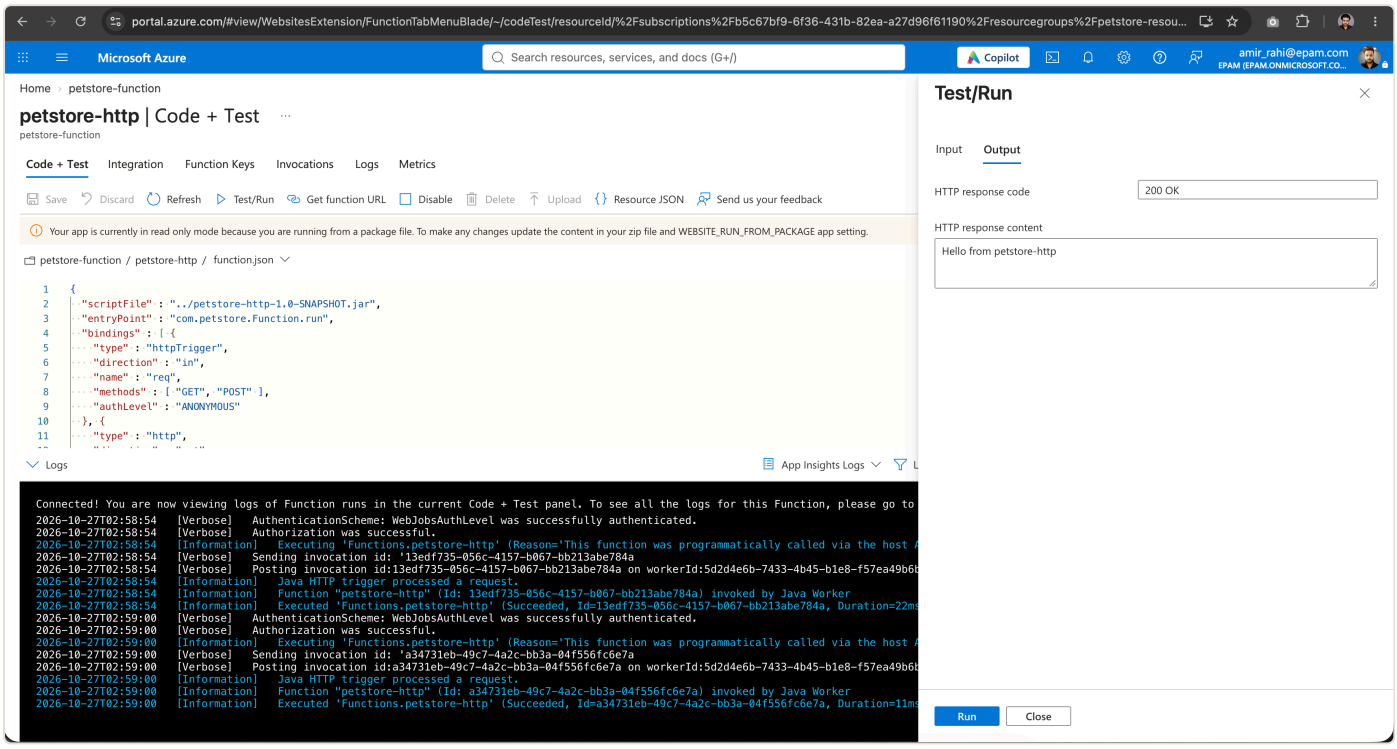

Figure 42. Code + Test for `petstore-http`. The Test/Run output is HTTP 200 with the body `Hello from petstore-http`. The log shows the host authenticated, the Java worker invoked the function, and the execution succeeded.

20. Microsoft Entra External ID

Customer sign-in uses a dedicated Microsoft Entra External ID tenant, separate from the workforce directory that owns the Petstore subscription. The paid store is registered there as `petstore-customer-app`. Customers authenticate with the `signupsignin` user flow. The Spring Boot application on the developer machine exchanges the authorization code, and the React demo published to App Service calls that local application through a public HTTPS tunnel. Secret values are not printed.

Requirement	Result in this setup
Dedicated External ID tenant	`petstore-tenant`, primary domain `petstorecustomers.onmicrosoft.com`
Paid store application	`petstore-customer-app`, client ID `d61edc16-5f84-4246-8bb1-2353a3ec516d`
User flow, identity provider, attributes	`signupsignin` with email and password local accounts
Optional claims	`email`, `given_name`, and `family_name`, requested with scopes `openid profile email`
OAuth 2.0 / OpenID Connect in Spring Boot	Authorization code with PKCE, token exchange, and a server session
Local authentication test	Completed against the local Spring Boot process before the cloud test

Requirement	Result in this setup
Azure deployment with External ID	React demo on `intra-web-application`, Java backend kept local
Production redirect URI	Three web redirect URIs, including the App Service origin and the tunnel origin

Dedicated tenant for customer identities

The customer directory is `petstore-tenant`. Its tenant ID is `9f8d97a4-9691-45df-8fb9-8c215ff7b7be` and its primary domain is `petstorecustomers.onmicrosoft.com`. The license shown on the overview is Microsoft Entra ID Free. Sign-in for this directory is served from `ciamlogin.com`, which is the External ID endpoint, rather than from the workforce login host. The tenant overview records 2 users and 2 applications.

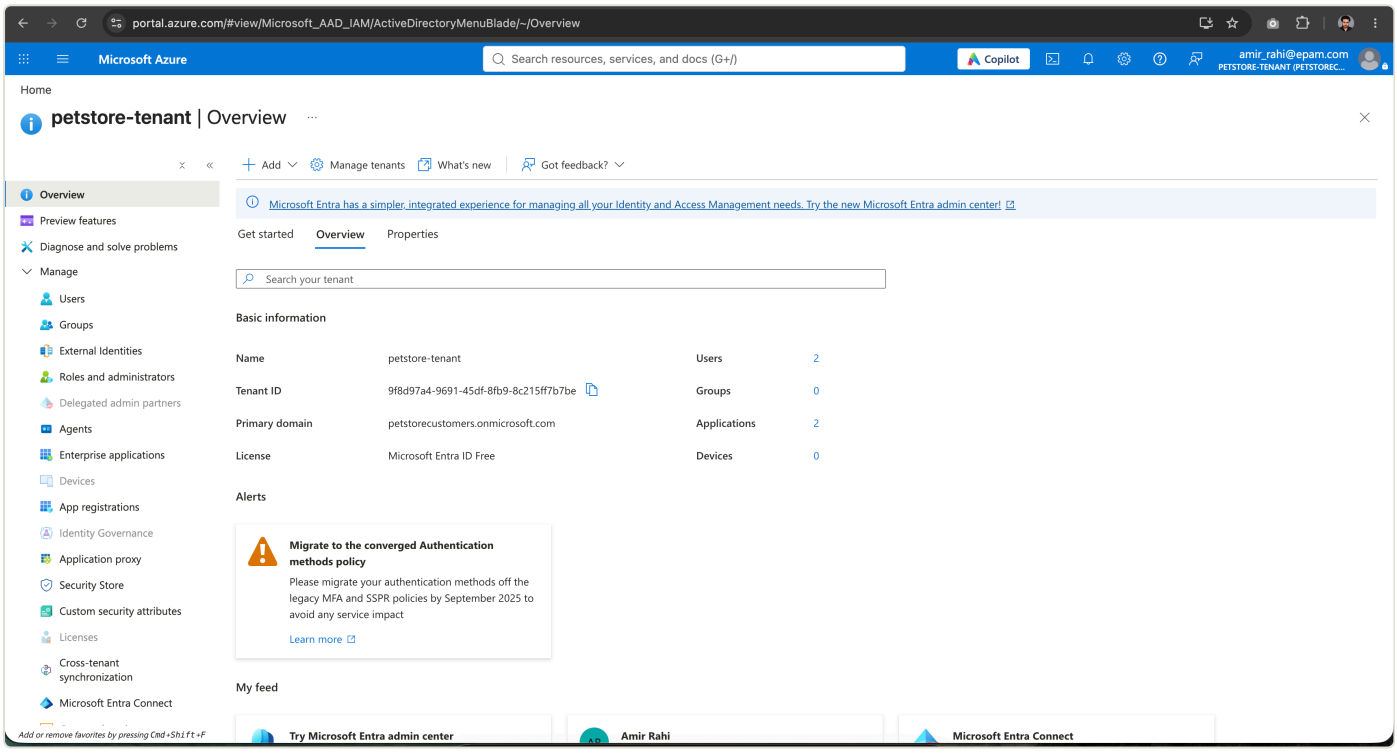

Figure 43. Overview of `petstore-tenant`. Name, tenant ID, primary domain `petstorecustomers.onmicrosoft.com`, 2 users, and 2 applications.

The two directory users are both members. `amirrahi29@petstorecustomers.onmicrosoft.com` is the local customer account used in the sign-in test. `amir_rahi_epam.com#EXT#@petstorecustomers.onmicrosoft.com` is the external administrator invited from the workforce account. Customer identities therefore live in this tenant, and the workforce user is only an external member for administration.

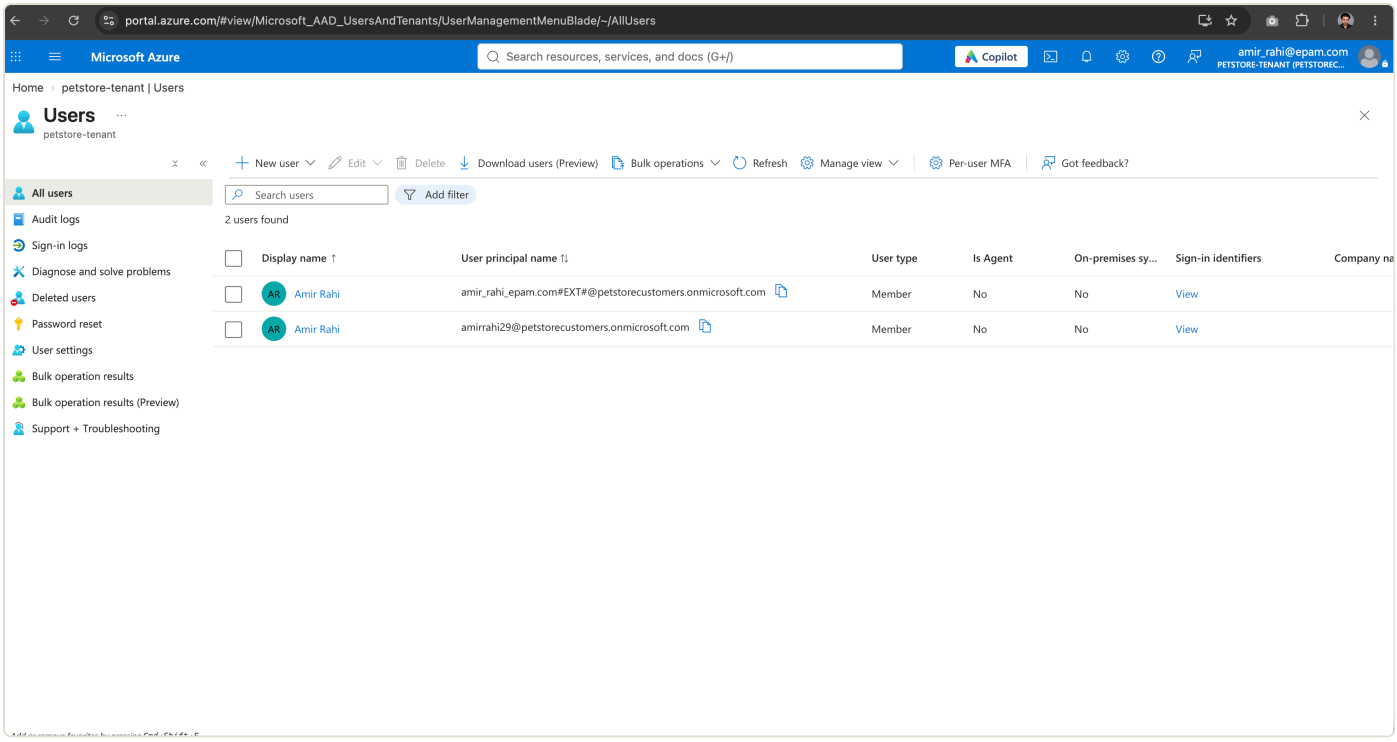

Figure 44. Users in `petstore-tenant`. Two members: the external administrator and `amirrahi29@petstorecustomers.onmicrosoft.com`.

Paid store application registration

The store application in this tenant is `petstore-customer-app`. It was created on 27 October 2026. The application (client) ID is `d61edc16-5f84-4246-8bb1-2353a3ec516d` and the object ID is `77cf9188-81c6-42fb-98a4-1564e07a24ca`. Supported account type is this organization only. The second registration, `b2c-extensions-app`, is the platform application Entra creates for user-attribute storage and is left unchanged.

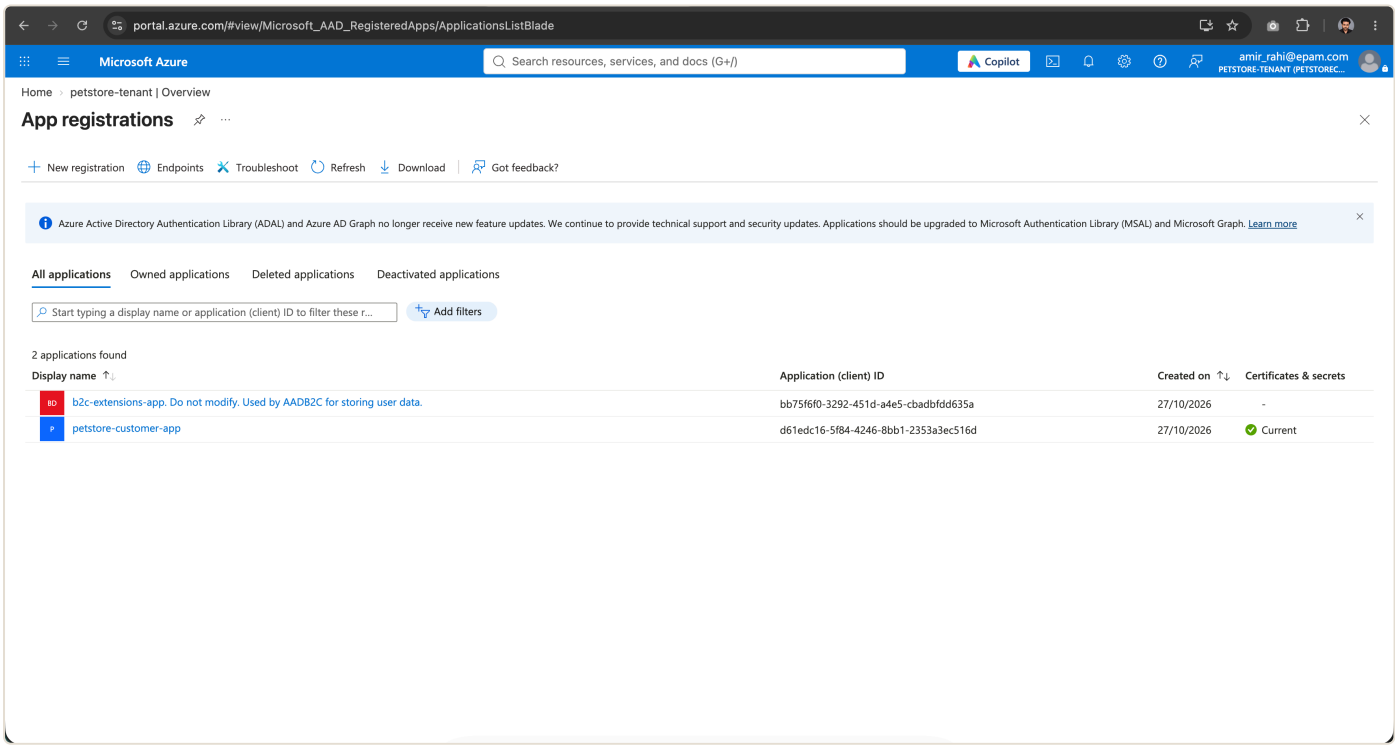

Figure 45. App registrations in `petstore-tenant`. `petstore-customer-app` is listed with its client ID, created on 27 October 2026, and certificates and secrets marked Current.

The application overview shows the registration is Activated. Client credentials are 0 certificates and 2 secrets. A secret is required because this is a web application: the Spring Boot token request sends that secret with the authorization code. Redirect URIs are 3 web, 0 single-page, and 0 public client. Those

three web URIs are the local callback `http://localhost:8083/login/oauth2/code/azure`, the deployed site origin `https://intra-web-application-agb3cearbmcef5ej.westus3-01.azurewebsites.net/`, and the tunnel origin `https://captured-feature-hearing-scholar.trycloudflare.com` with no trailing slash.

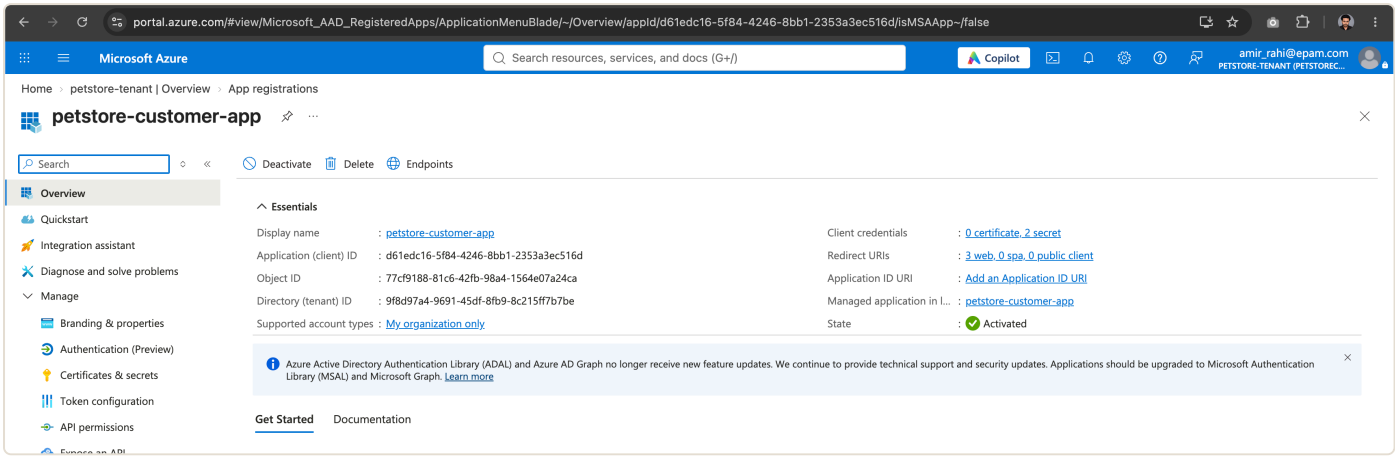

Figure 46. Overview of `petstore-customer-app`. Client ID, directory tenant ID, state Activated, 2 secrets, and 3 web redirect URIs.

User flow, identity provider, and attributes

The customer journey is the `signupsignin` user flow. The authorize request selects it with `p=signupsignin` and sends `prompt=login`, so Entra asks for credentials instead of silently reusing a session. The identity provider for this flow is the local email-and-password account in the External ID tenant. The first page collects the email address. The second page collects the password for that account. Email is both the sign-in identifier and a collected attribute. Given name, surname, and display name are the profile attributes associated with the same user flow.

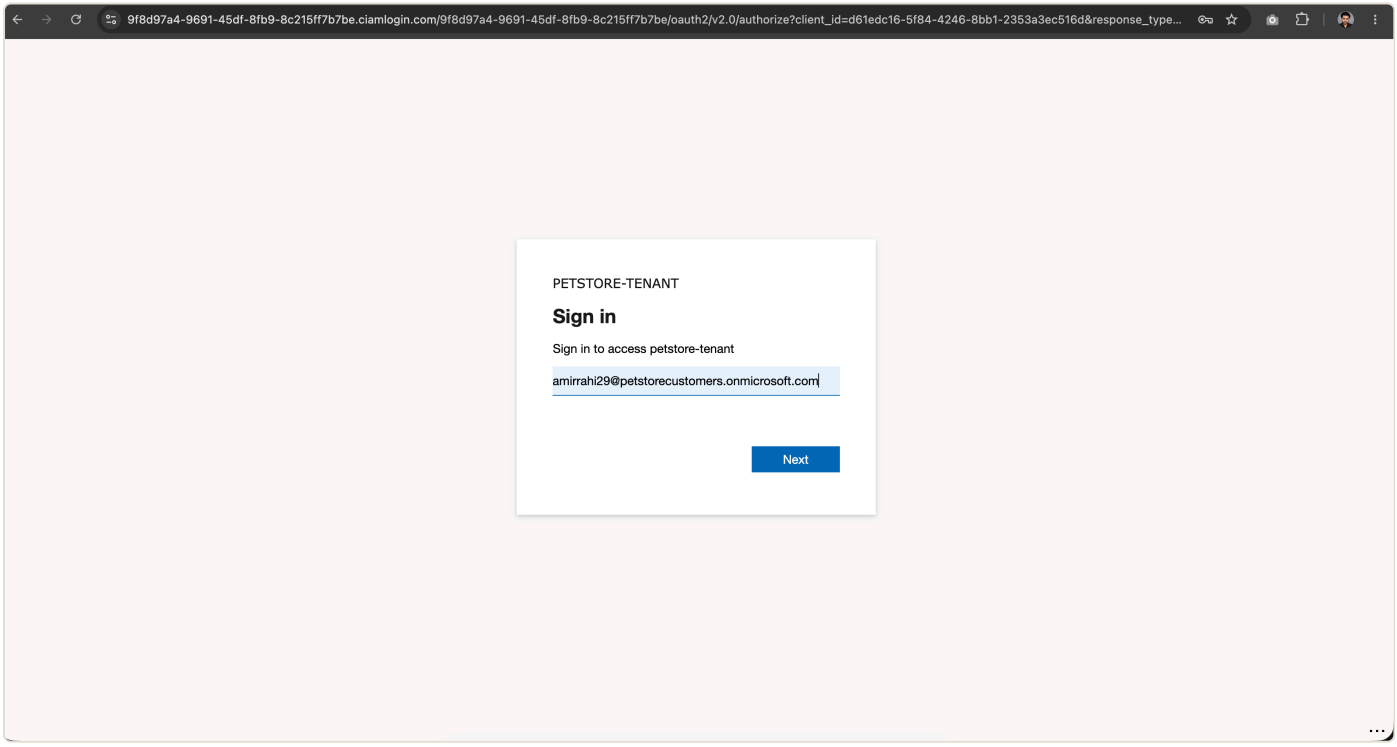

Figure 47. External ID sign-in hosted at `ciamlogin.com` for PETSTORE-TENANT. The customer email `amirrahi29@petstorecustomers.onmicrosoft.com` is entered. The address bar is the v2.0 authorize endpoint for client `d61edc16-5f84-4246-8bb1-2353a3ec516d`.

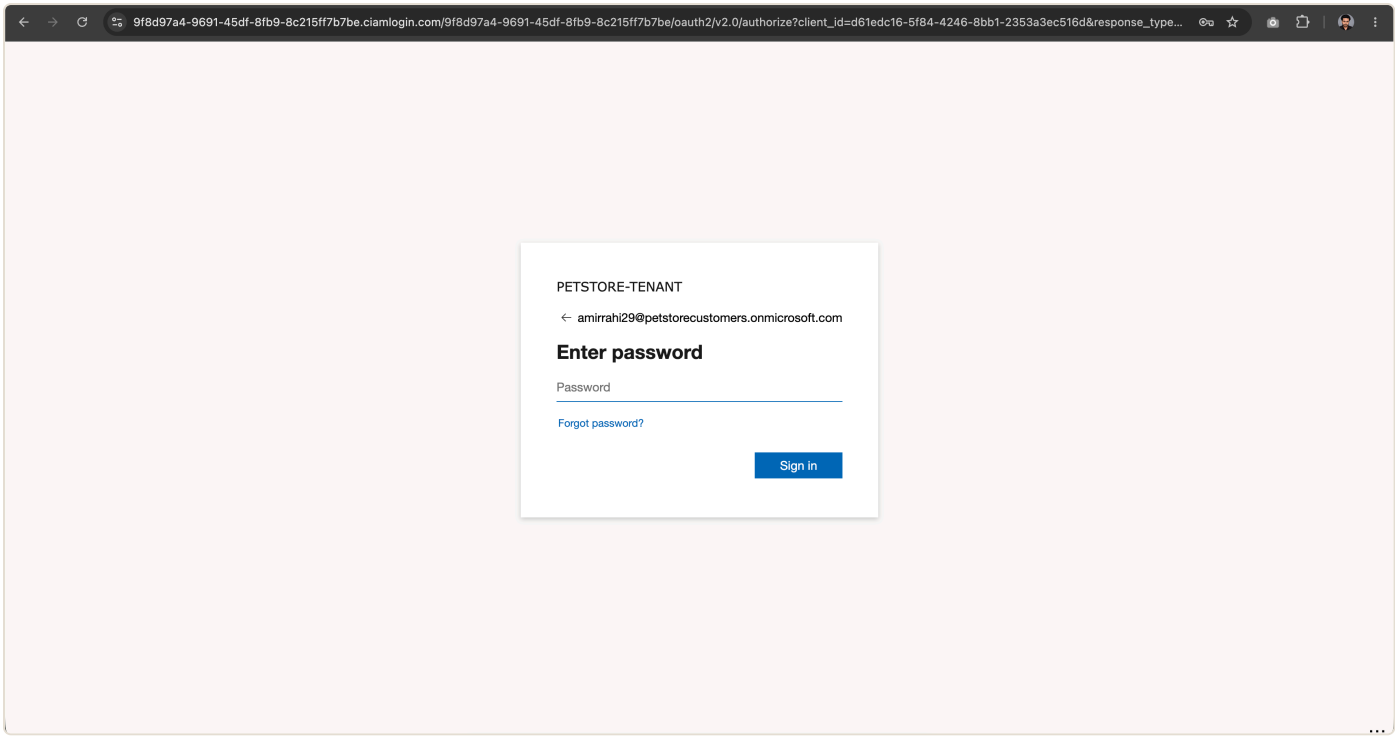

Figure 48. Password step for the same account. Entra accepted the email and is collecting the local-account password. Forgot password is offered by the user flow.

Optional claims

The customer application is configured so the ID token can carry `email`, `given_name`, and `family_name`. The sign-in request asks for those claims by sending the scopes `openid`, `profile`, and `email`. That scope list is part of the authorize URL in Figure 47 and Figure 48. The Spring Boot callback reads the ID token after the code exchange and keeps the browser session only when the token is present. The portal Token configuration blade is not one of these screenshots. The app registration menu in Figure 46 includes Token configuration, and the authorize request shows the scopes that pull the optional profile claims.

OAuth 2.0 and OpenID Connect in Spring Boot

The Intra ID backend is a Java 17 Gradle Spring Boot application, `com.intra.id`, running locally with `bootRun` on port 8080. It is the confidential client for `petstore-customer-app`. `GET /login/entra` starts the authorization-code flow with a PKCE S256 challenge and redirects the browser to the tenant authorize endpoint. Entra returns the code to the tunnel origin, which is the registered redirect URI. The application posts that code, the PKCE verifier, the client ID, and the client secret to the v2.0 token endpoint. The secret is read from local configuration and is not stored in the repository. On a successful ID token the server marks the HTTP session signed in and sends the browser back to the frontend. `GET /api/entra/items` returns the demo rows Collar 12, Leash 8, and Bowl 5 only for that session. `POST /logout` invalidates the session. The session cookie is `SameSite=None` and `Secure`, and cross-origin calls from the App Service hostname are allowed with credentials, so the deployed page can use the local session.

Local authentication test

The same flow was exercised locally before the cloud test. The React dev server on port 3000 sent the browser to the Spring Boot sign-in endpoint, Entra completed the user flow, and the demo list was returned to the local page. Logout returned the browser to the sign-in form because `prompt=login` forces a fresh credential prompt. Those local pages are not in this screenshot set. Figures 47 through 50 are the later run from the deployed site, which uses the same application registration, the same user flow, and the same token exchange.

Deployment to Azure with External ID

The customer demo frontend is the React application in `Intra-ID/intra-frontend`. GitHub Actions workflow `intra-frontend.yml` on `main` of `amirrahi29/Petstore-Microservice-App-SpringBoot-React` builds that application and deploys it to App Service `intra-web-application` in `petstore-resource-group`, West US 3, plan `ASP-petstoreresourcegroup-86dc`. Publishing for that app is Code. The app was also set to the Java 17 stack while the Spring Boot package was being prepared. The package that is live is the React frontend: the site serves the Pet Store page, and the Java Intra ID process was not deployed. It stays on the developer machine at port 8080. A later frontend publish pointed the page at the public tunnel so sign-in and the items API leave the App Service host and reach that local process.

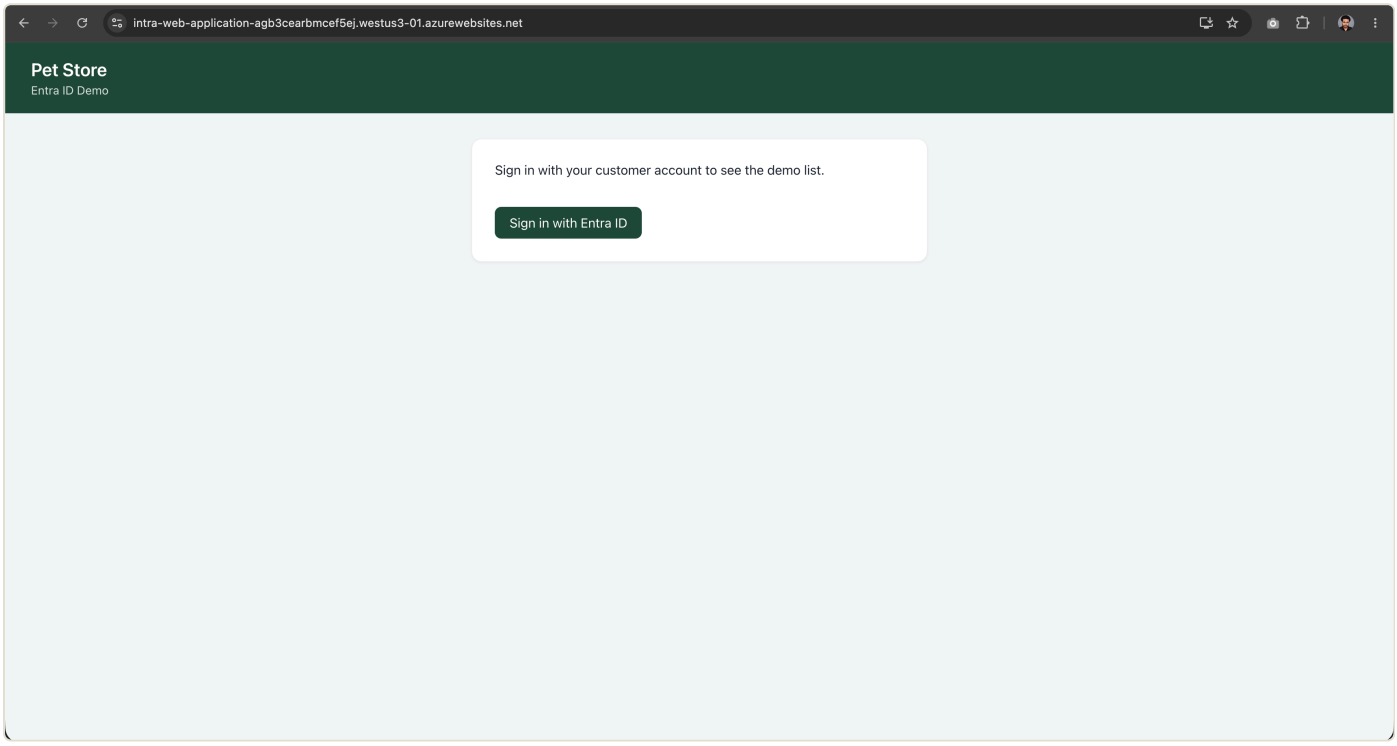

Figure 49. `https://intra-web-application-agb3cearbmcef5ej.westus3-01.azurewebsites.net` before sign-in. The page is the deployed React demo, with the header Pet Store / Entra ID Demo and the button Sign in with Entra ID.

Production redirect URI

Entra accepts a redirect only when it matches a URI on the application. Figure 46 records three web redirect URIs. The local development callback remains `http://localhost:8083/login/oauth2/code/azure`. The deployed site origin `https://intra-web-application-agb3cearbmcef5ej.westus3-01.azurewebsites.net/` is registered for the cloud frontend. The local Java process is not reachable from that site on `localhost`, so a public HTTPS tunnel fronts port 8080. An ngrok tunnel at `https://discount-tipped-hurray.ngrok-free.dev` answered with the free-plan browser warning page, which would replace the OAuth redirect. The working tunnel is `https://captured-feature-hearing-scholar.trycloudflare.com`, registered with `https://` and no trailing slash, and that exact origin is the `redirect_uri` sent by the Spring Boot application. After the password in Figure 48 is accepted, Entra returns the browser to that URI, the local application creates the session, and the browser lands on the App Service site already signed in.

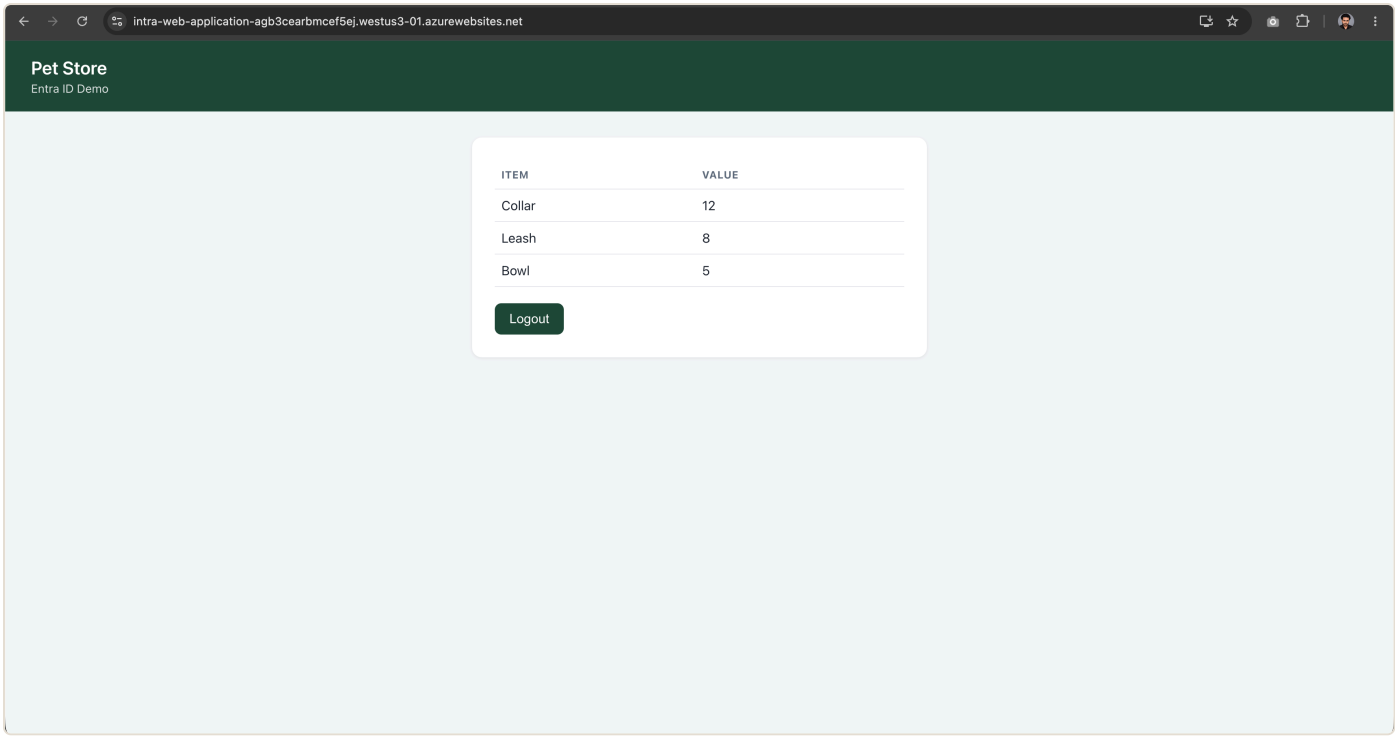

Figure 50. The same App Service hostname after the External ID sign-in. The session is authenticated: the demo list shows Collar 12, Leash 8, and Bowl 5, and Logout is available. The sign-in button is gone.

21. Order item reserver

The order item reserver is a separate Java 17 Spring Boot application, `order-item-reserver`. It listens on the Service Bus queue `order-item-reservations`, uploads to the `orderitems` storage account with exponential backoff, and posts a failure notification to the Logic App `order-item-reserver-email-fallback`. The five screenshots groups below are the portal record of that work. Sections 1 through 20 are unchanged.

Modify and deploy the application

Work item: modify and deploy the application. App Service `order-item-reserver` is the deployed reserver. It sits in `petstore-resource-group`, West US 3, on Linux plan `ASP-petstoreresourcegroup-868c` (B1). Publishing model is Container. Runtime status is Healthy, and the container image comes from `petstoreazurecontainerregistry.azurecr.io`.

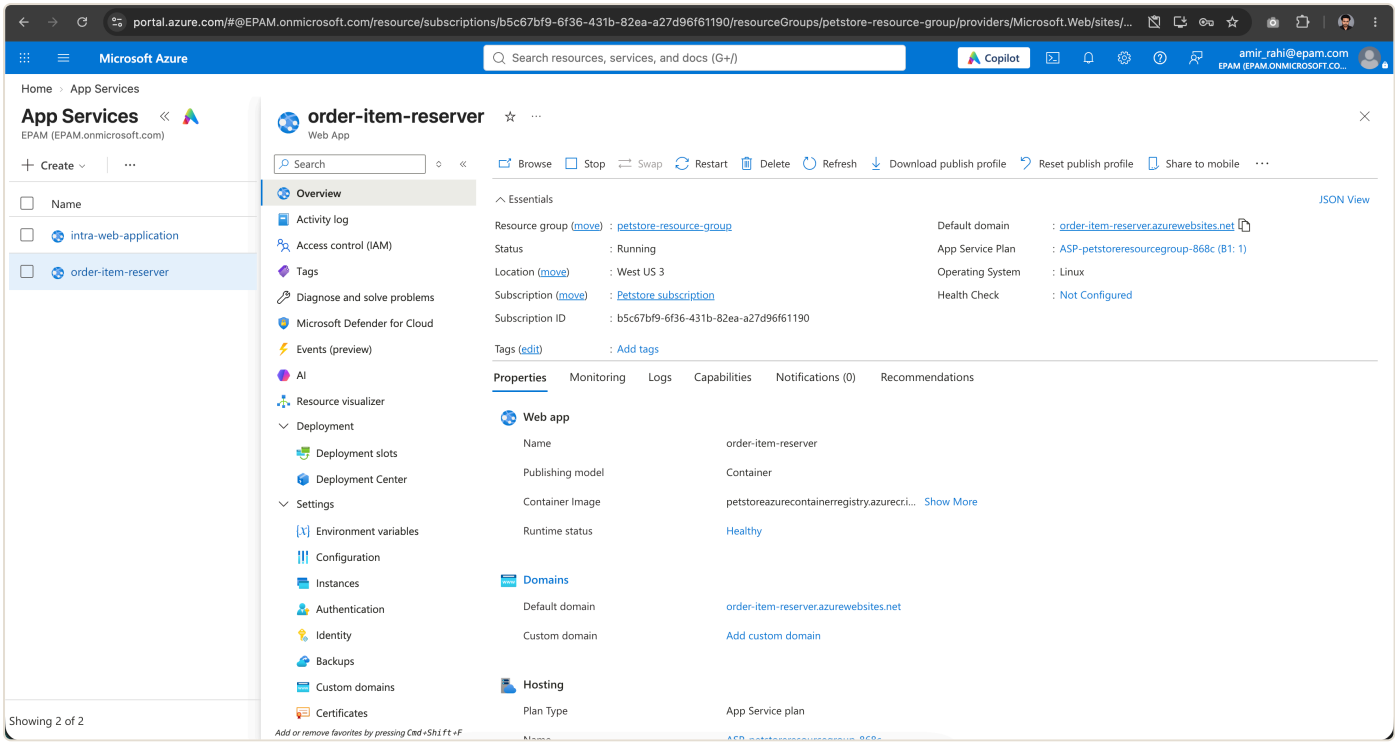

Figure 51. Overview of web app `order-item-reserver`. Status Running, runtime Healthy, default domain `order-item-reserver.azurewebsites.net`.

The same deployment answers on the public hostname. Opening the root path returns the plain text `order-item-reserver is running`, which is the home controller added for this release. That response is HTTP 200, so the container that was pushed is the build that contains the controller.

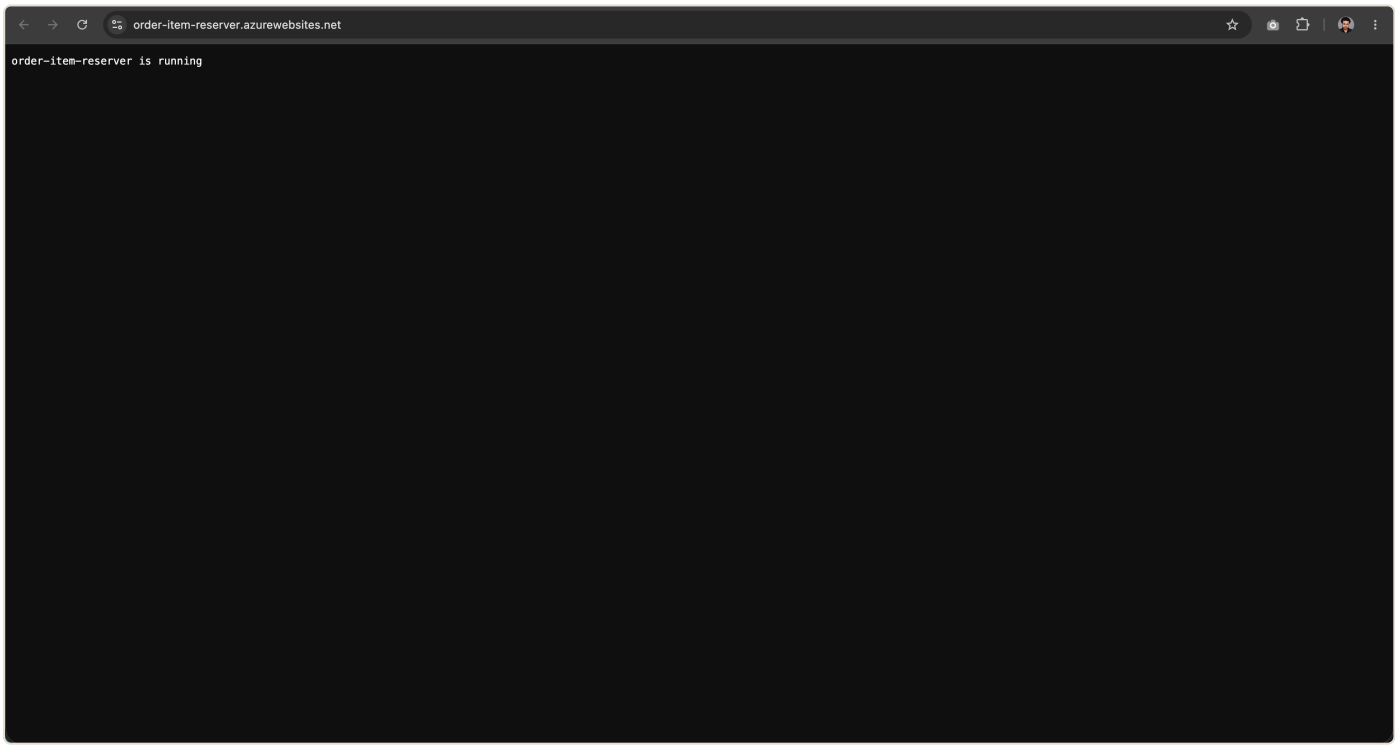

Figure 52. `https://order-item-reserver.azurewebsites.net/` after the new image was deployed. The page body is `order-item-reserver is running`.

Work item: modify and deploy the application. GitHub Actions workflow `order-item-reserver` on `main` built that image and pushed it to the registry. Run 40477315156, job `push`, succeeded in 56 seconds. Checkout, Azure login, ACR login, Docker Buildx, and Build and push all completed.

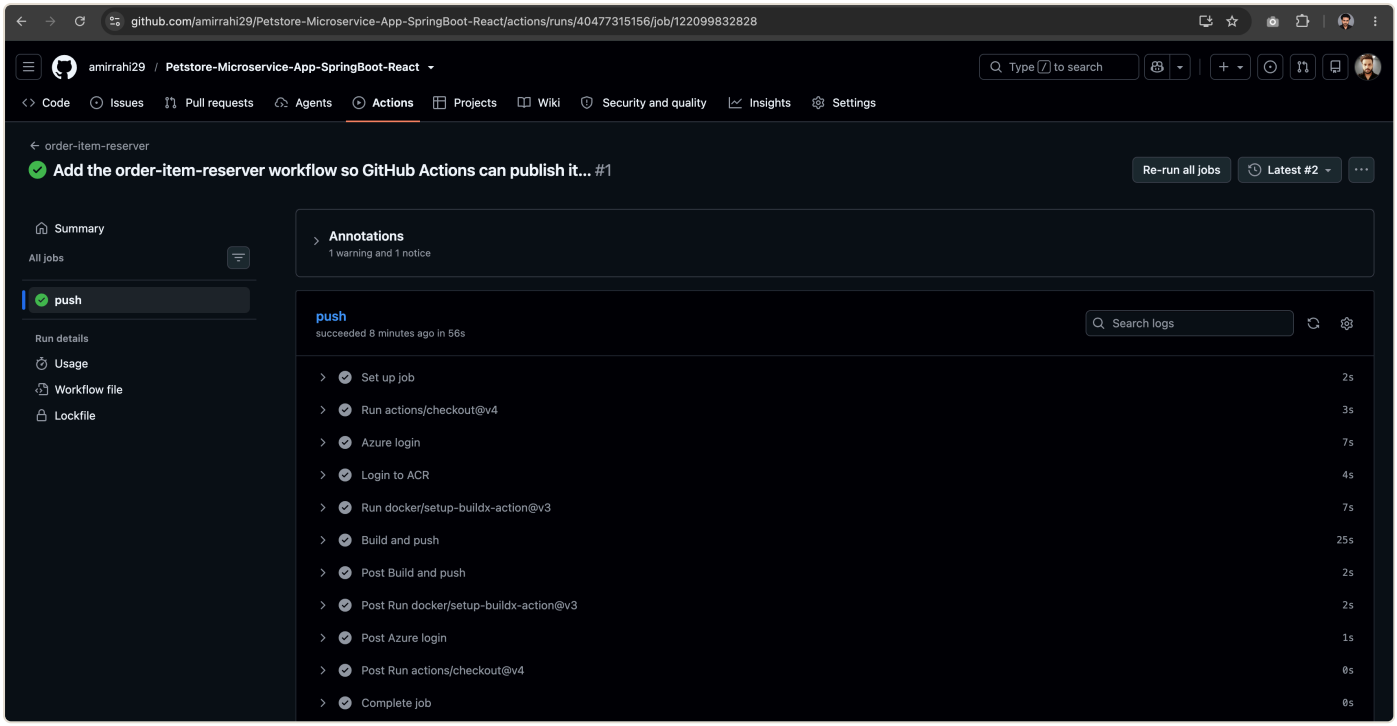

Figure 53. Actions run "Add the order-item-reserver workflow so GitHub Actions can publish it… #1". The `push` job is green, including Build and push.

After that workflow published the image, the App Service overview still shows the same site Running and Healthy, with the container image from `petstoreazurecontainerregistry`. This is the deployed application the queue consumer runs in.

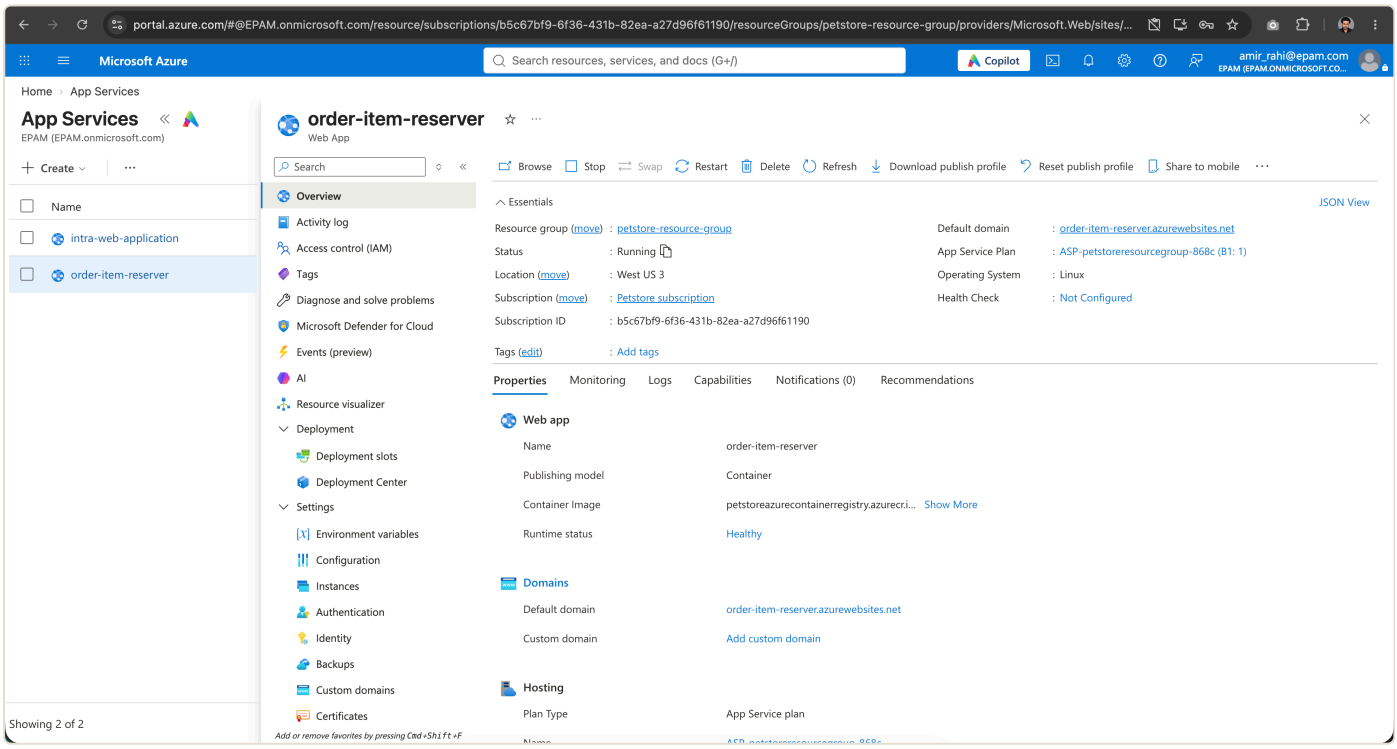

Figure 54. `order-item-reserver` after the image publish. Publishing model Container, runtime status Healthy, location West US 3.

Service Bus messaging and the order item reserver

Work item: configure Azure Service Bus for application messaging. Namespace `petstore-sbns` is in `petstore-resource-group`, West US 3, pricing tier Basic, provisioning Succeeded. The host name is `petstore-sbns.servicebus.windows.net`. Local authentication is enabled. The request chart shows 14 successful requests and no server errors.

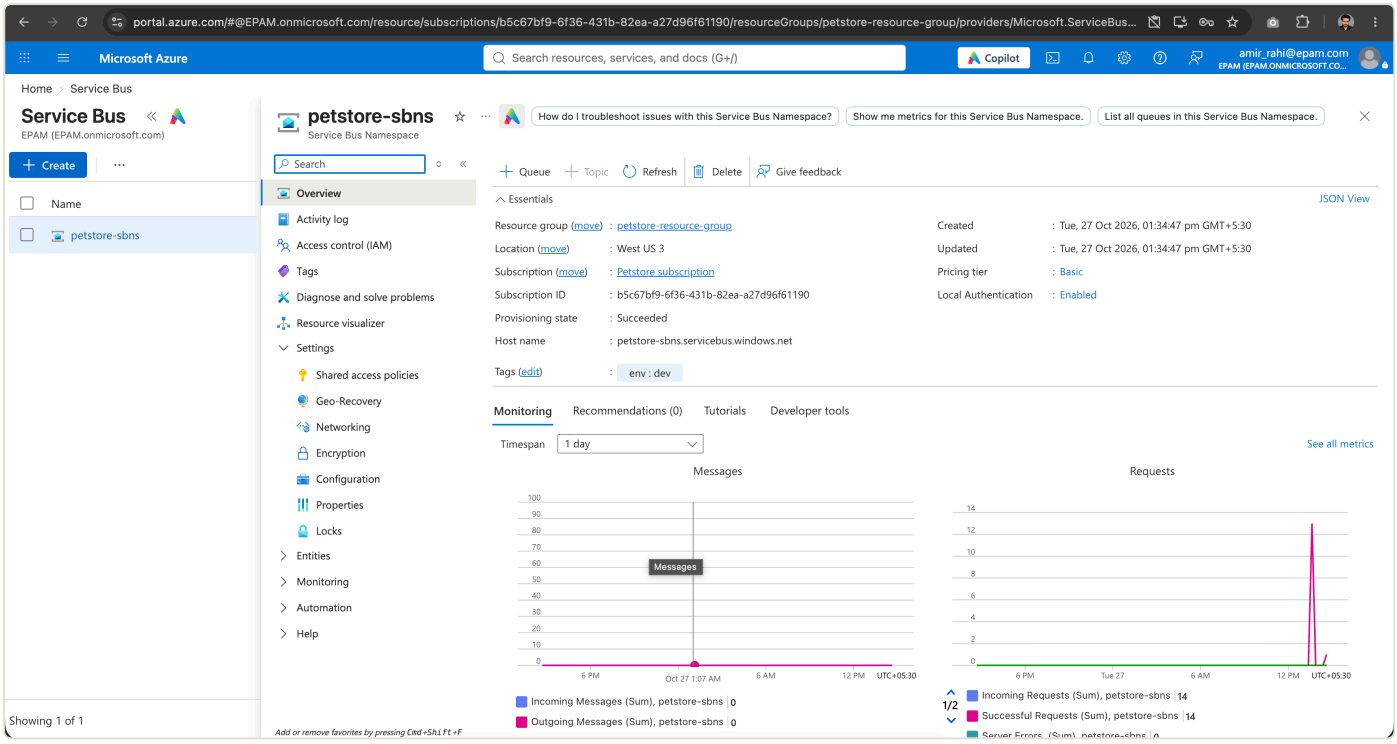

Figure 55. Overview of `petstore-sbns`. Tag `env: dev`. Incoming and outgoing message sums are 0 on the day chart; successful requests peak at 14.

The namespace has one queue, `order-item-reservations`, and its status is Active. Message count, active messages, dead-letter messages, and scheduled messages are 0 on this view. This is the queue name configured for the reserver.

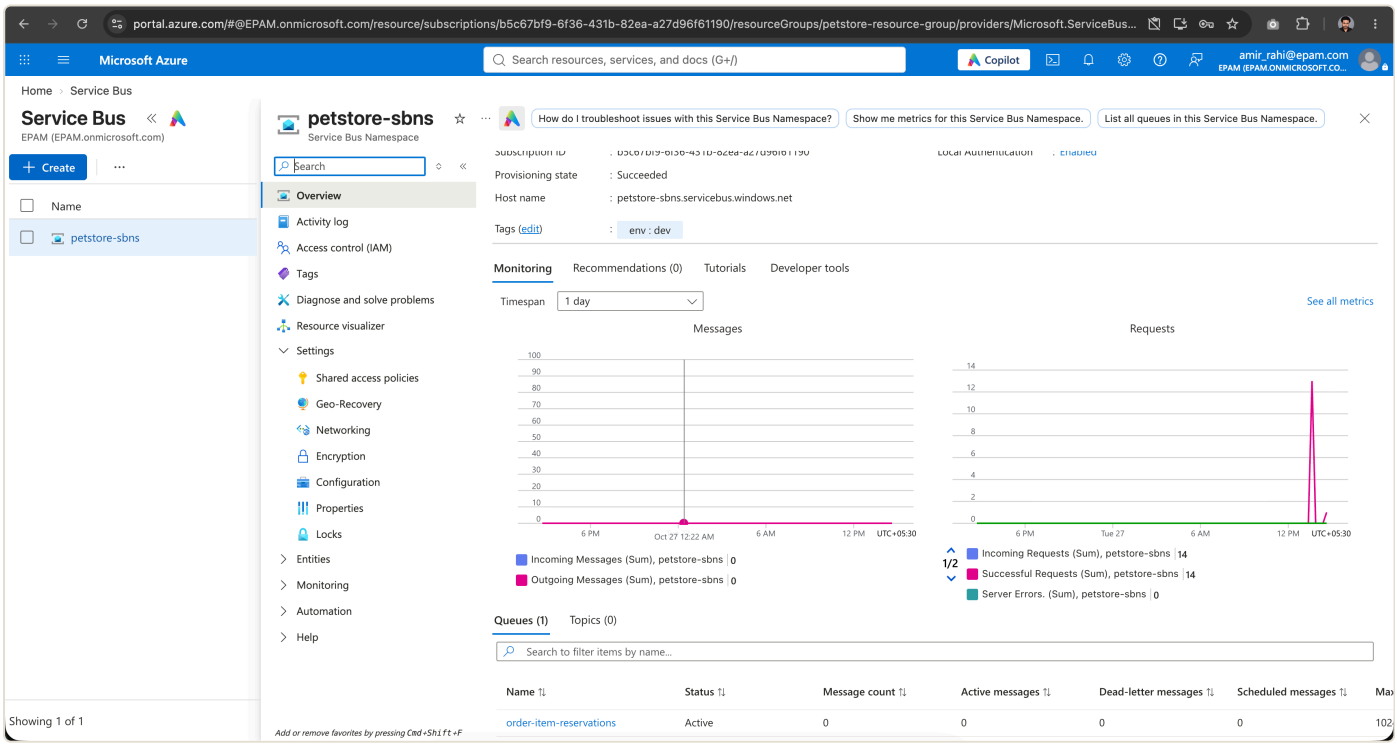

Figure 56. Queues on `petstore-sbns`. `order-item-reservations` is Active with 0 messages.

Work item: create and deploy the order item reserver. The deployed Spring Boot application is the consumer of this queue. The queue itself is Active in namespace `petstore-sbns`. Its URL is `https://petstore-sbns.servicebus.windows.net/order-item-reservations`. Max delivery count is 10, message time to live is 14 days, and the lock duration is 1 minute. Sessions, partitioning, and duplicate detection are disabled. At the time of this screenshot the queue held 0 active, scheduled, and dead-letter messages.

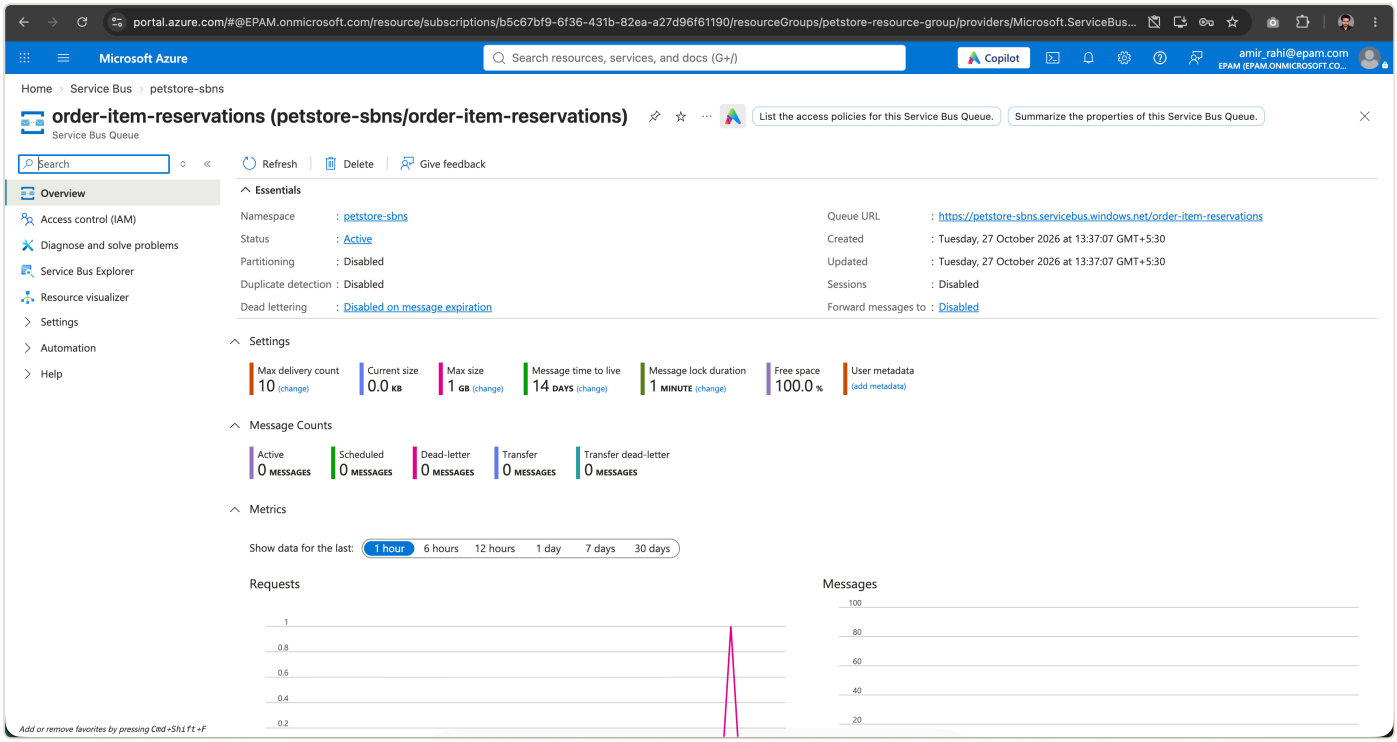

Figure 57. Queue `order-item-reservations`. Status Active, max delivery count 10, lock duration 1 minute, size 0.0 KB of 1 GB.

A one-hour metric on the same queue shows one successful request and no server errors, user errors, or throttling. Incoming and outgoing message sums on that chart are 0. The namespace accepted the call that the reserver uses to reach this queue.

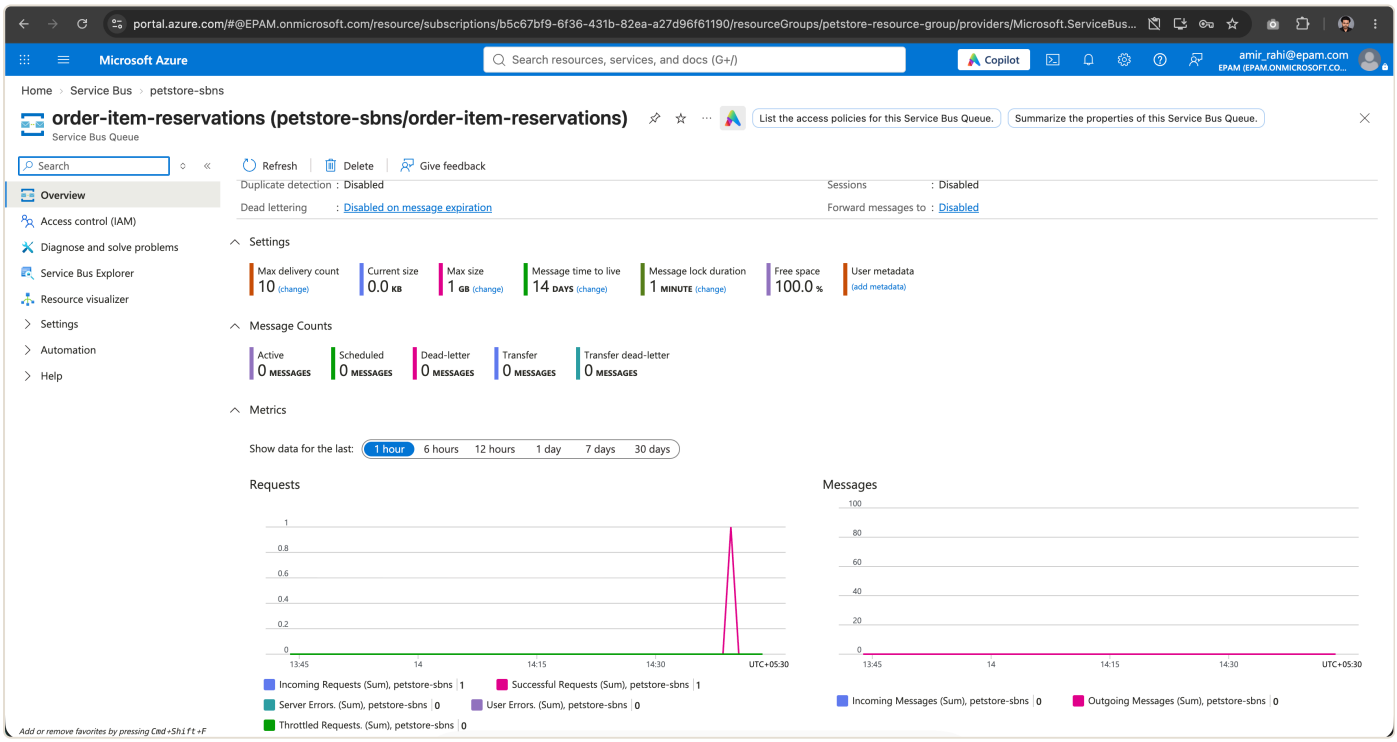

Figure 58. One-hour metrics for `order-item-reservations`. Successful requests 1. Server errors, user errors, and throttled requests are 0.

Blob upload retry

Work item: implement a retry policy for blob storage uploads. Storage account `orderitems` is the account the reserver uploads to. It is a Standard general-purpose v2 account in West US 3, locally redundant, provisioning Succeeded, disk Available, created 27 October 2026 at 13:45. Secure transfer and storage account key access are enabled. Blob and container soft delete are enabled for 7 days. The

application client uses exponential backoff when an upload is retried; this overview is the account that client is pointed at.

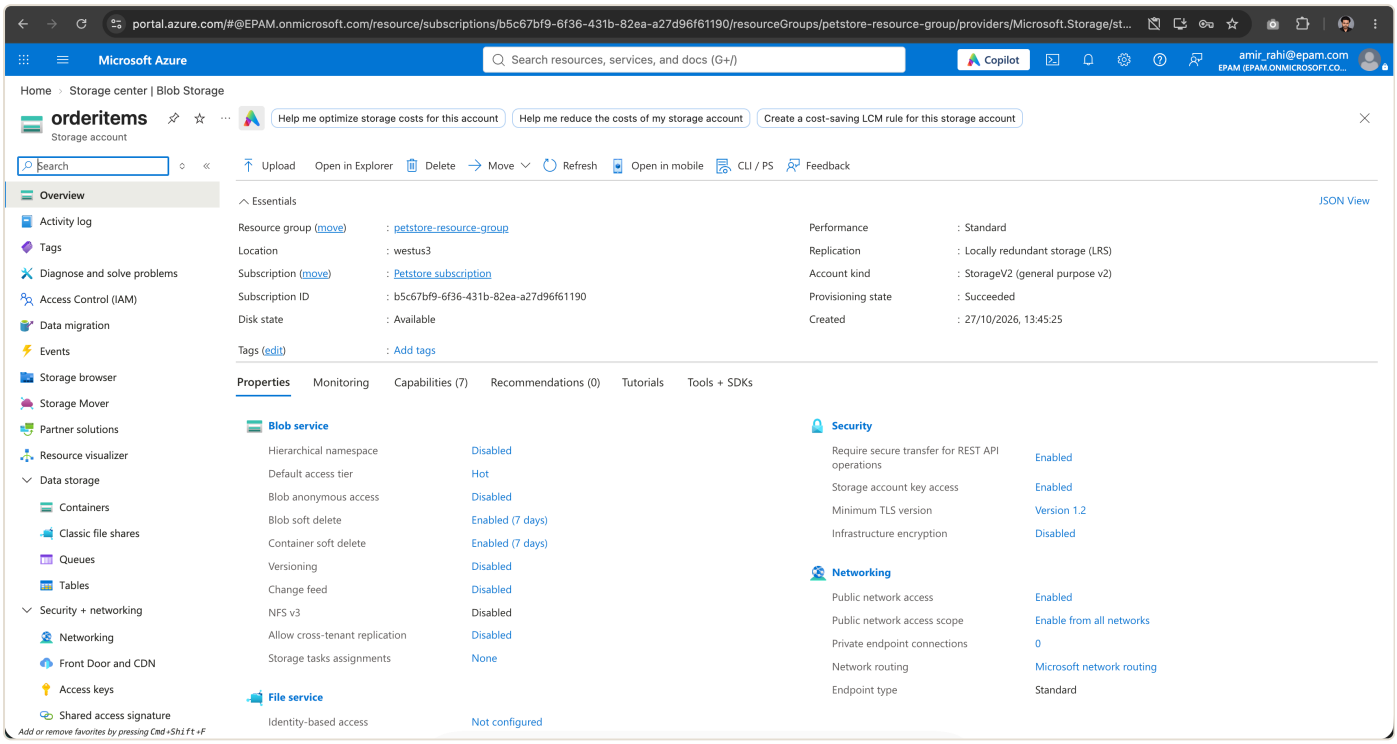

Figure 59. Overview of storage account `orderitems` in `petstore-resource-group`. Hot access tier, anonymous access disabled, public network access enabled from all networks.

Container `order-items` on that account holds one block blob, `Screenshot 2026-10-27 at 2.23.50 PM.png`, 2.13 MiB, access tier Hot, lease Available, last modified 27 October 2026 at 14:46. Authentication for the container blade is the account key. A stored blob here is the upload target used with the exponential retry client.

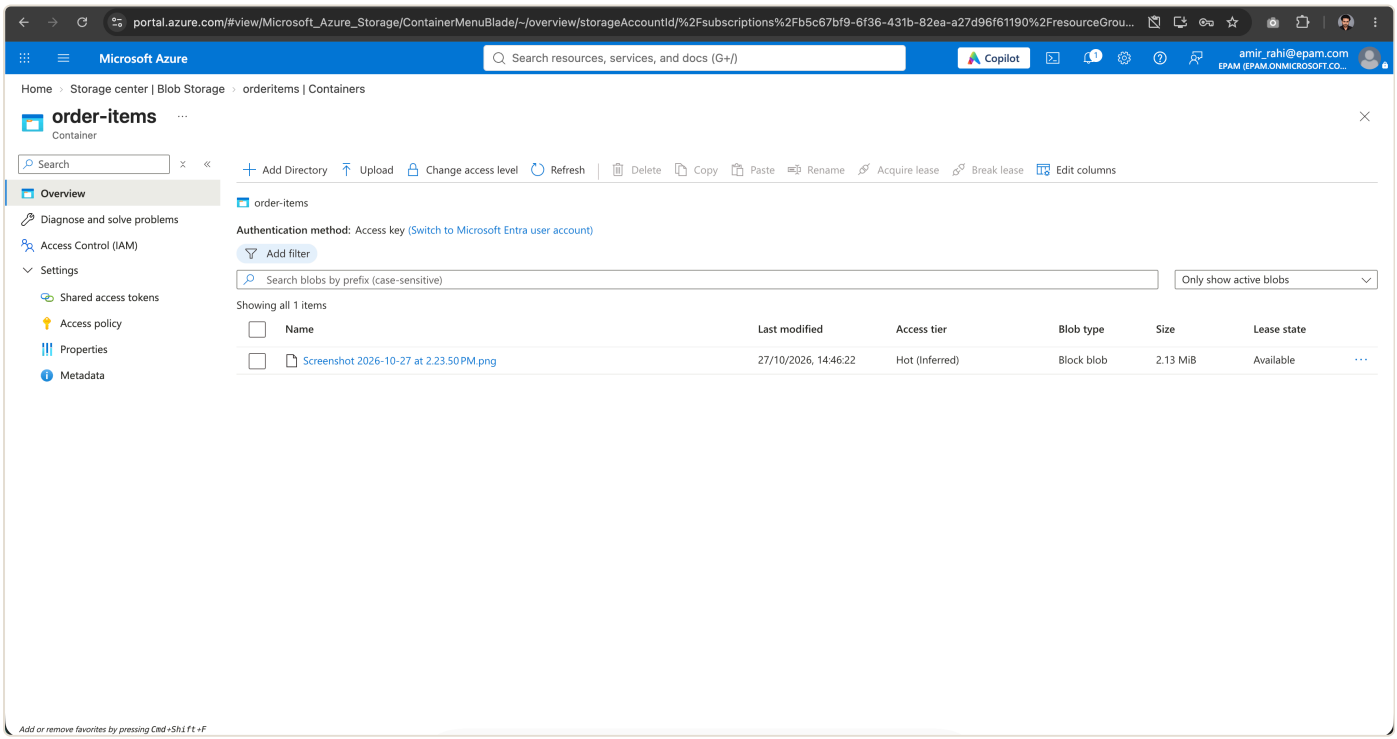

Figure 60. Container `order-items`. One block blob, 2.13 MiB, lease Available.

Fallback Logic App email

Work item: create a fallback Logic App for email notification. Logic App `order-item-reserver-email-fallback` starts with "When an HTTP request is received" and then "Send an email (V2)". The designer shows the workflow saved. The mail action is addressed to `amirrahi29@gmail.com`, with subject "amir subject" and body "amir message body", and the connection is Outlook.

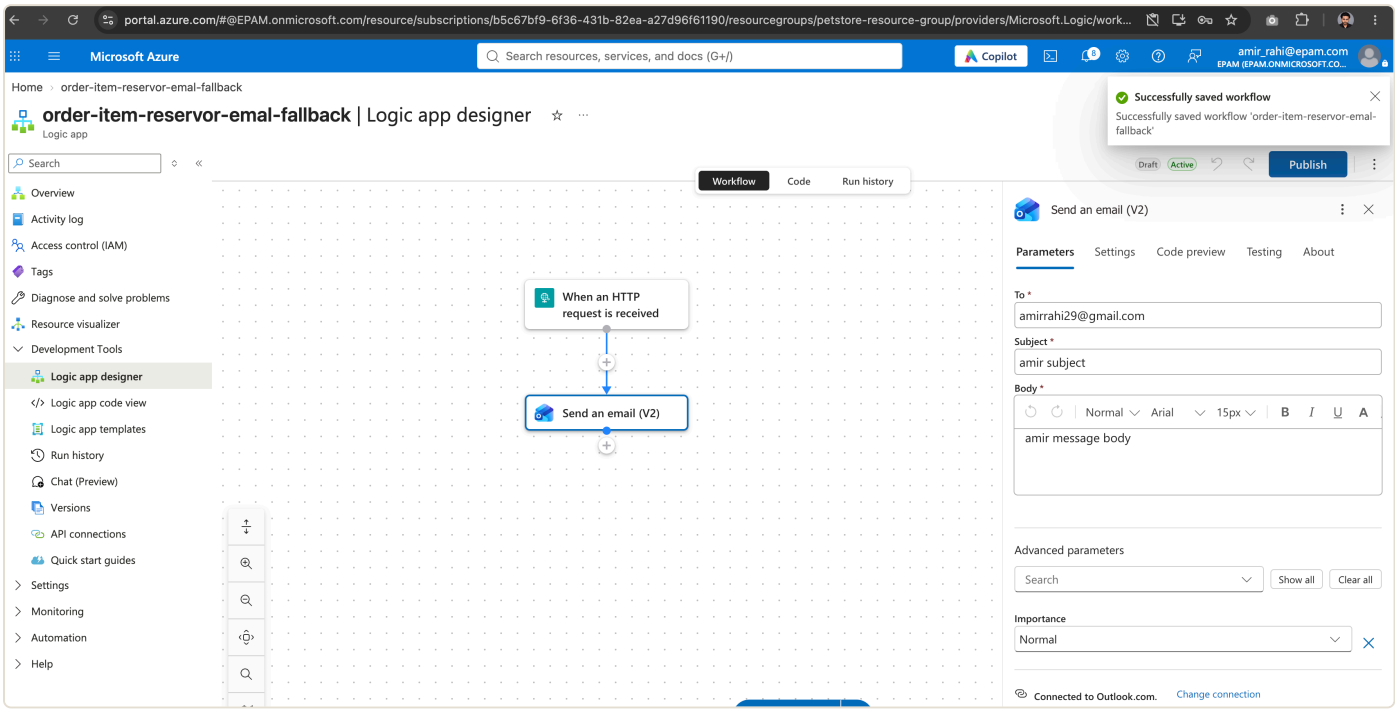

Figure 61. Designer for `order-item-reserver-email-fallback`. The toast says the workflow was saved. Publish is available.

A draft run of that workflow on 27 October 2026 at 2:12 pm succeeded in 772 milliseconds. Both the HTTP trigger and Send an email (V2) show a green check. The email action output status code is 200. This is the fallback path the reserver calls when reservation processing fails.

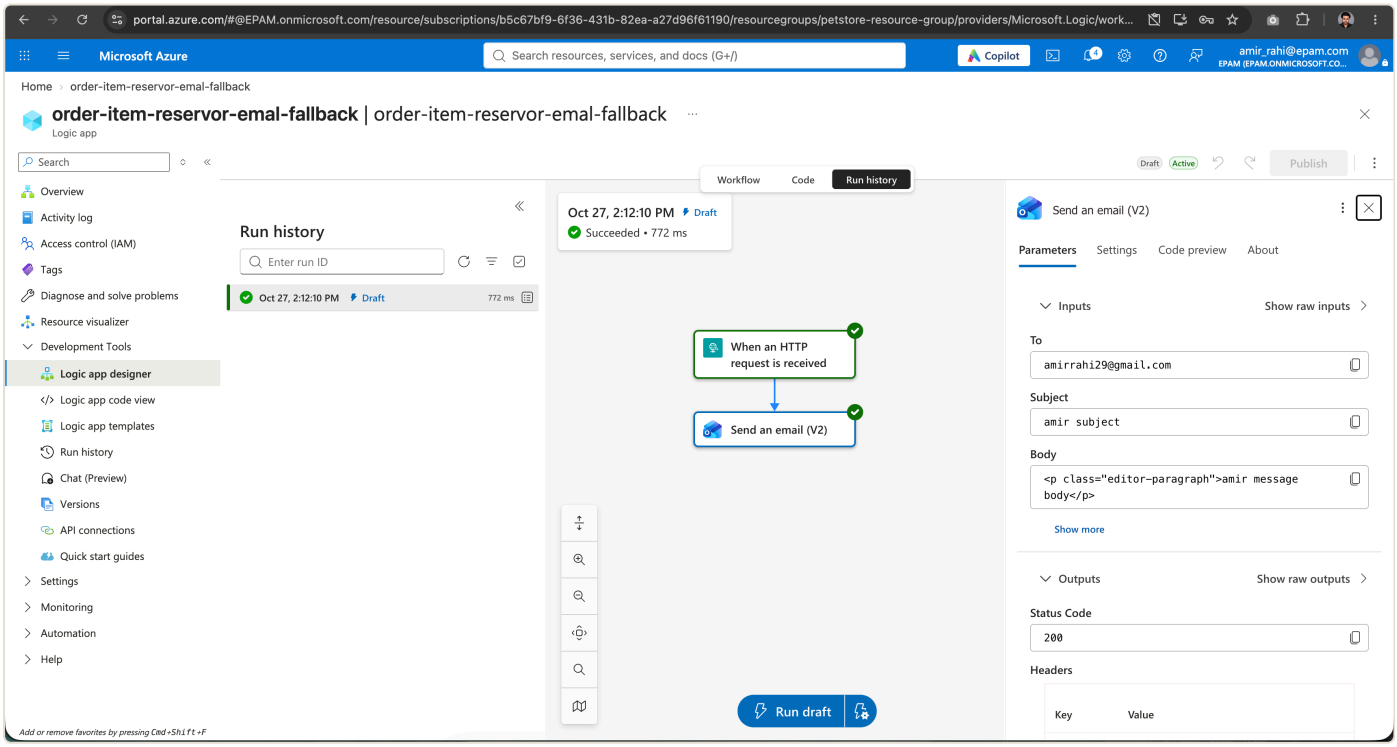

Figure 62. Run history. The 2:12 pm draft run succeeded. Send an email (V2) returned status 200 to `amirrahi29@gmail.com`.

The message from that run arrived in Gmail. The subject is “amir subject”, the sender is Amir Rahi via outlook.com, and the body is “amir message body”. Gmail placed it in Spam and noted that it resembles messages previously identified as spam. The send itself completed, which matches the status 200 in Figure 62.

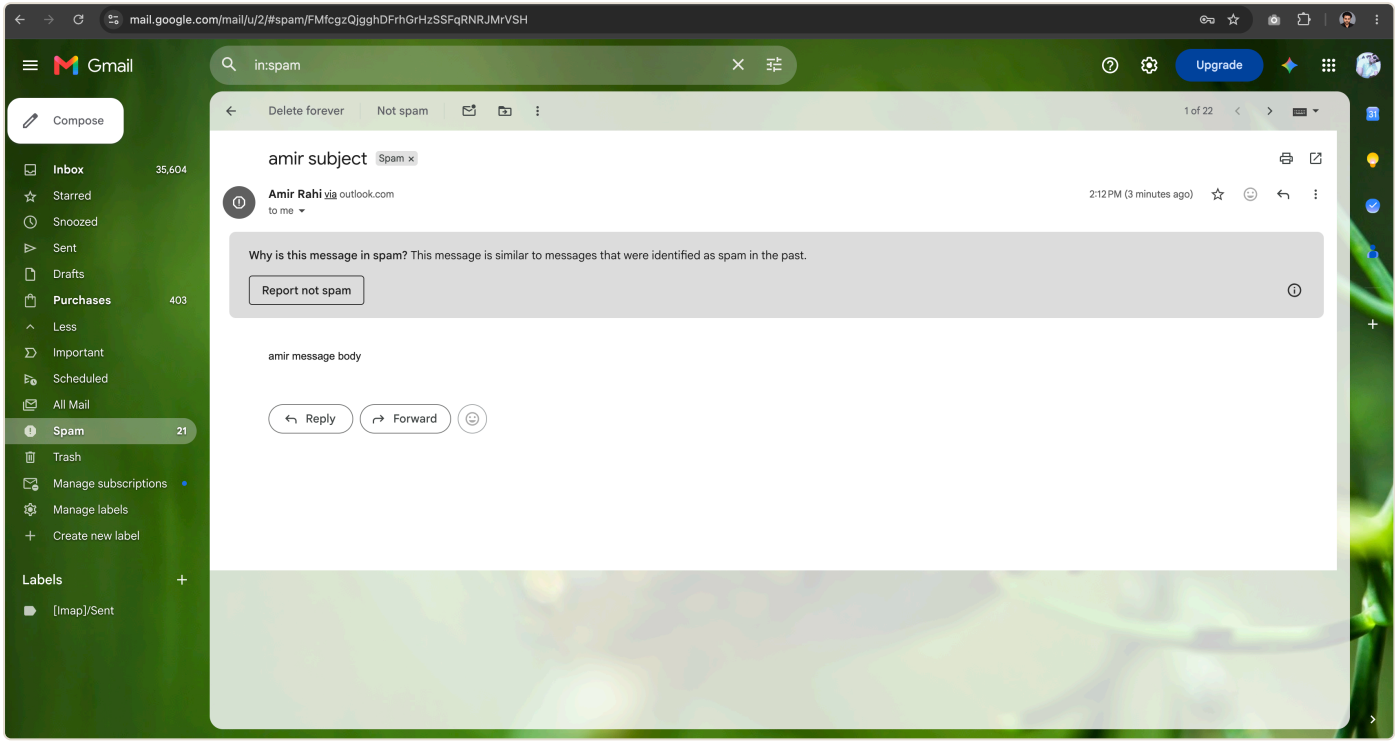

Figure 63. Gmail, Spam, message “amir subject” received at 2:12 pm from the Logic App email action.

22. Azure Container Apps

Container App `petstore-container-app` is a second host for the order item reserver image. It runs in `DefaultResourceGroup-USW3`, West US 3, environment `managedEnvironment-DefaultResource-8855`. The four records below are deploy, ingress, scale, blue-green traffic, and the order API address used as application configuration. Sections 1 through 21 are unchanged.

Deploy a container app

The first overview shows the app Running in West US 3 before ingress was turned on. The resource group is `DefaultResourceGroup-USW3`. The environment type is Workload profiles, and the development stack is Generic. Application URL says Ingress disabled, so there was no public address yet.

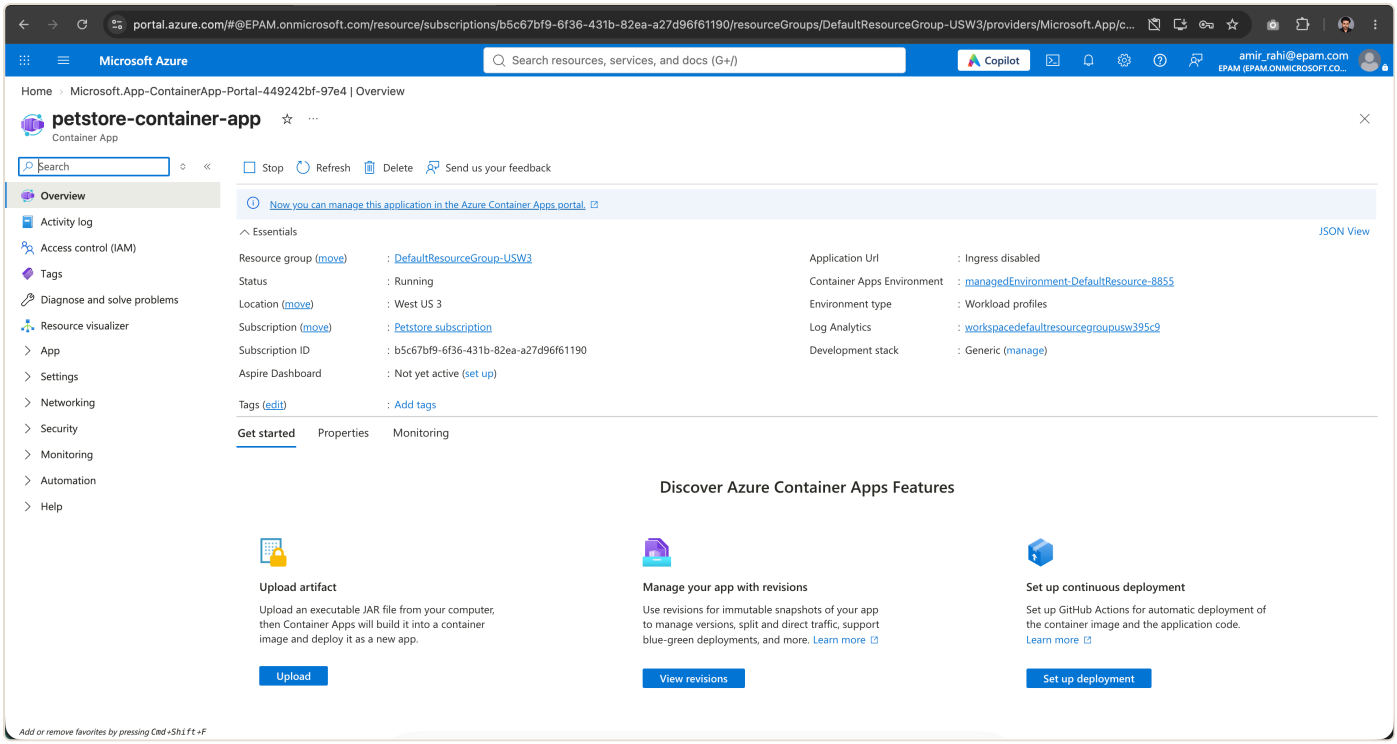

Figure 64. Overview of `petstore-container-app`. Status Running, location West US 3, ingress disabled.

Ingress was then enabled for external HTTP traffic. The target port is `8084`, which is the port the order item reserver container listens on. Transport is HTTP/1. Insecure connections are off, so callers use HTTPS. The endpoint is `https://petstore-container-app.jollyhill-a23e8734.westus3.azurecontainerapps.io`.

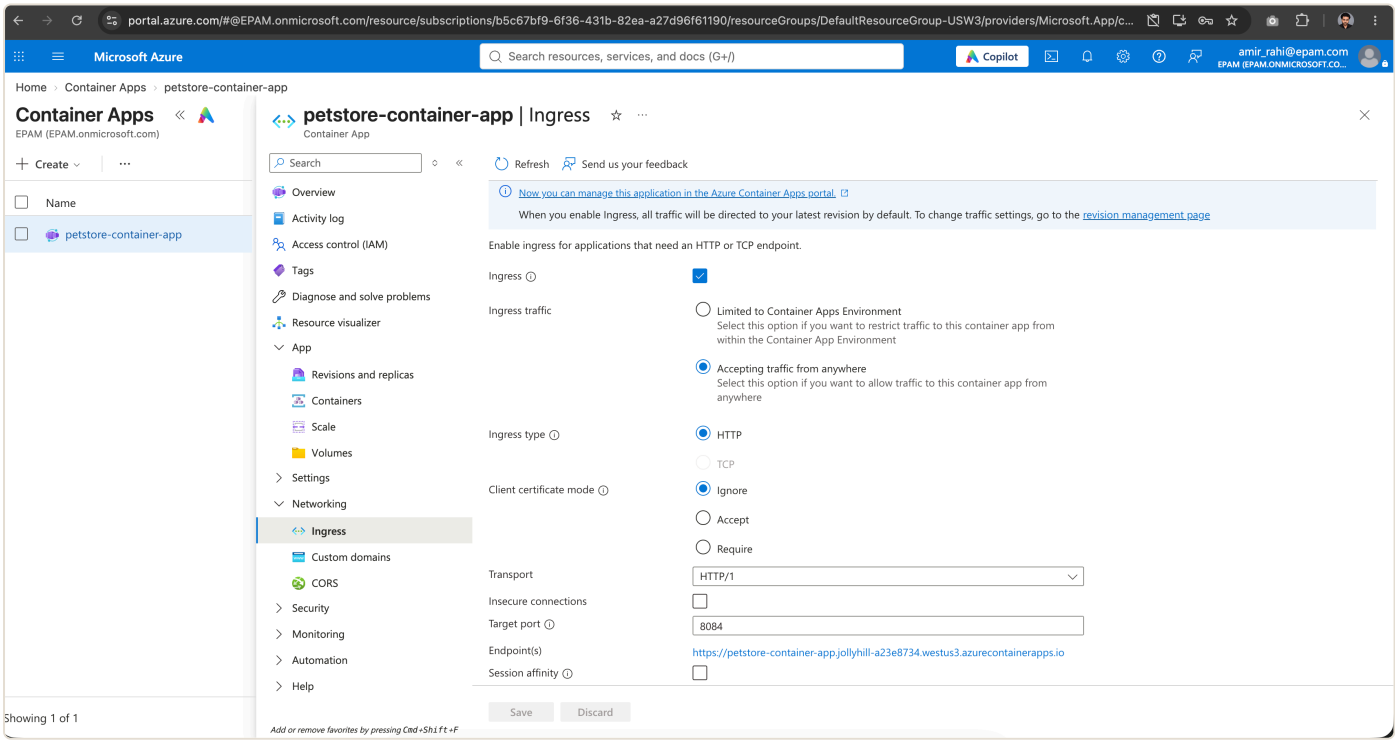

Figure 65. Ingress for `petstore-container-app`. External traffic, HTTP, target port 8084.

Opening that hostname returns the plain text `order-item-reserver is running`. That is the same home response as the App Service copy of this application, so the container behind the ingress is the reserver image.

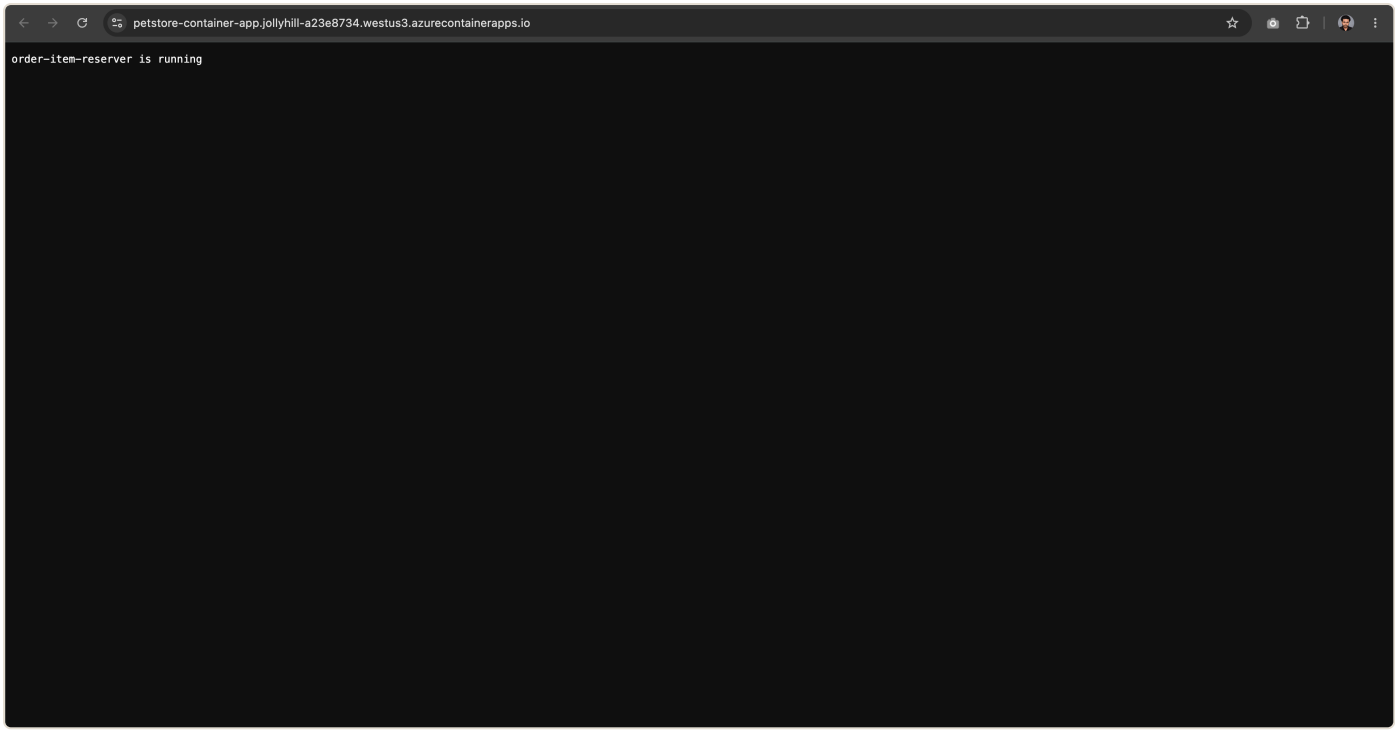

Figure 66. `https://petstore-container-app.jollyhill-a23e8734.westus3.azurecontainerapps.io` after ingress. The page body is `order-item-reserver is running`.

Auto-scaling

Scale is set on revision `petstore-container-app--0000001`. Minimum replicas are 1 and maximum replicas are 3, so the app stays up and can add two more replicas. The HTTP scale rule is named `http-requests`, type HTTP scaling, and the concurrent request threshold is 10. Cooldown is 300 seconds and the polling interval is 30 seconds. The screenshot shows one replica running.

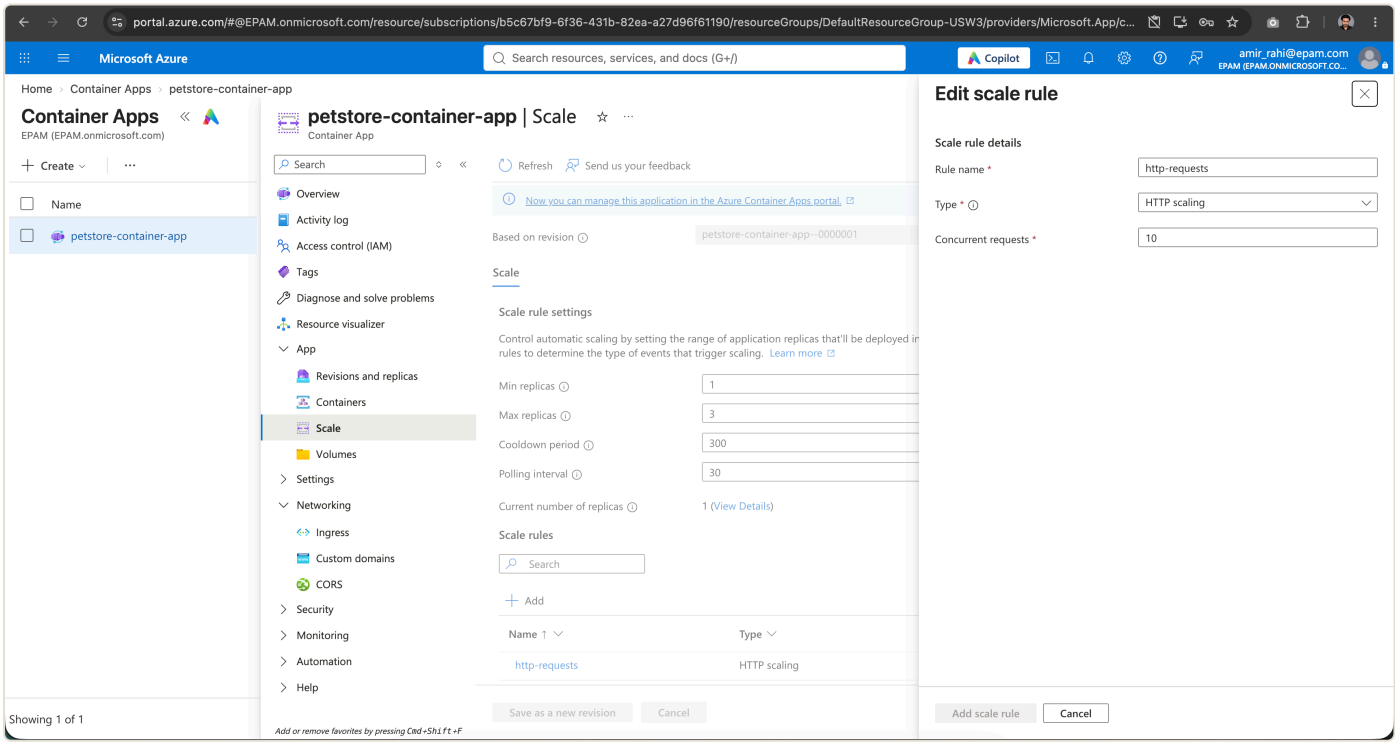

Figure 67. Scale for `petstore-container-app`. Rule `http-requests`, 10 concurrent requests, min 1, max 3.

Blue-green deployment

Container Apps has no separate blue-green switch. Two revisions carry the traffic. `petstore-container-app--0000001`, created 27 October 2026 at 15:40, is the blue revision. `petstore-container-app--green`,

created at 15:59, is the green revision. Both are Running and Active, each with one replica, and each receives 50 percent of traffic.

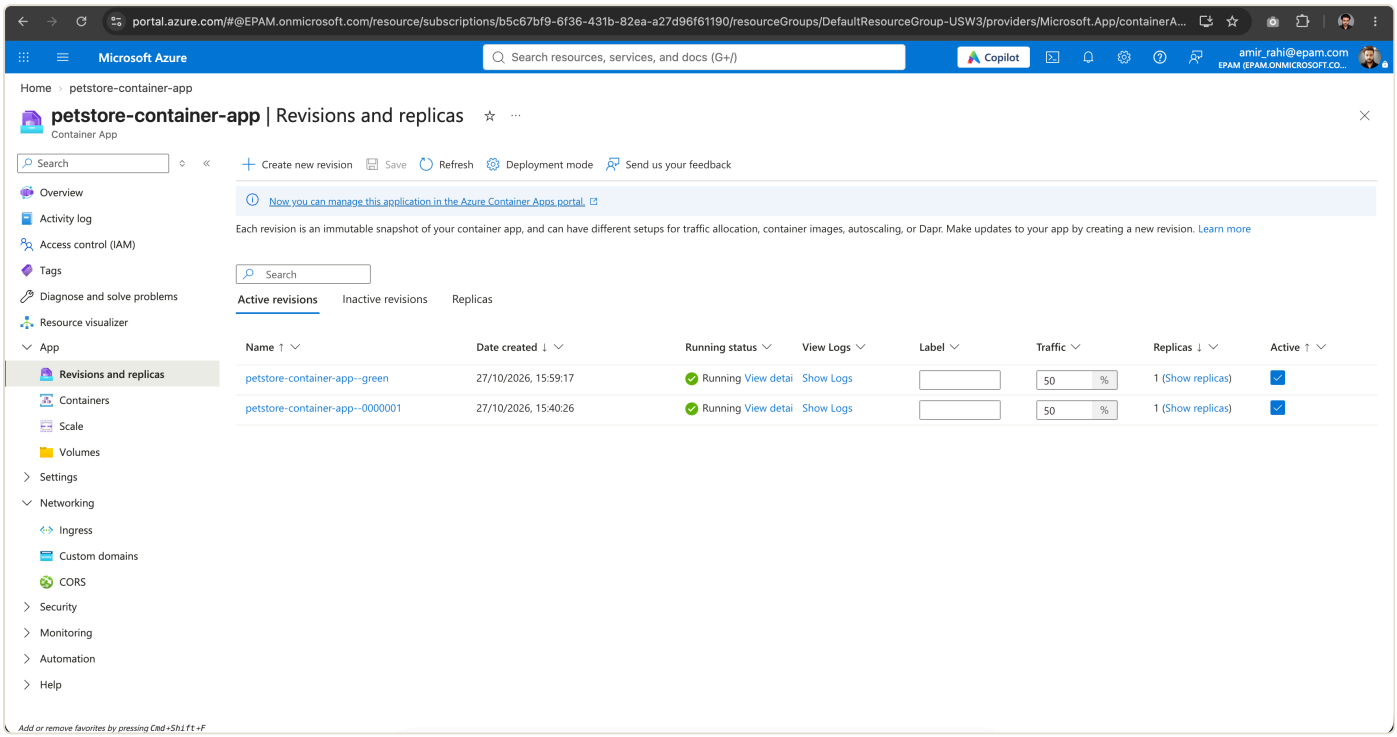

Figure 68. Revisions and replicas. `petstore-container-app--green` and `petstore-container-app--0000001` are both Running at 50 percent traffic.

Application configuration

The order API address used for this container app is a small web app, `petstore-dummy-order-api`, in West US 3. It has no database. Its root path returns a fixed JSON list of two sample orders: dog food, quantity 2, and cat toy, quantity 1. That URL is the value stored as `petstore.order-api-url` on revision `petstore-container-app--dummy-orders`.

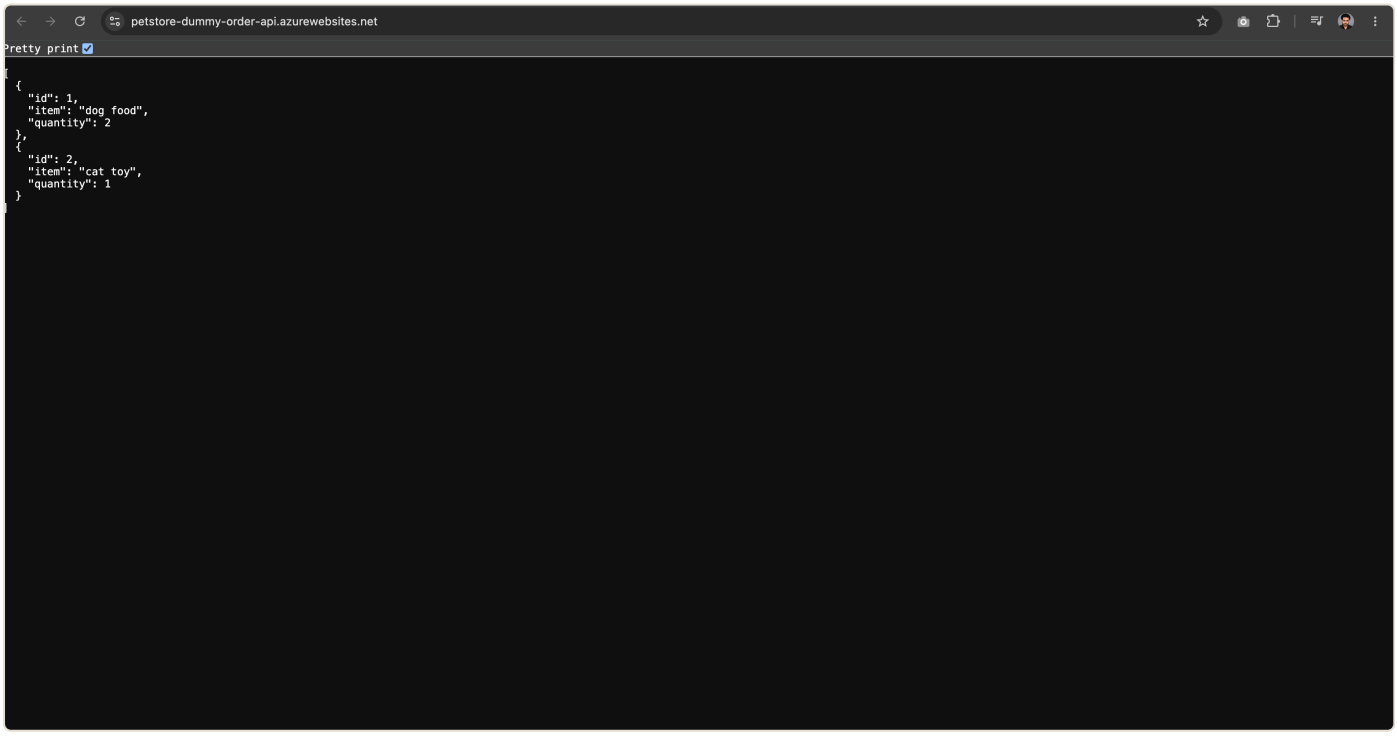

Figure 69. `https://petstore-dummy-order-api.azurewebsites.net`. The response is the sample order list.